Introduction to Neuroscience – Behavioral Neuroscience

Social Neuroscience

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2026

Outline

Diversity of social behaviors

Social identity

Aggression

Parenting

Prosocial behaviors

What next?

Nest building – Attracting mates



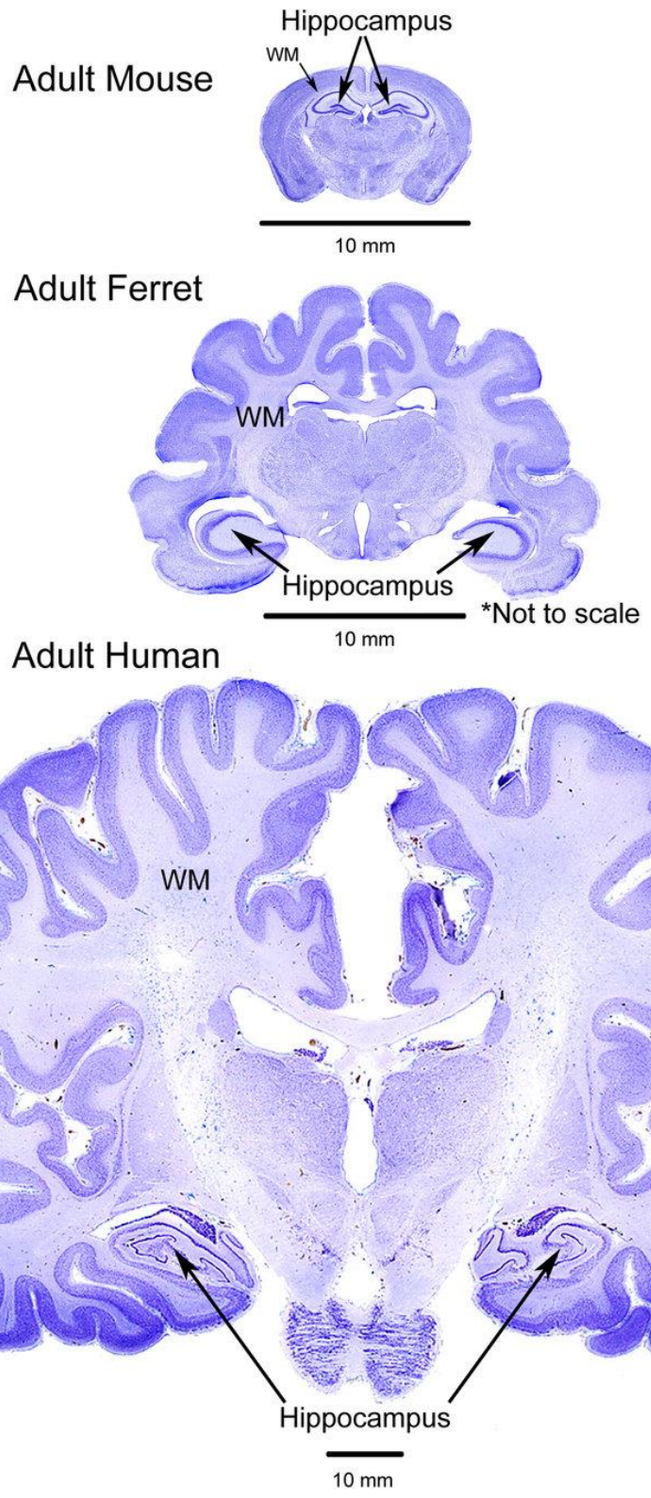
Coordinated Behavior – Hermit crab shell exchanges

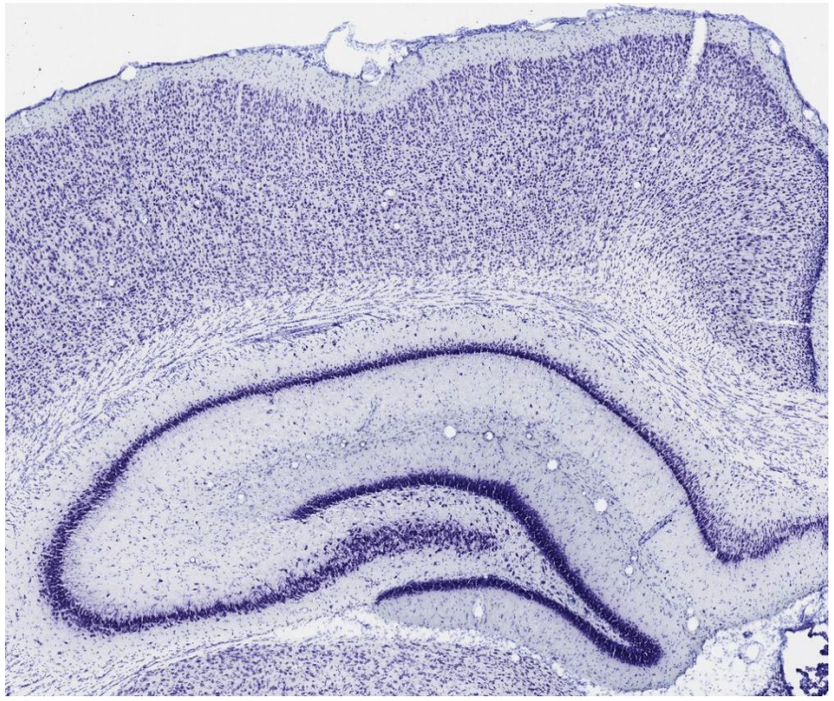


Social identity

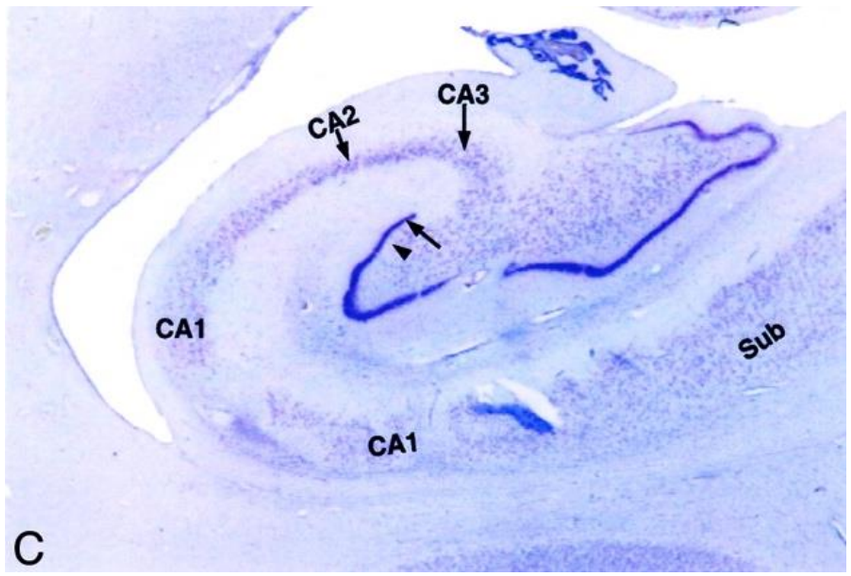


Social identity and memory – Hippocampus



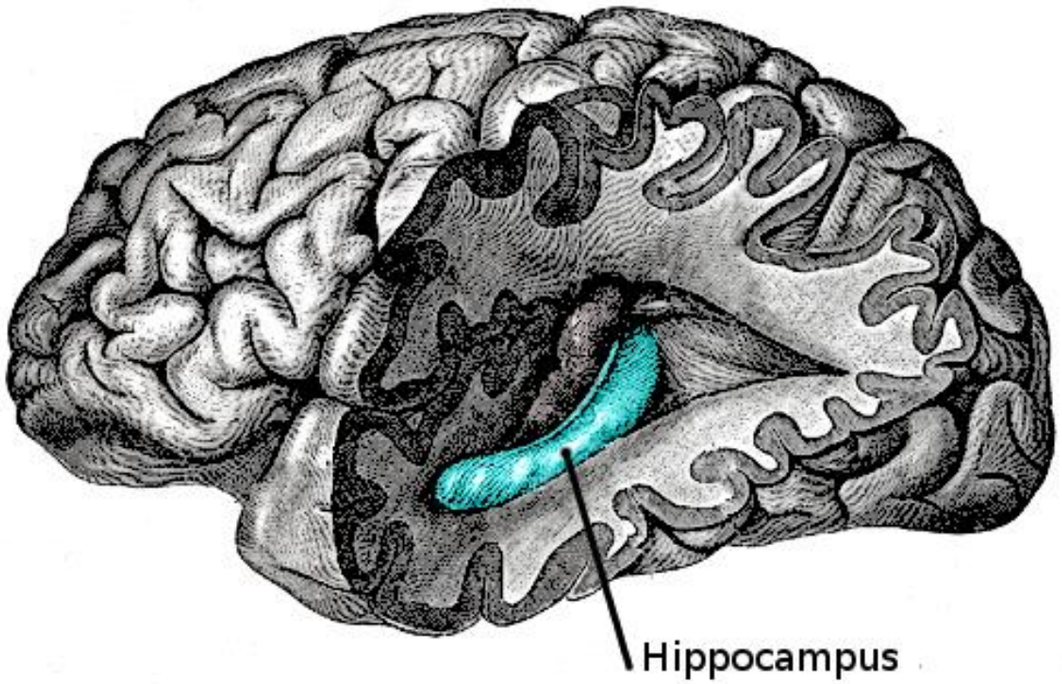


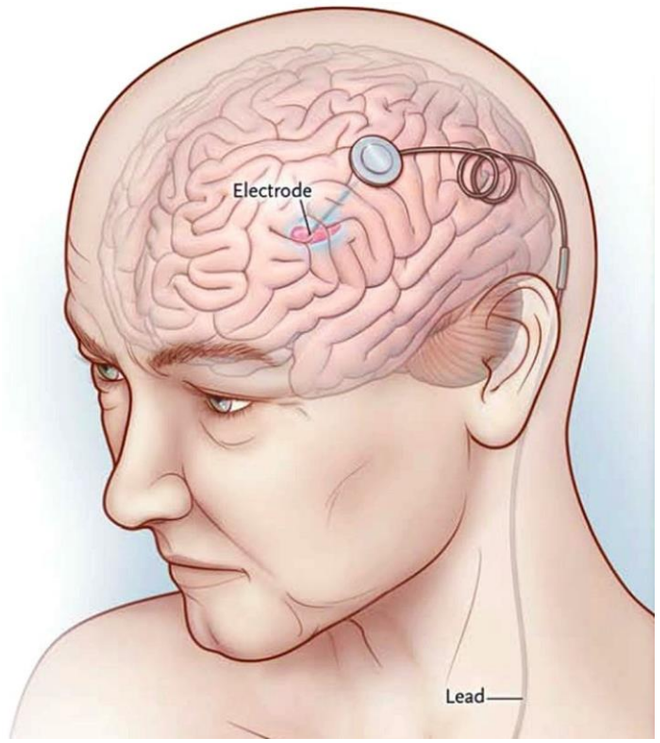
Mouse

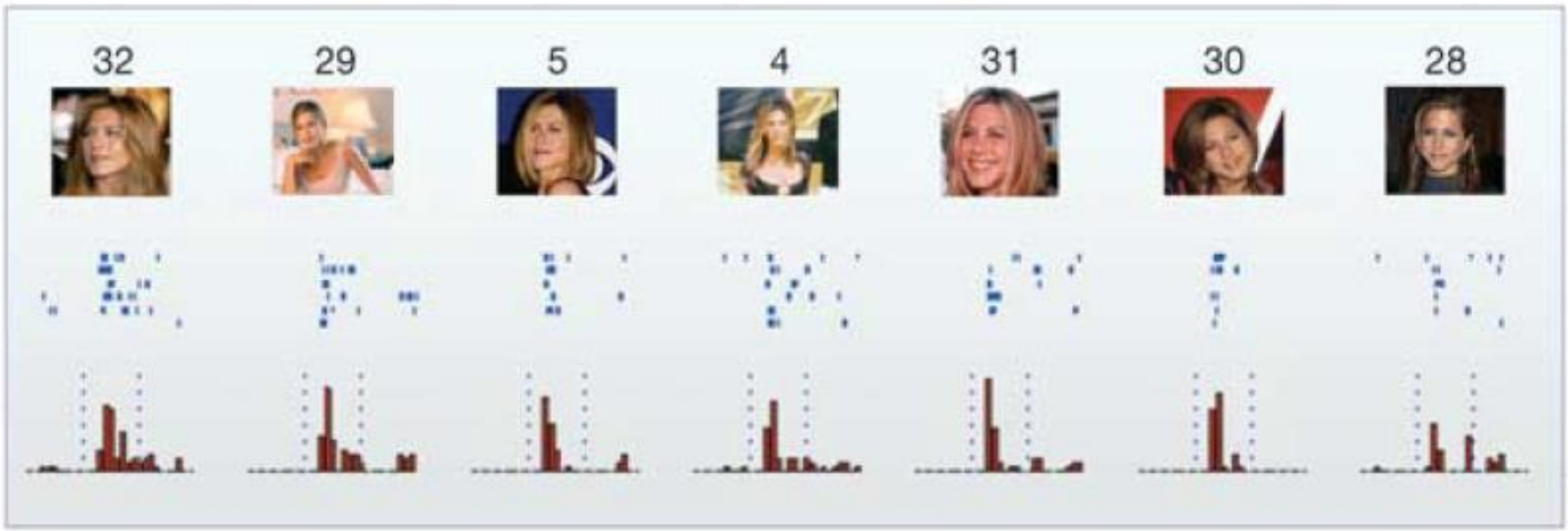


Human

Social Identity – Hippocampal representations of social identity

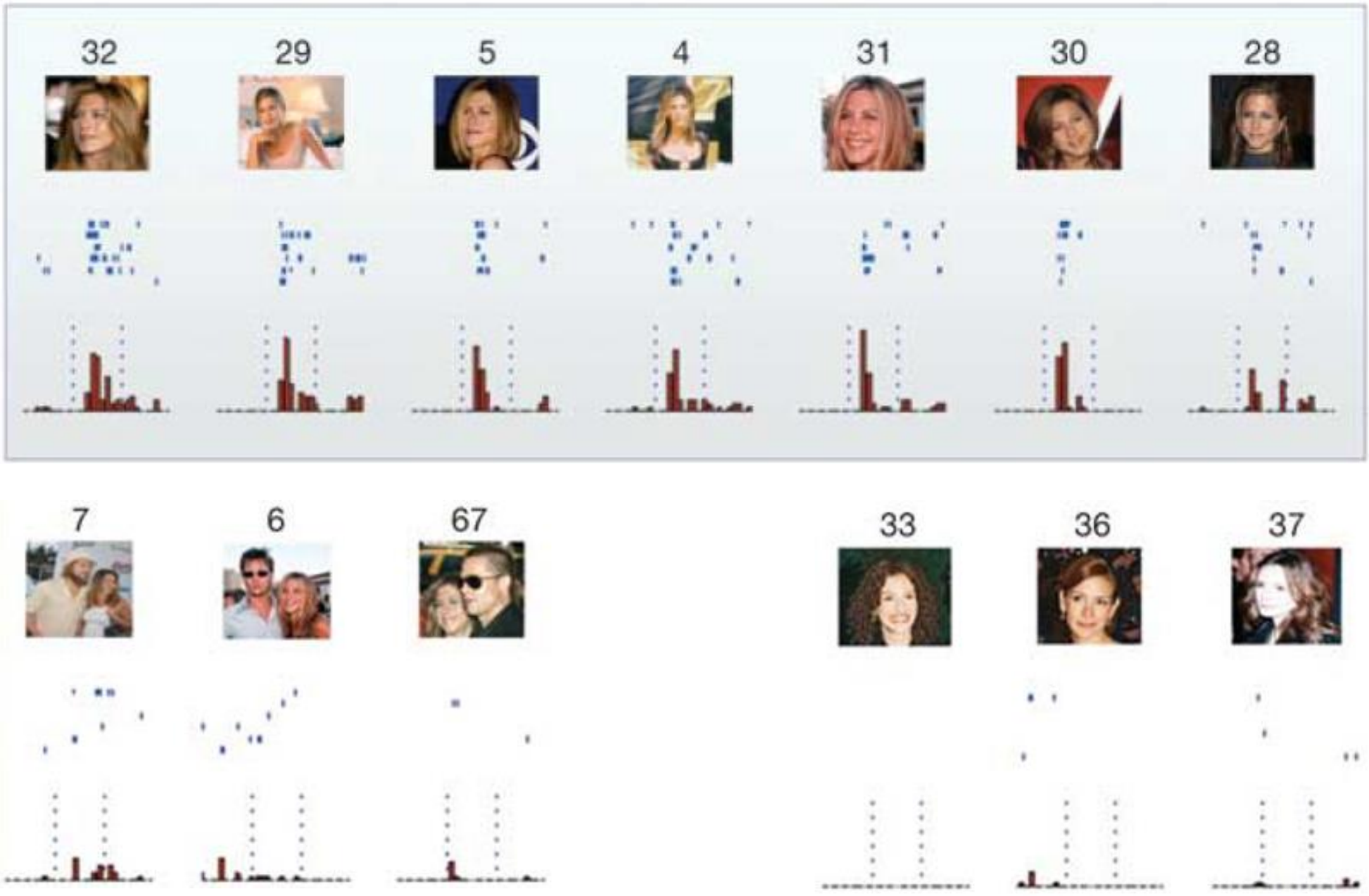






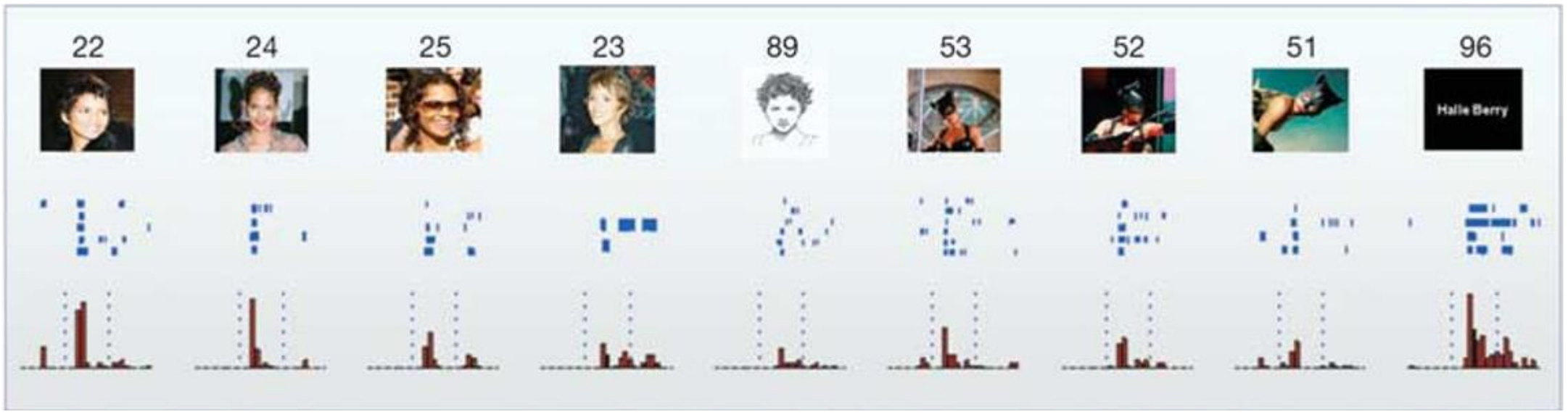
Fried and colleagues

Social Identity – Jennifer Aniston Cell



Fried and colleagues

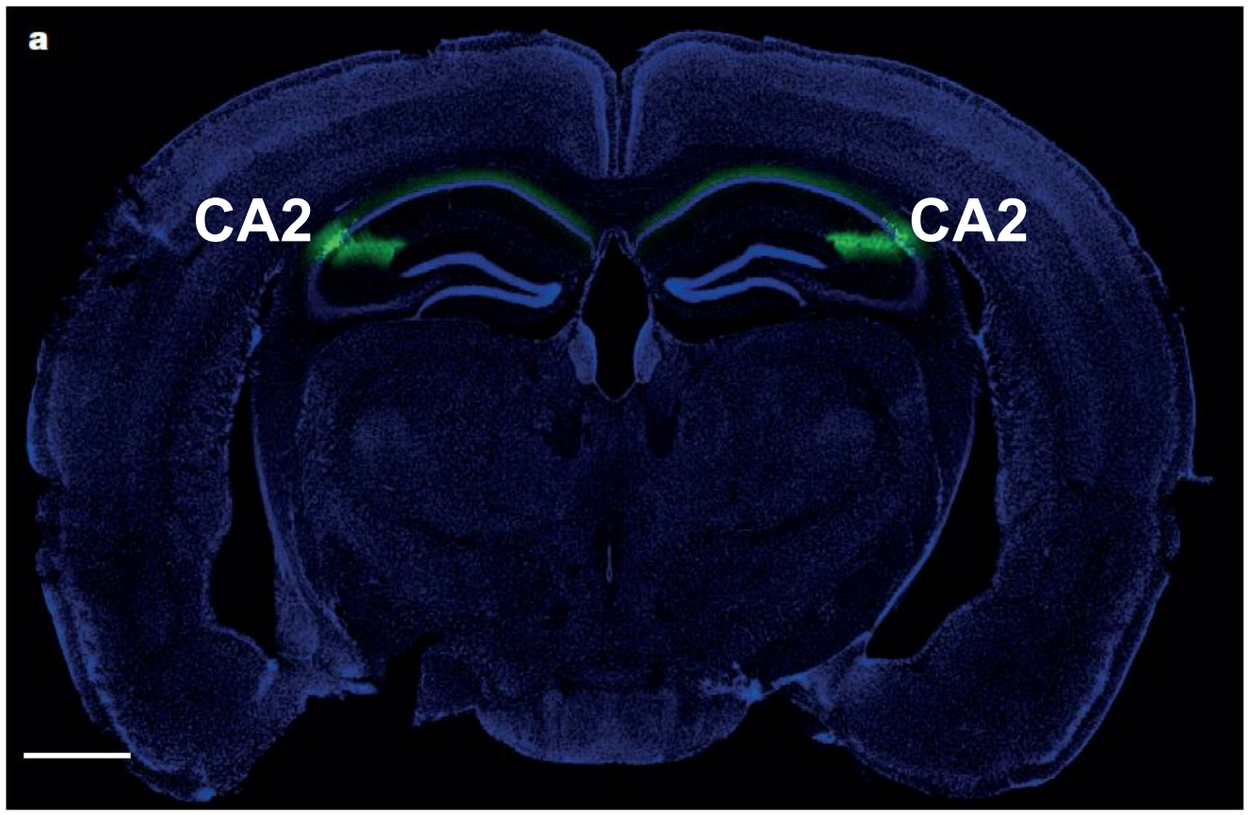
Social Identity – Halley Berry Cell

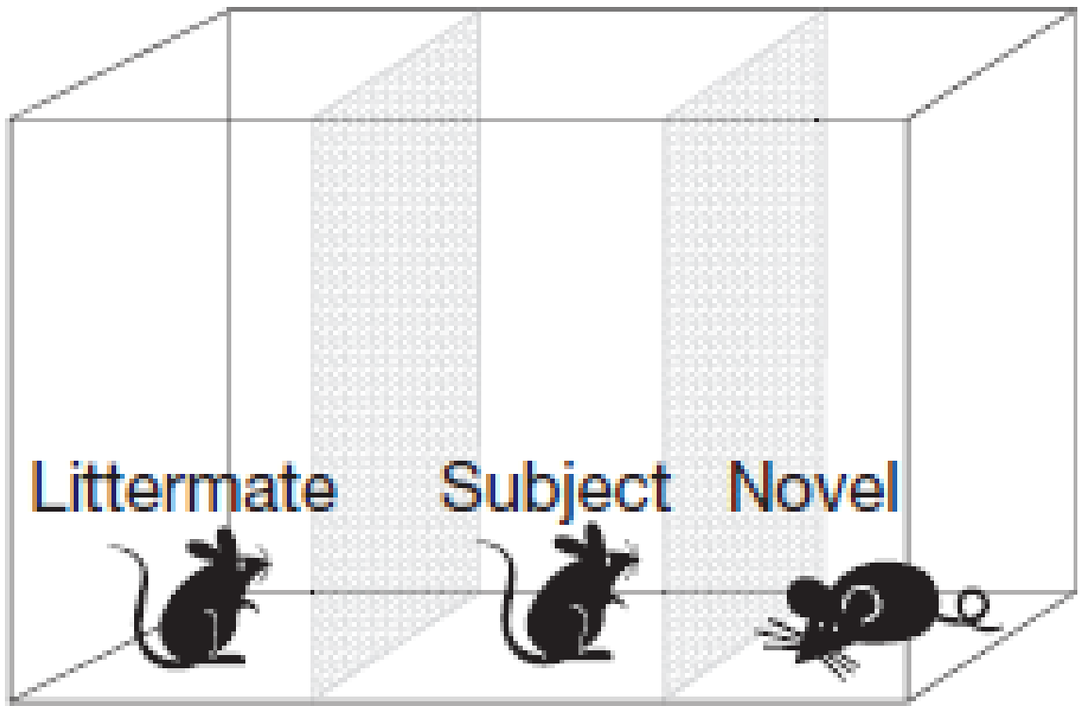


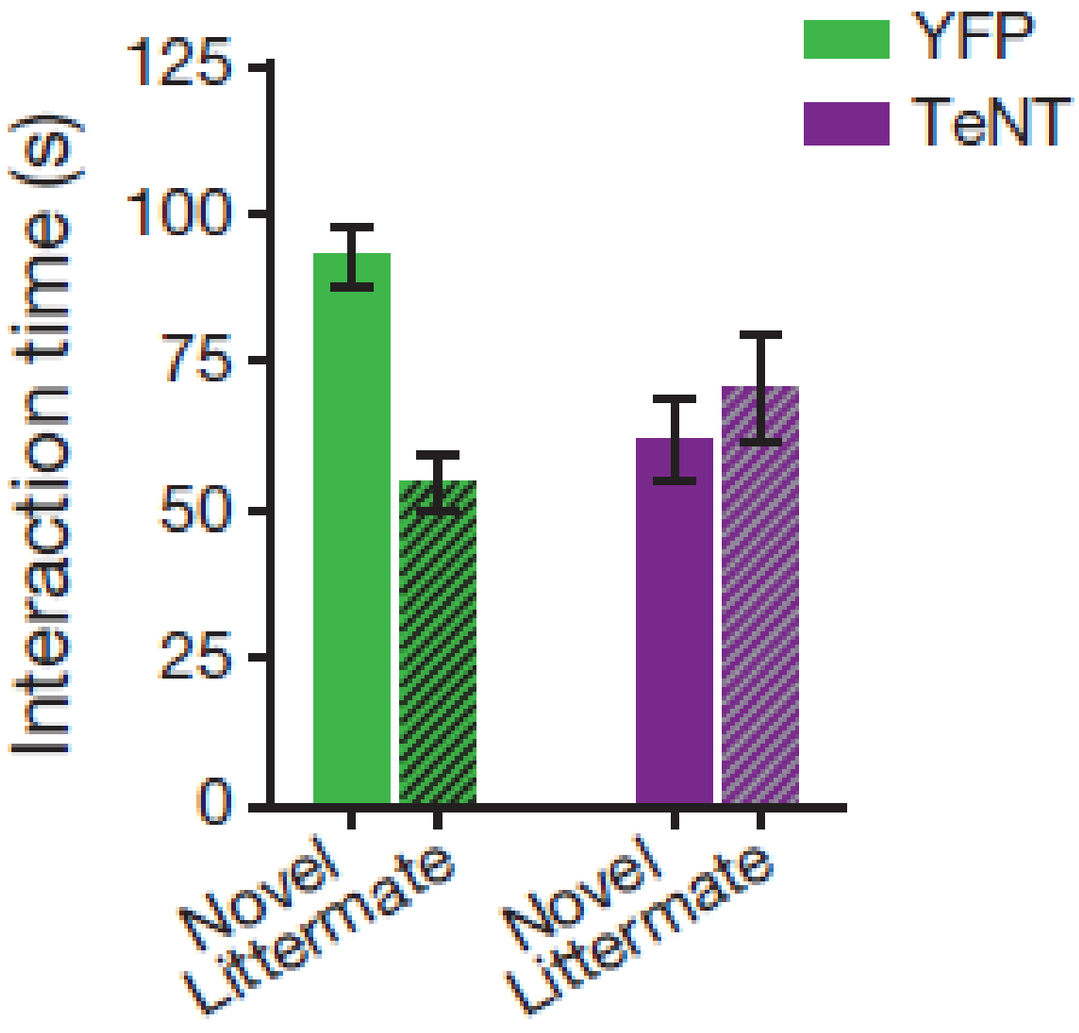
Also responded to the written name

Fried and colleagues

Social Memory – CA2 region of hippocampus

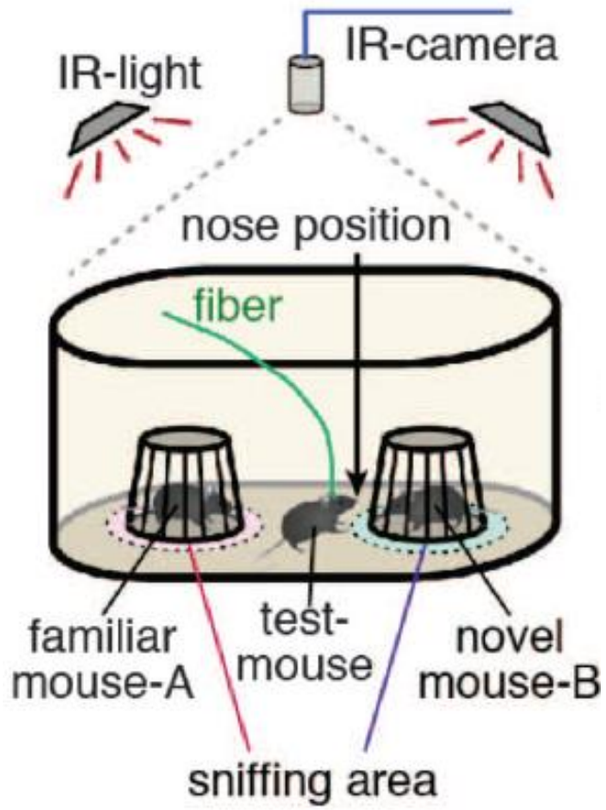


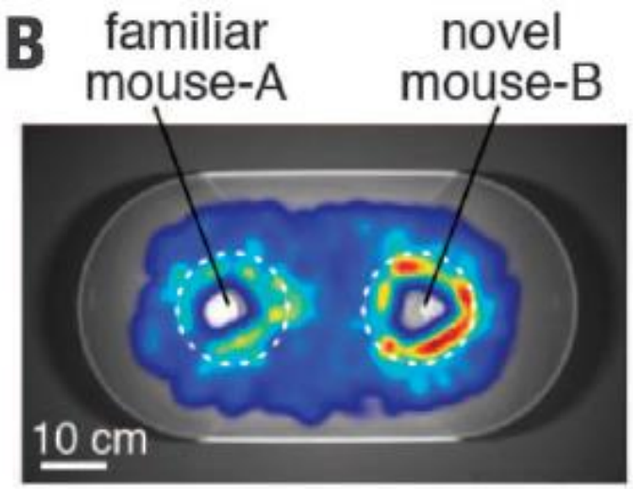


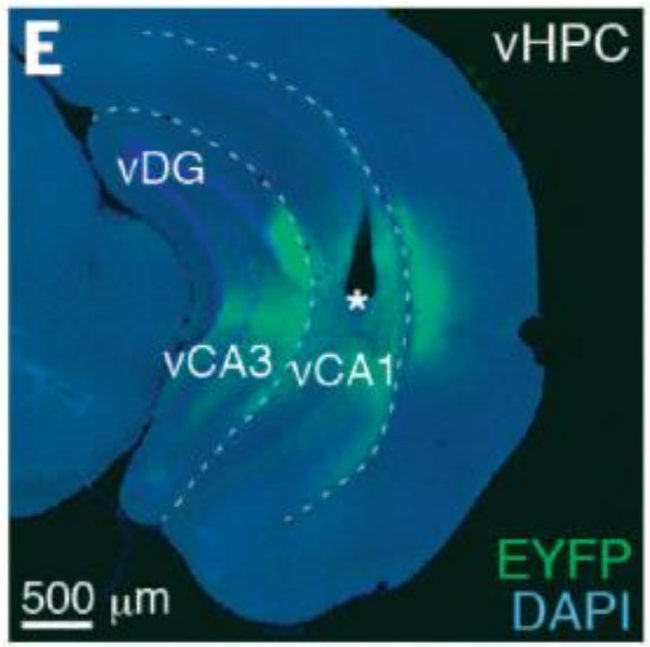


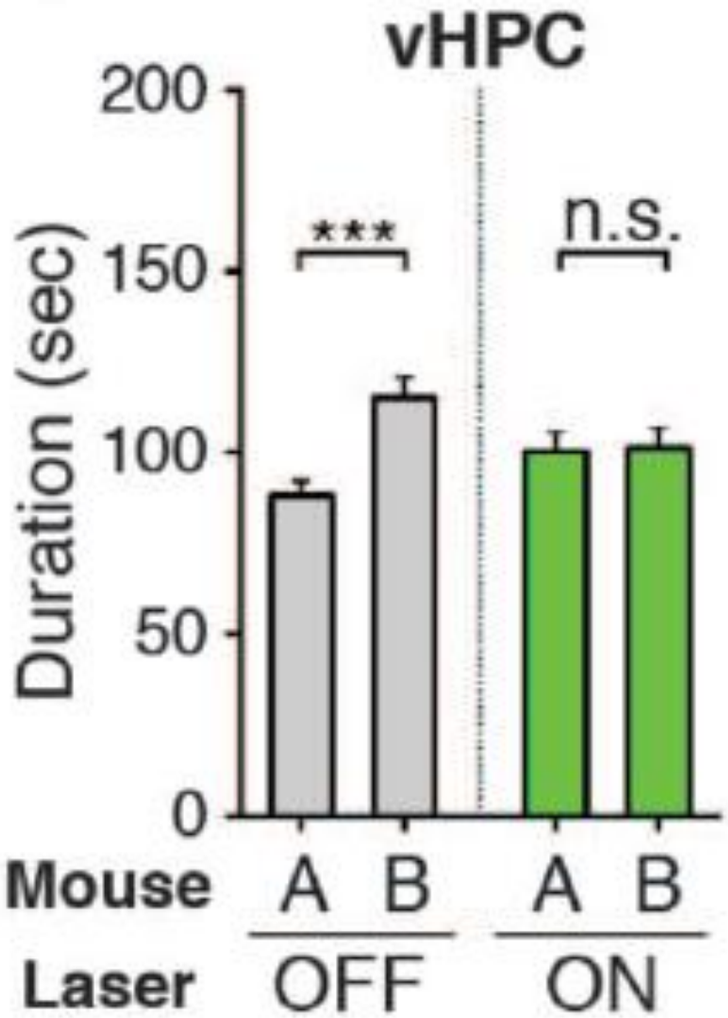
Sigelbaum and colleagues

Social Memory – vCA1 region of hippocampus



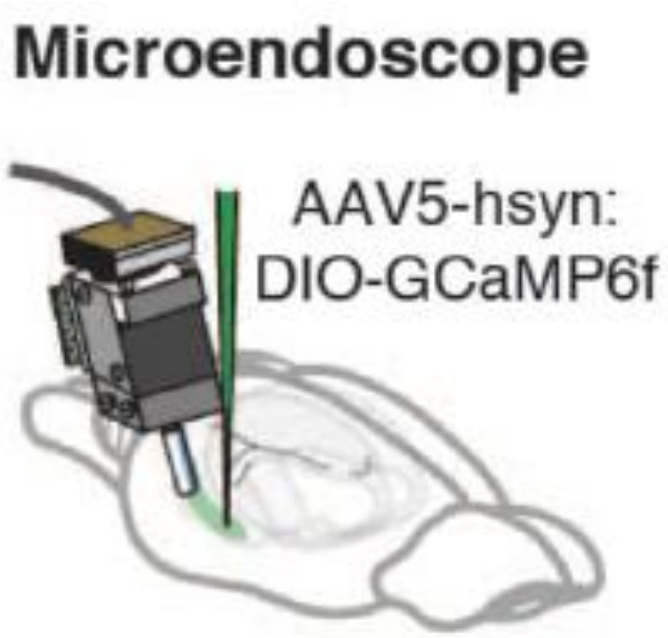


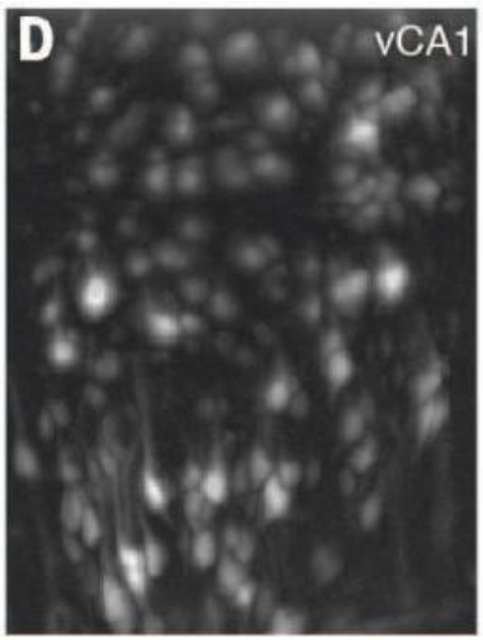


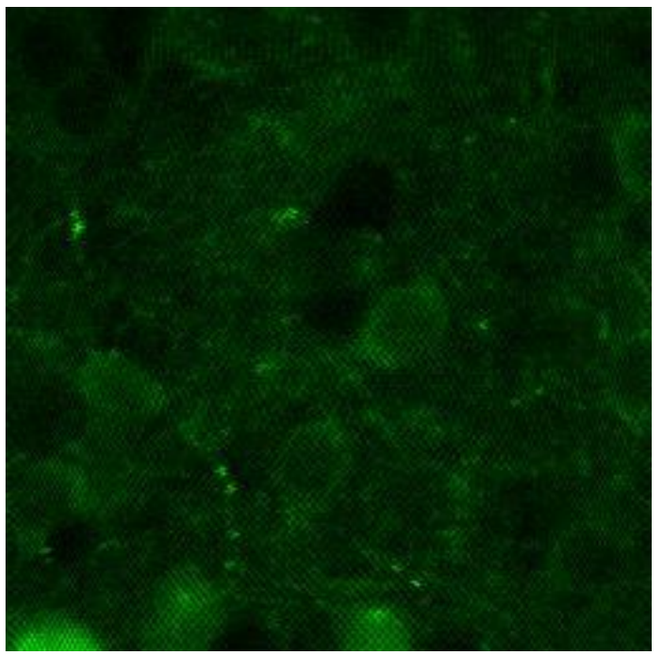


Tonegawa and colleagues

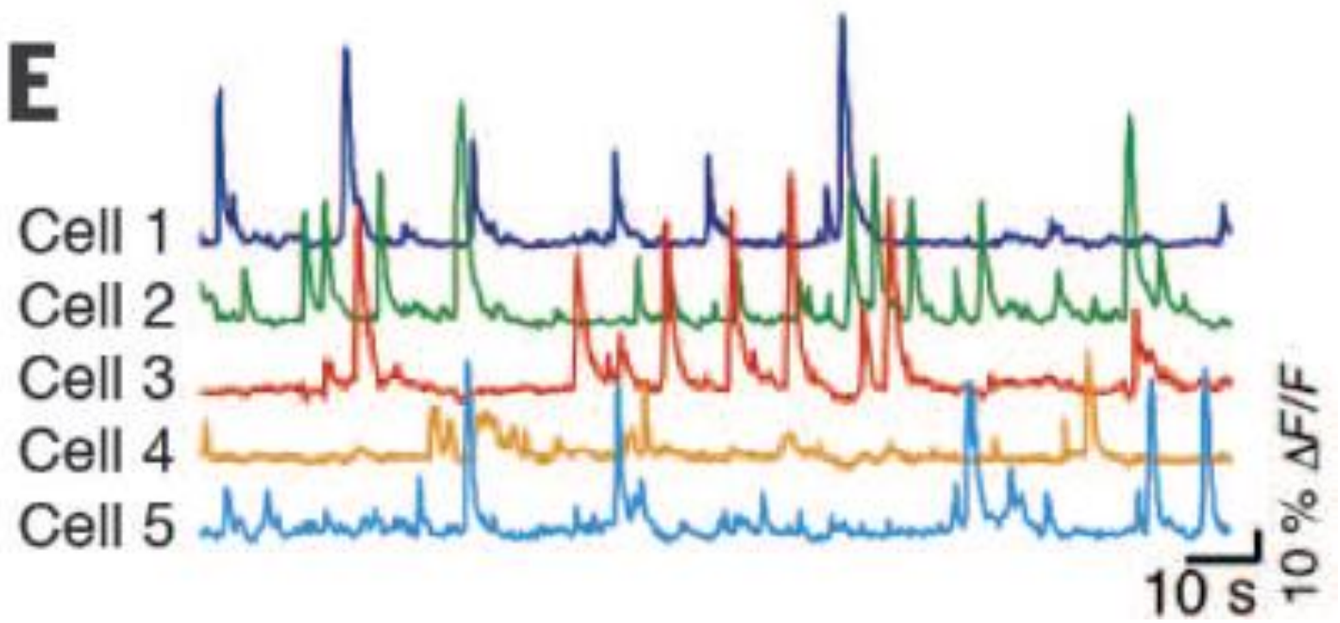
Social Memory – vCA1 region of hippocampus





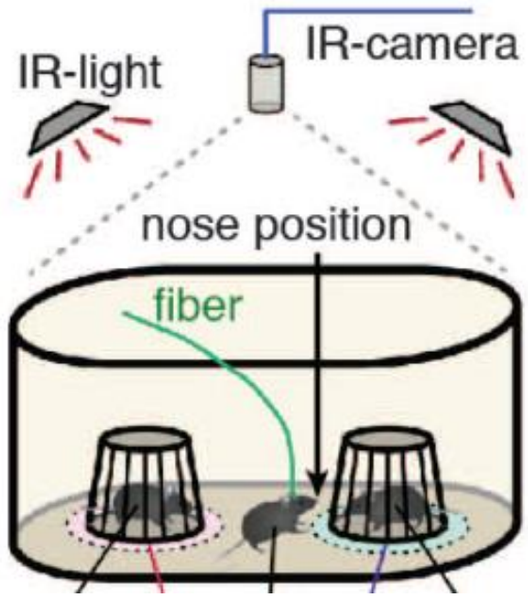


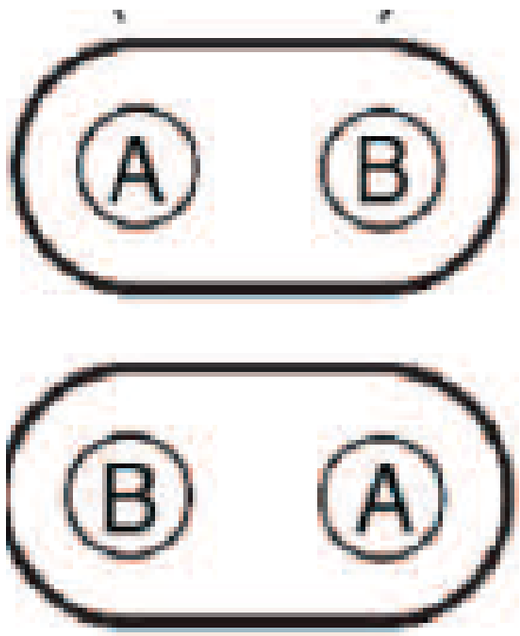
5x speed



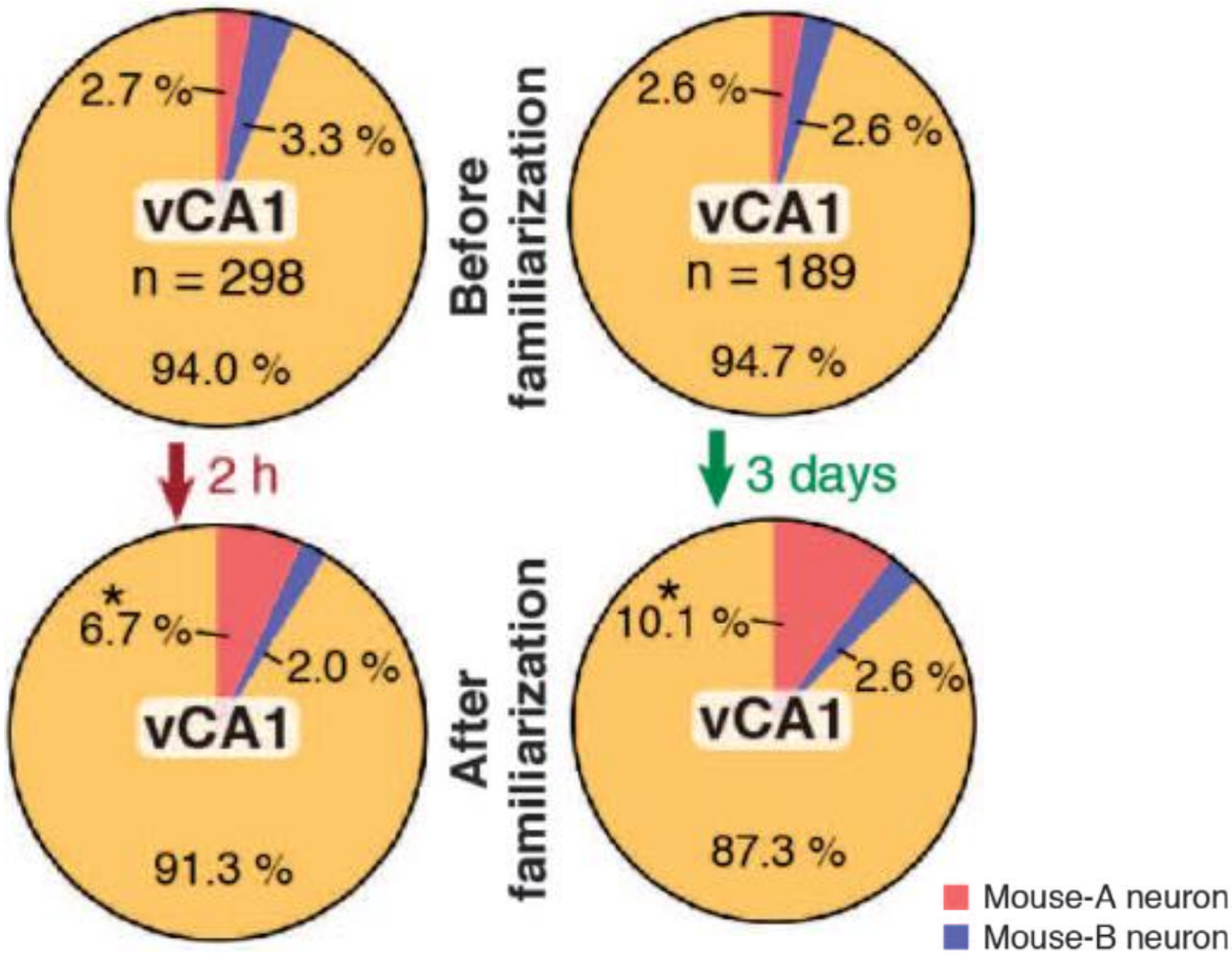
Tonegawa and colleagues

Social Memory – Number of neurons representing familiar mouse increases in vCA1



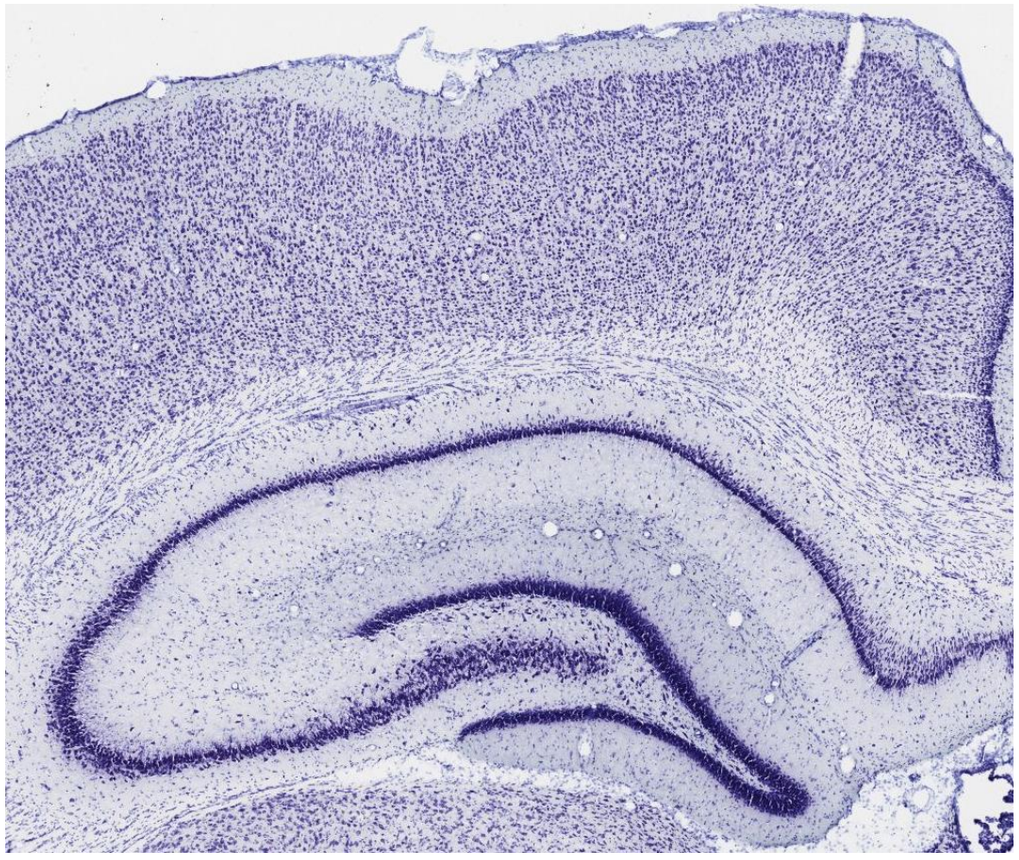


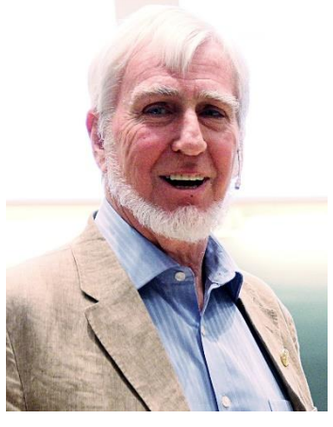
Then they familiarize with mouse A for 2 hours or 3 days



Tonegawa and colleagues

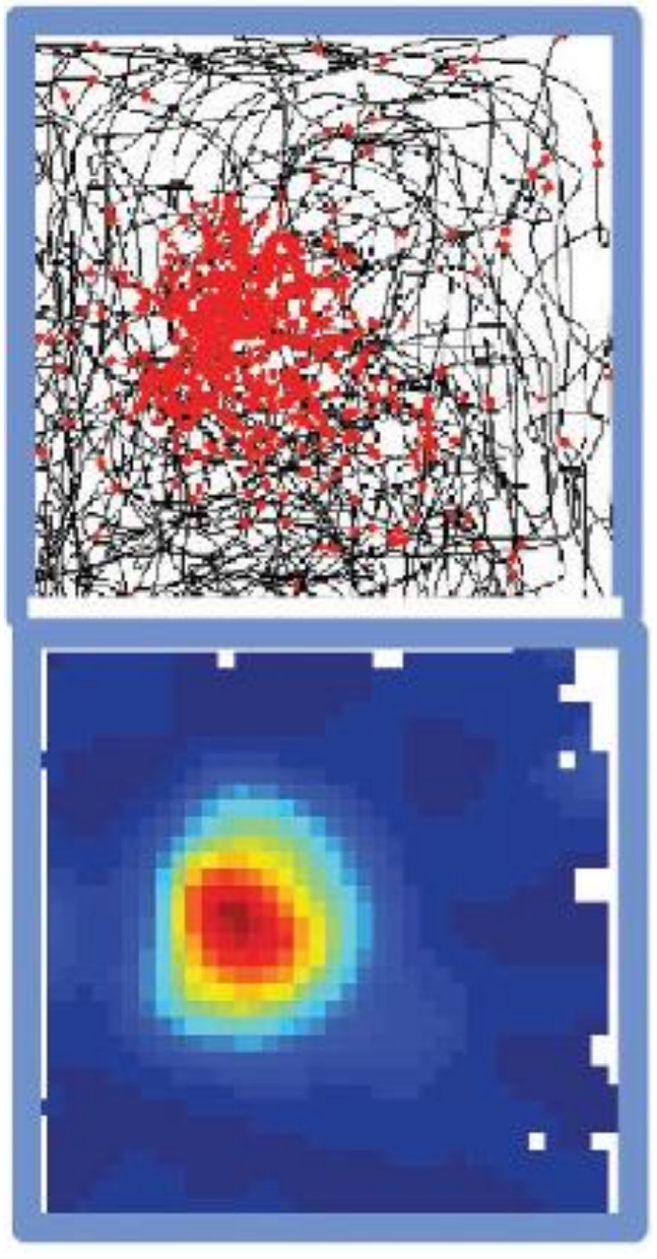
Hippocampus – Spatial Representation





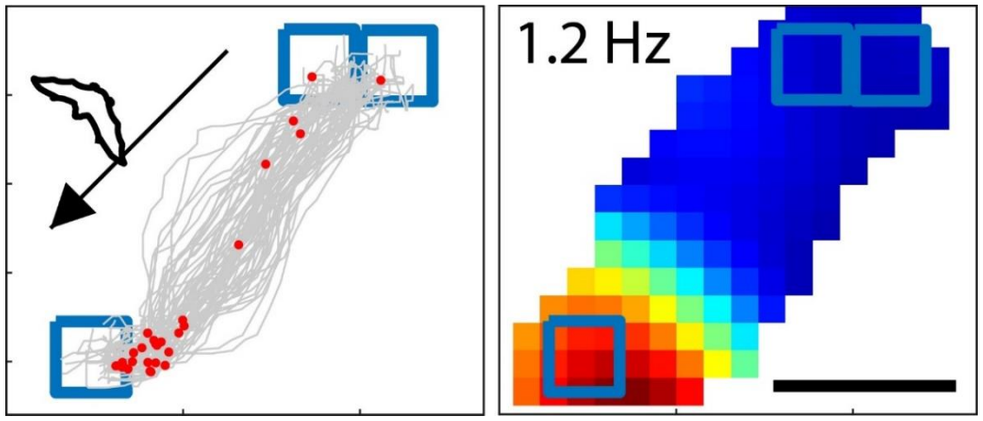


John O’Keefe

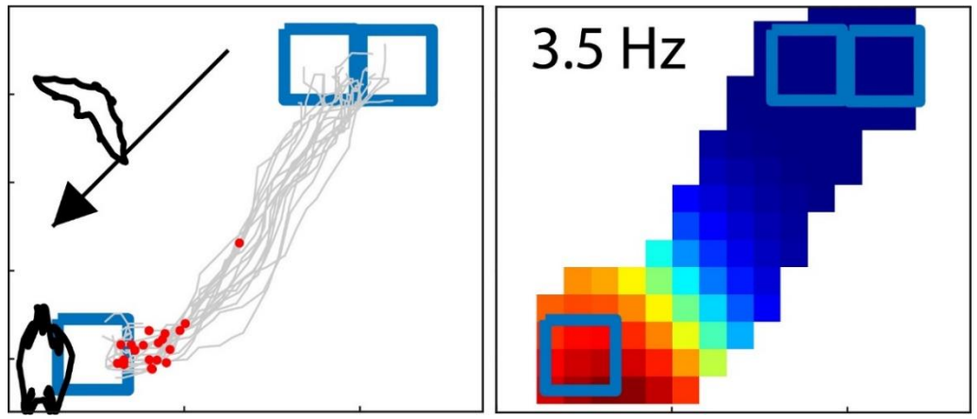


Social-Spatial Representation – dCA1 region of hippocampus

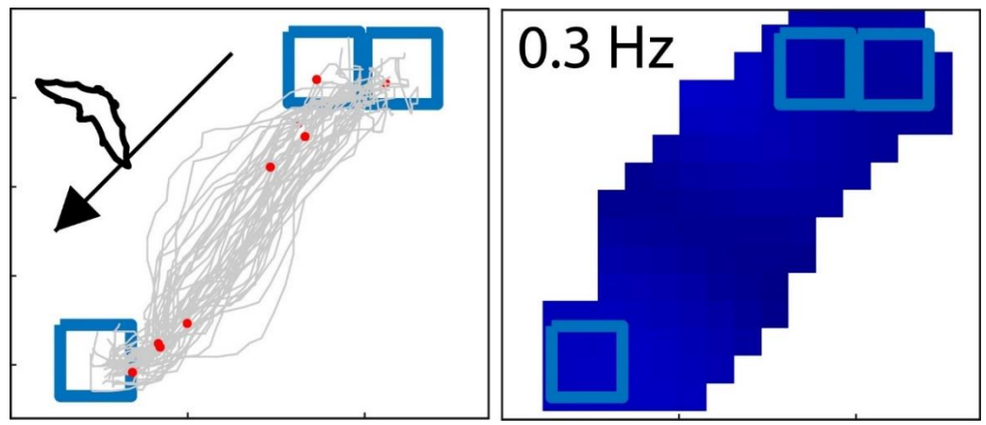
Place cell?



Meeting friends

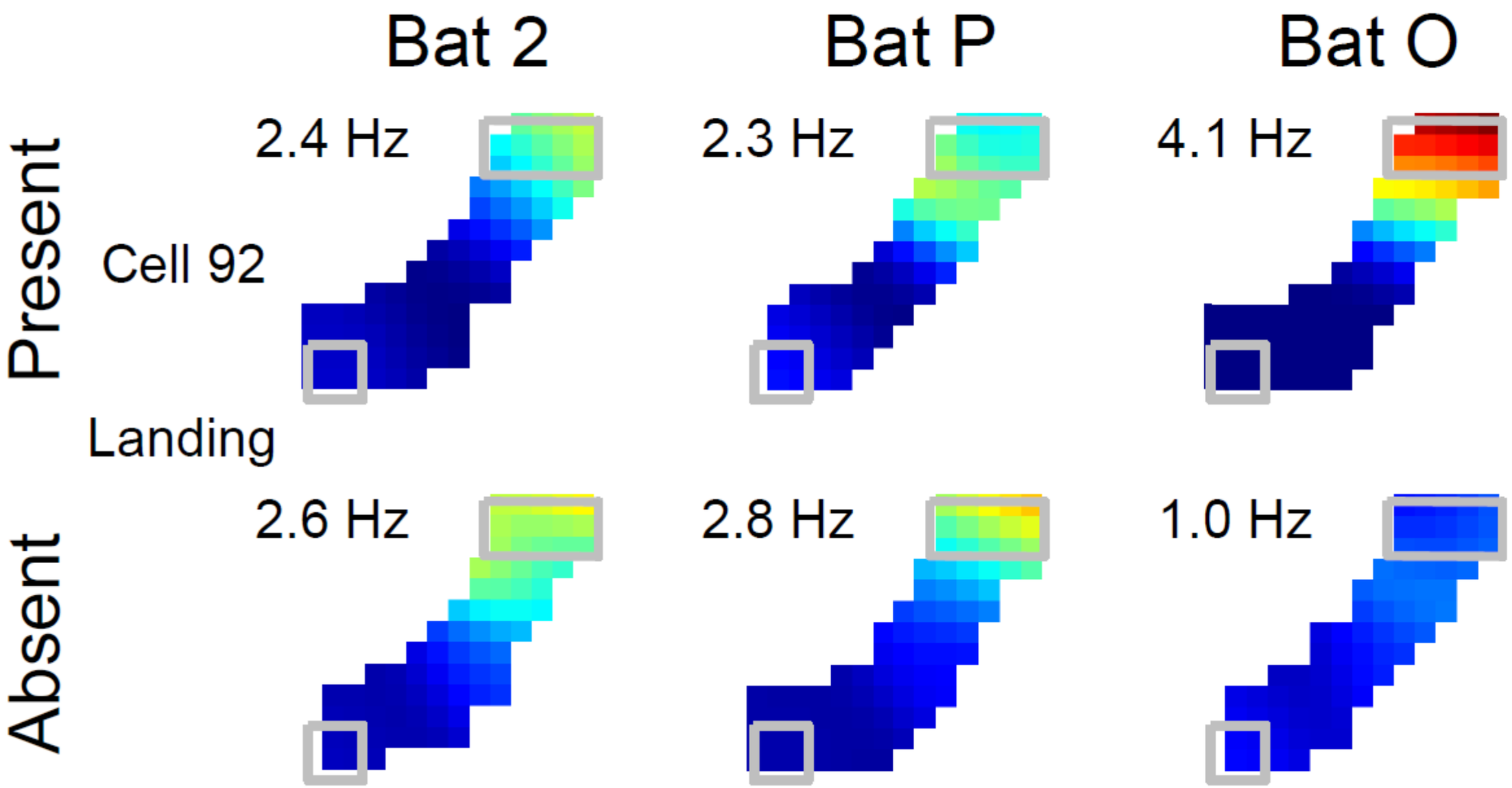


Being alone



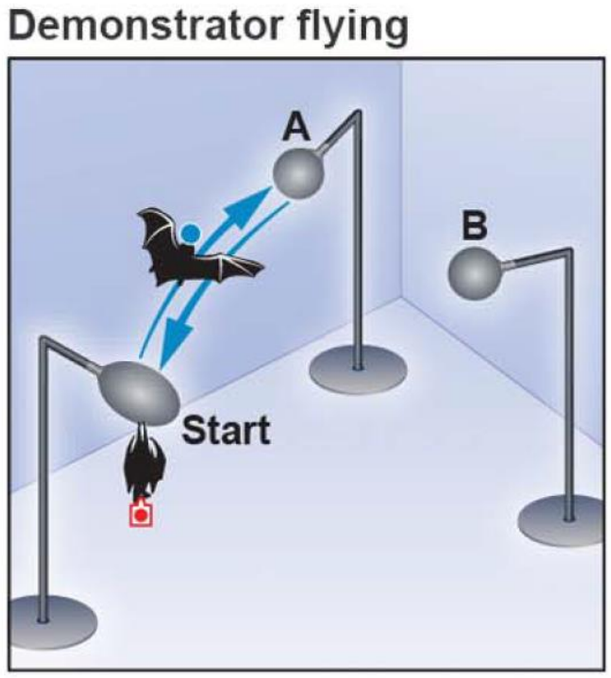
Ulanovsky and colleagues

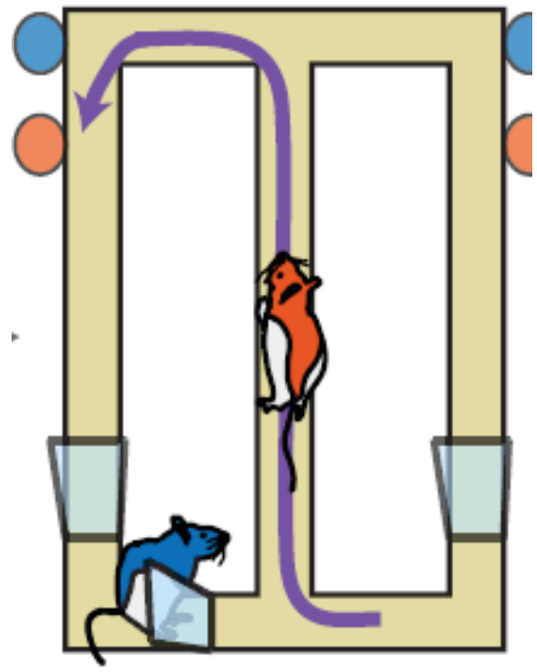
Social Identity – dCA1 region of hippocampus

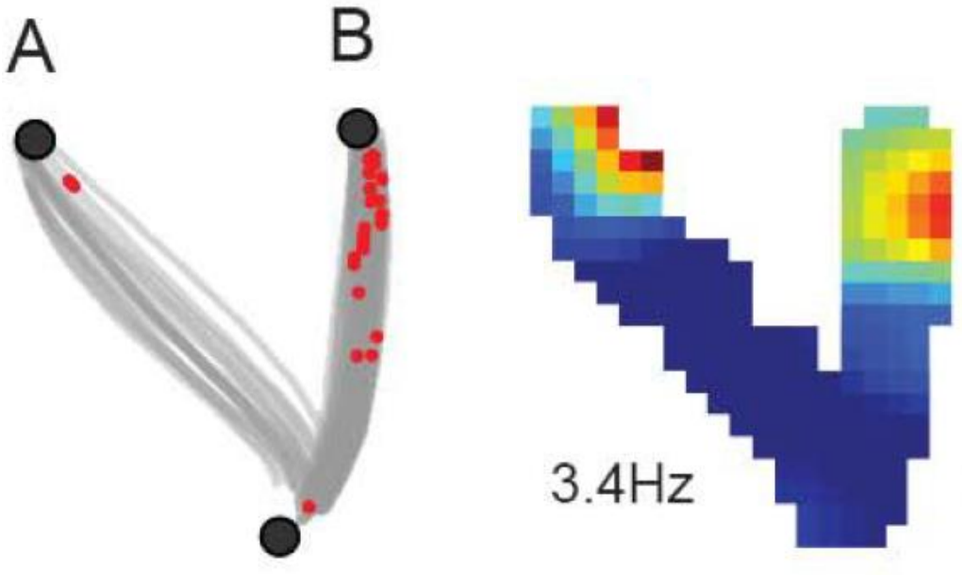


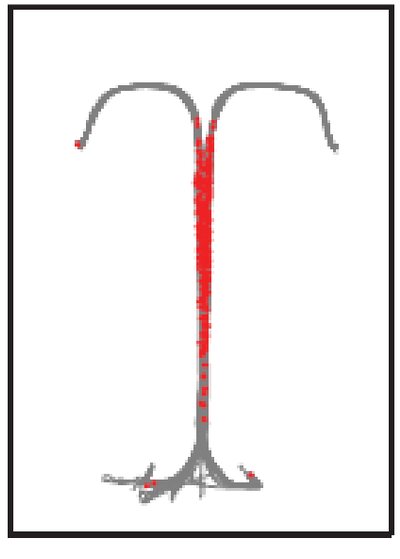
Ulanovsky and colleagues

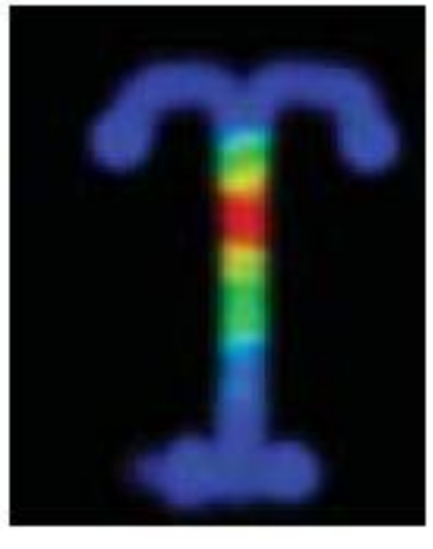
Tracking others locations – dCA1 region of hippocampus







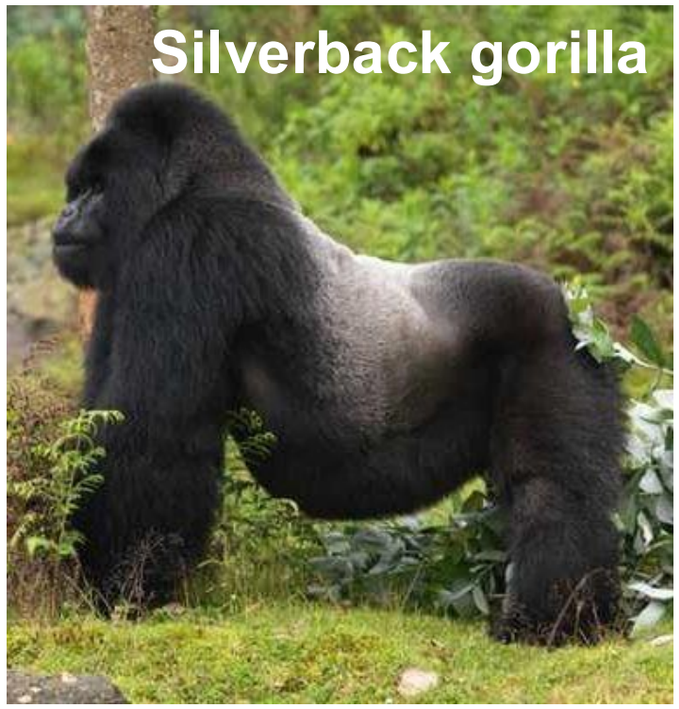




Ulanovsky and colleagues

Fujisawa and colleagues

Social Hierarchy – Animal societies have structure and hierarchy





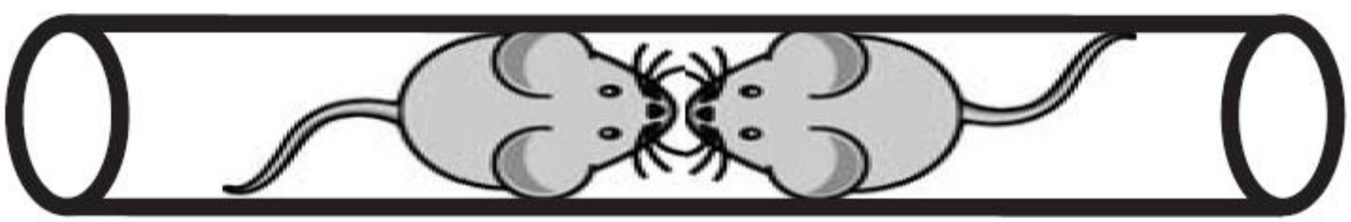
Type of adult bee	What they do	How many in a honey bee colony	How many in a bumble bee colony	What they look like in a honey bee colony	What they look like in a bumble bee colony
Queen	Lay eggs	1	1		
Worker	Take care of larvae, build and clean nest, forage	10,000-50,000	Less than 50 to over 400, depending on species		
Male	Leave nest to mate, then die	100-500	0-50, depending on species and season		

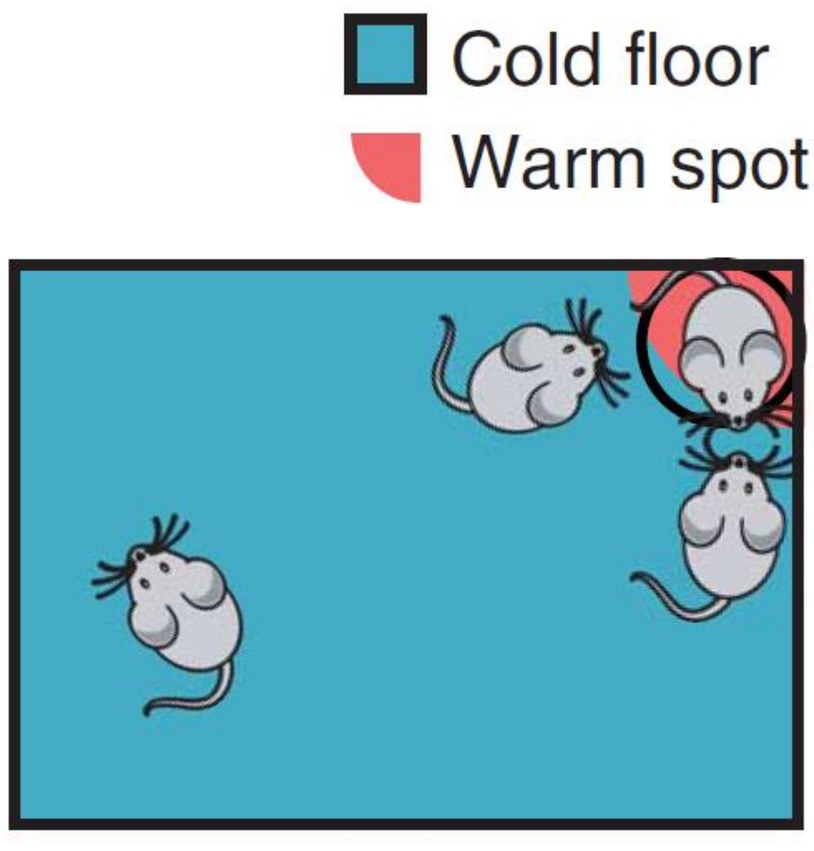
Social Hierarchy – Tube Test

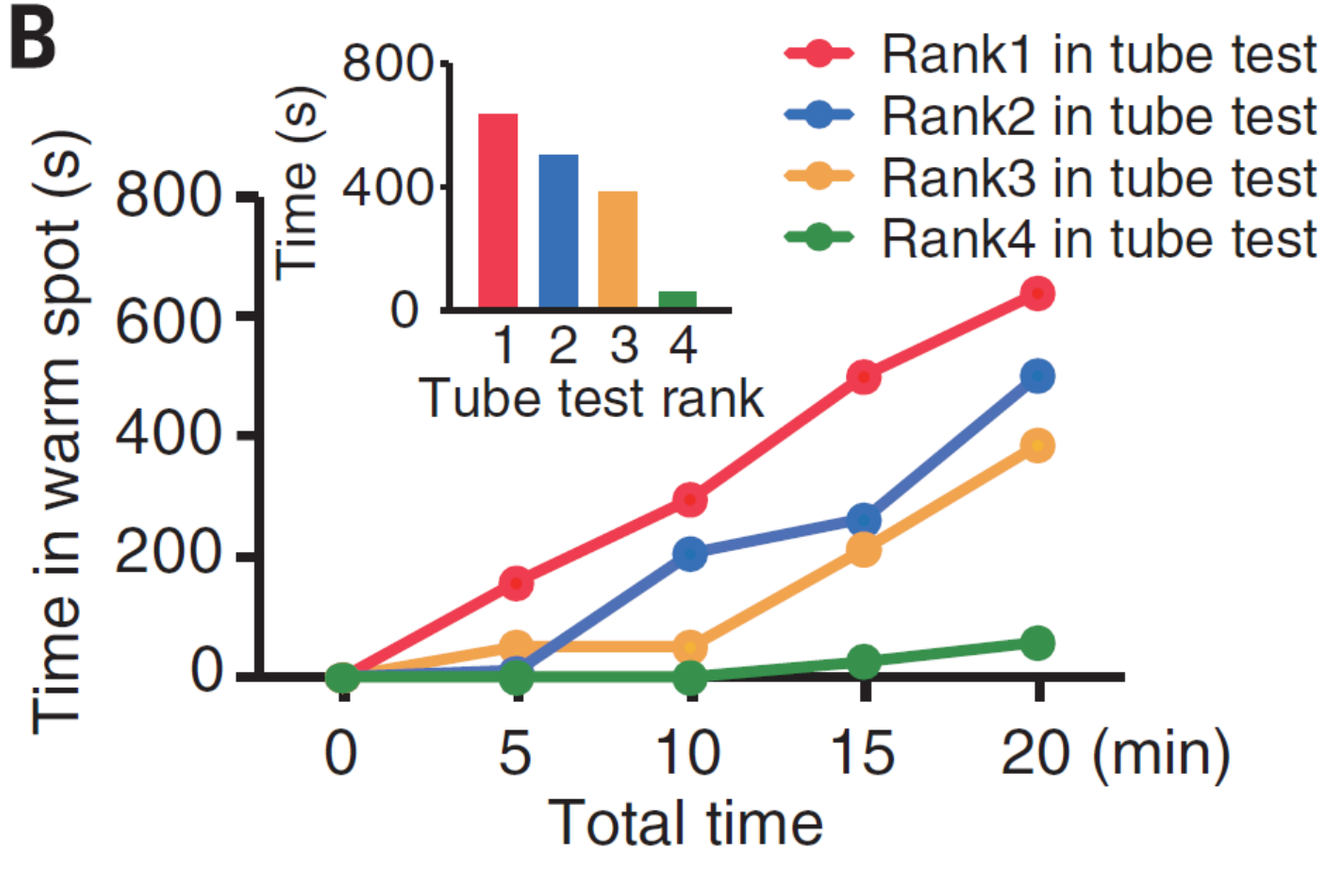


Hu and colleagues

Social Hierarchy – Consistent with other tests

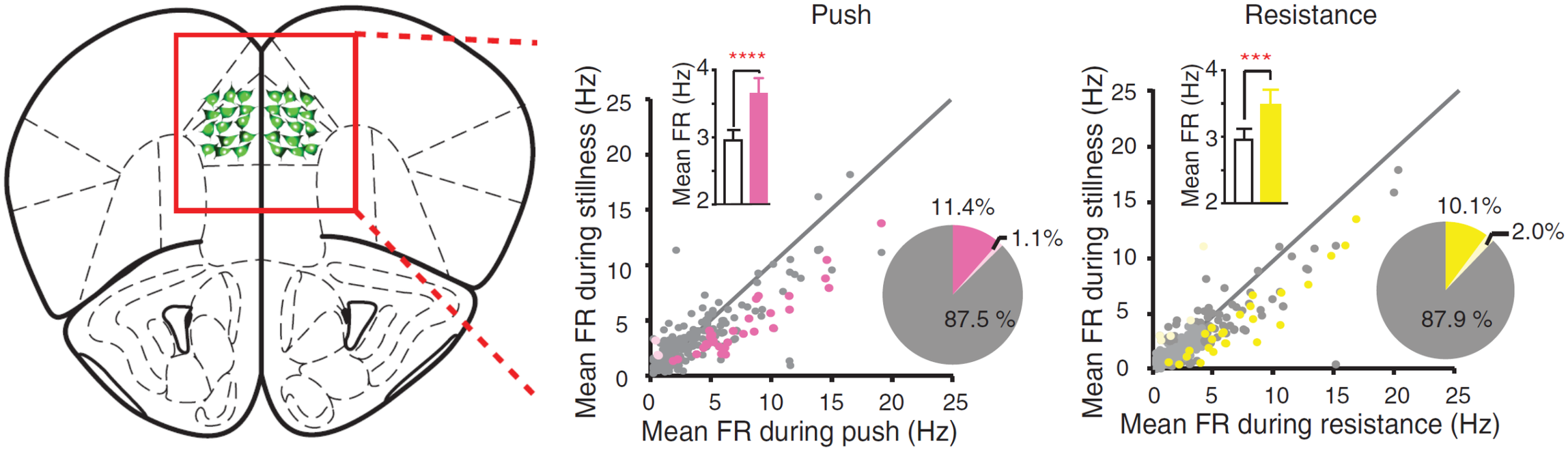






Hu and colleagues

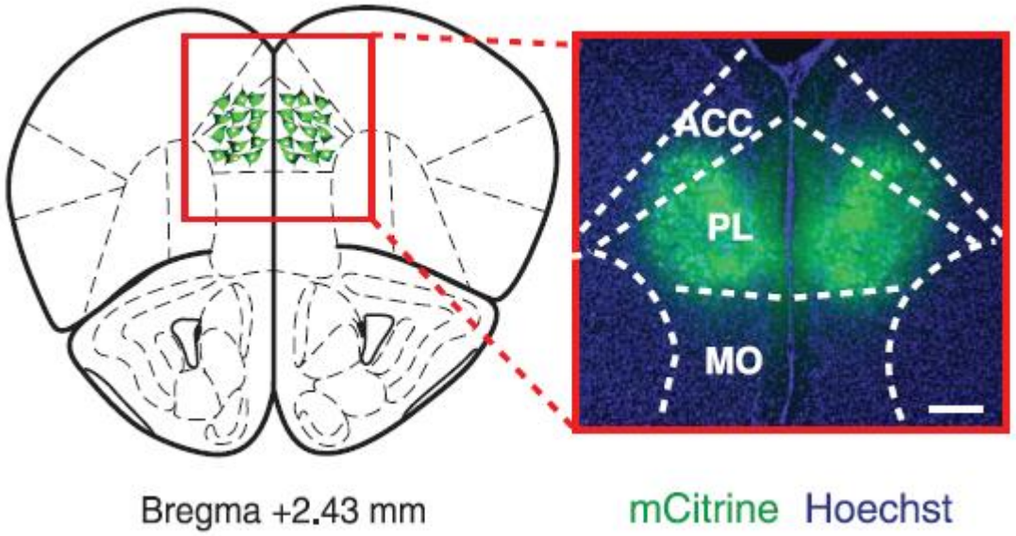
Social Hierarchy – Neural recording in medial prefrontal cortex



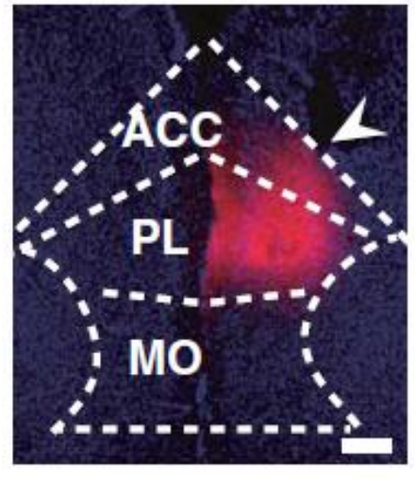
Neurons in the medial prefrontal cortex increased their firing rate during pushing or resisting

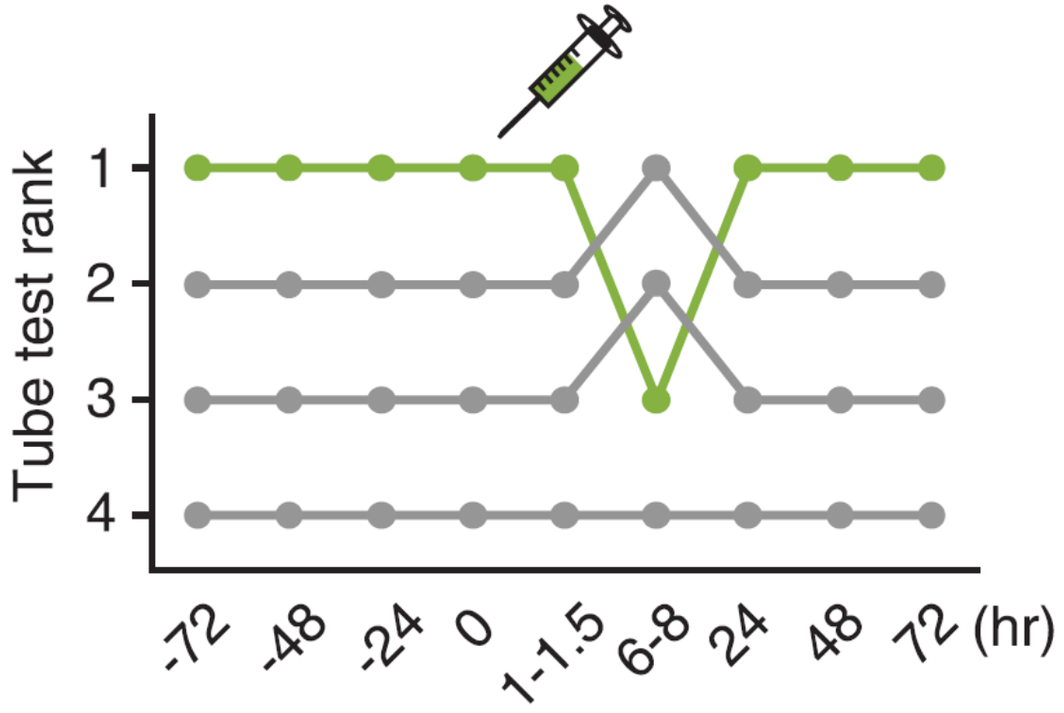
Hu and colleagues

Social Hierarchy – Modulation of medial prefrontal cortex (mPFC) neurons changes dominance

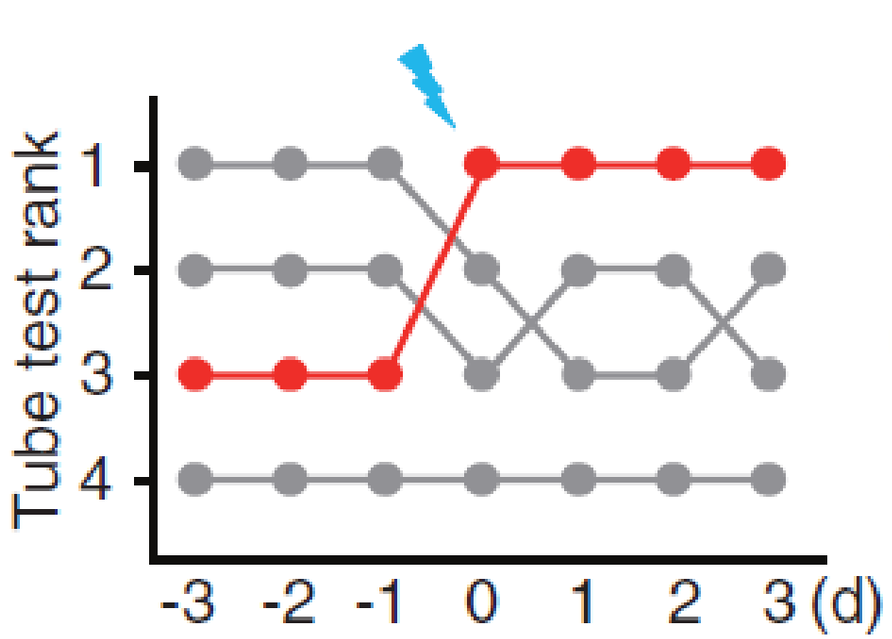


ACC/PL/MO-
sub regions of mPFC





Inhibiting cells in mPFC reduces rank



Activating cells in mPFC increases rank

Hu and colleagues

Questions

Aggression



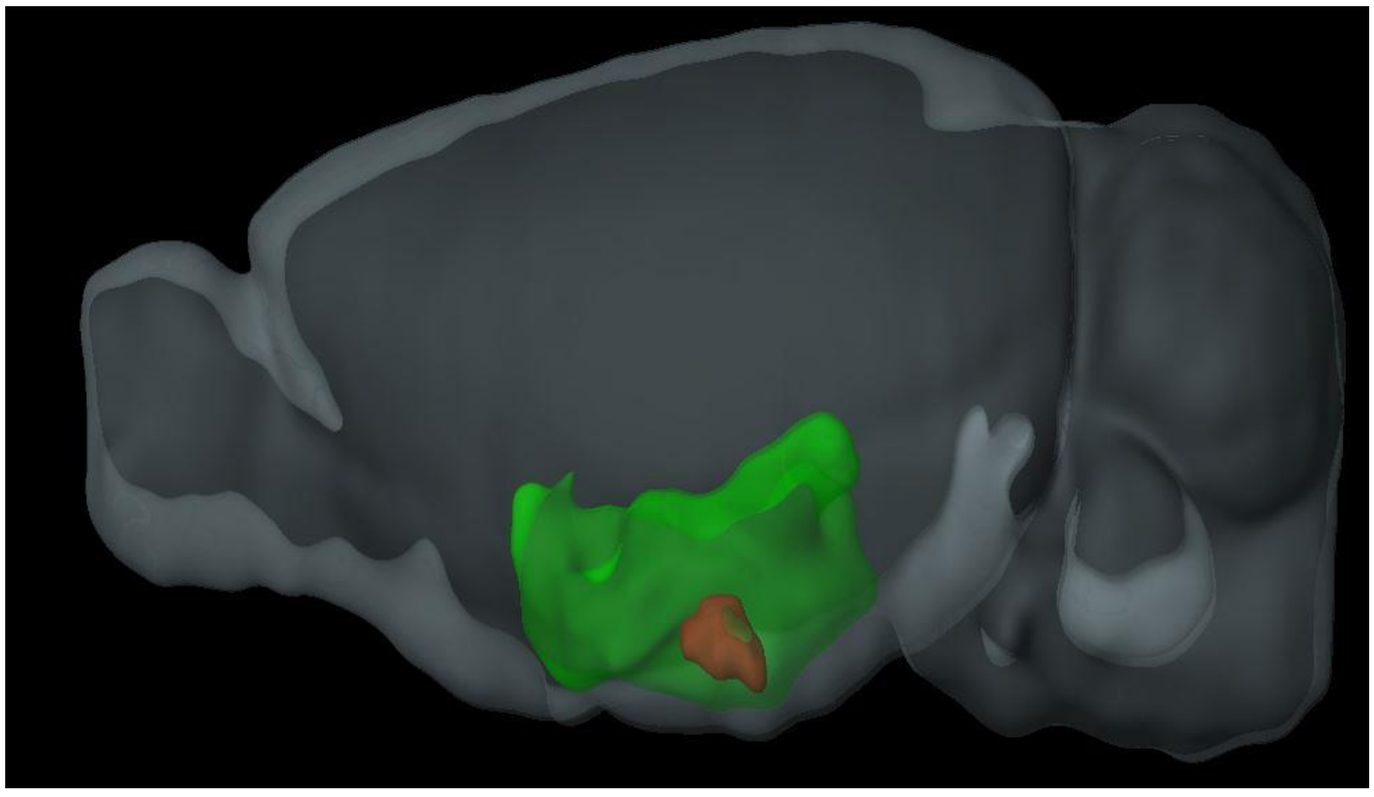




Aggression



Aggression – Hypothalamus

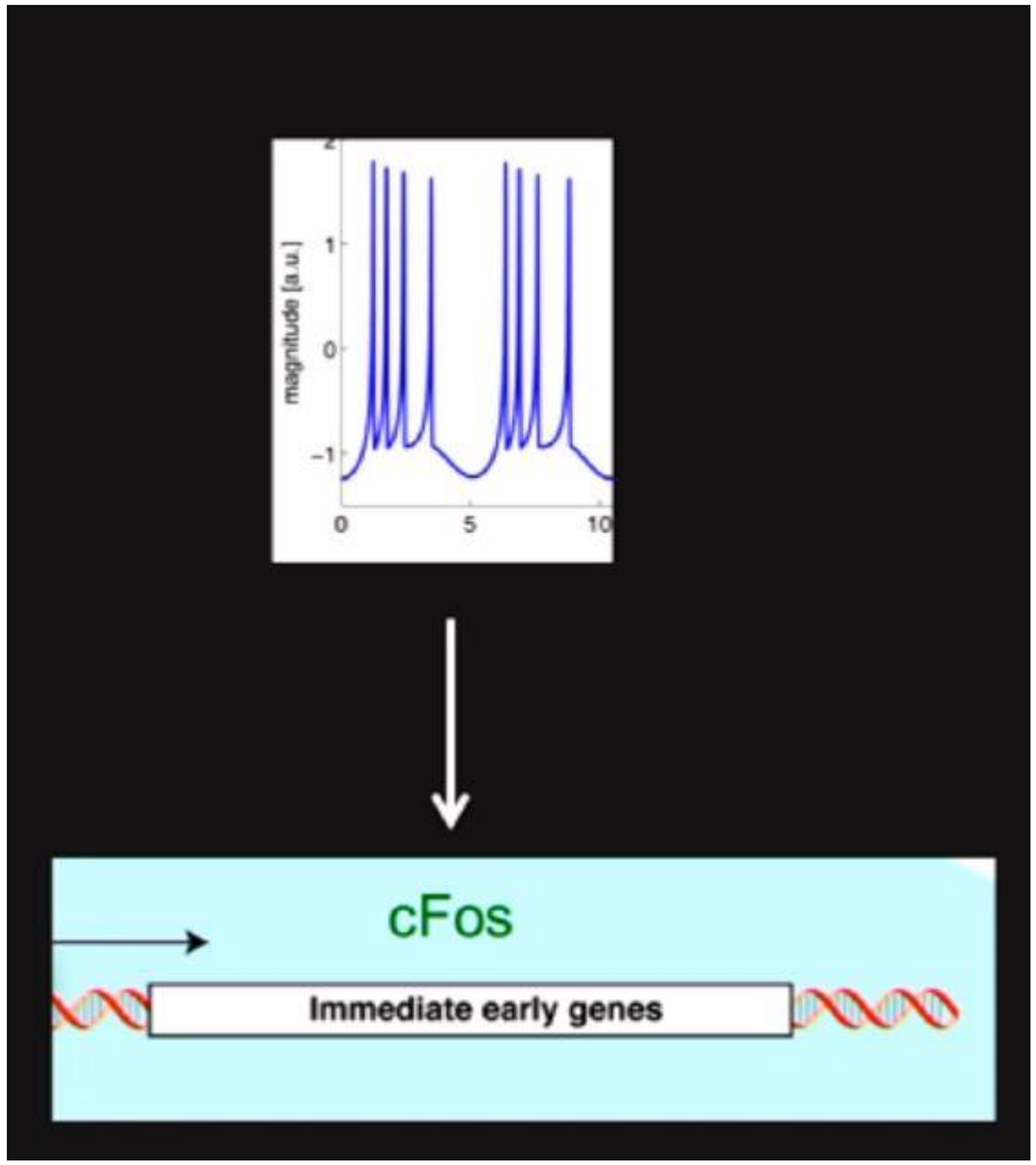


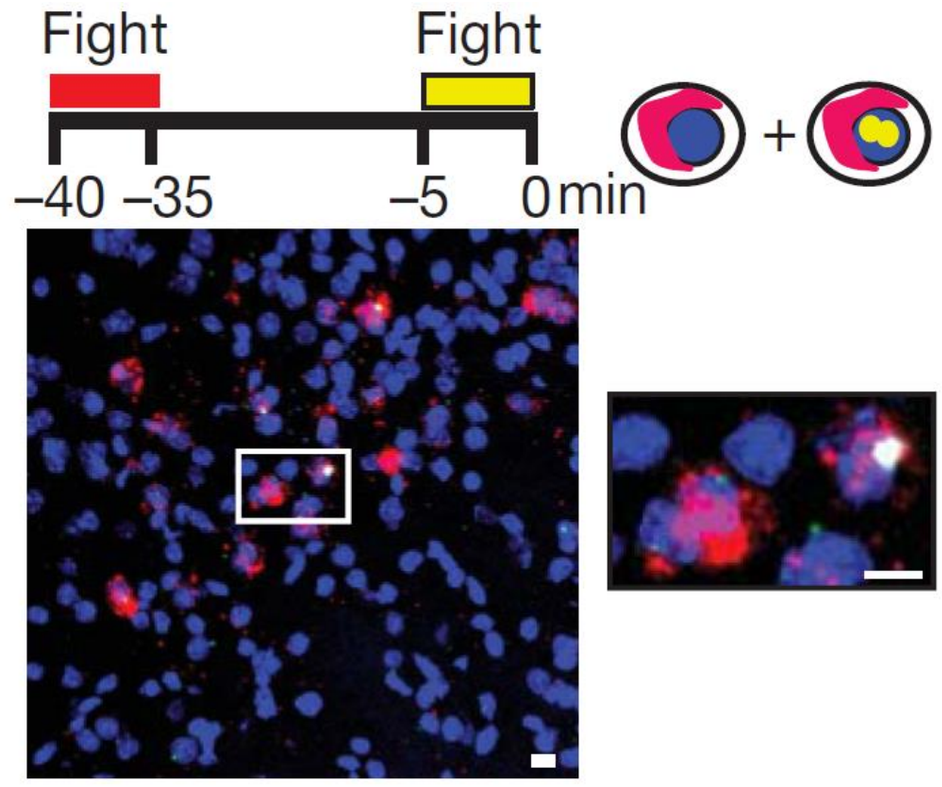
Hypothalamus

-Sleep
-Feeding
-Arousal
-Aggression
-Blood pressure
-Temperature Control
-Social hormones like oxytocin & vasopressin

Ventromedial nucelus

Aggression – Ventromedial hypothalamus

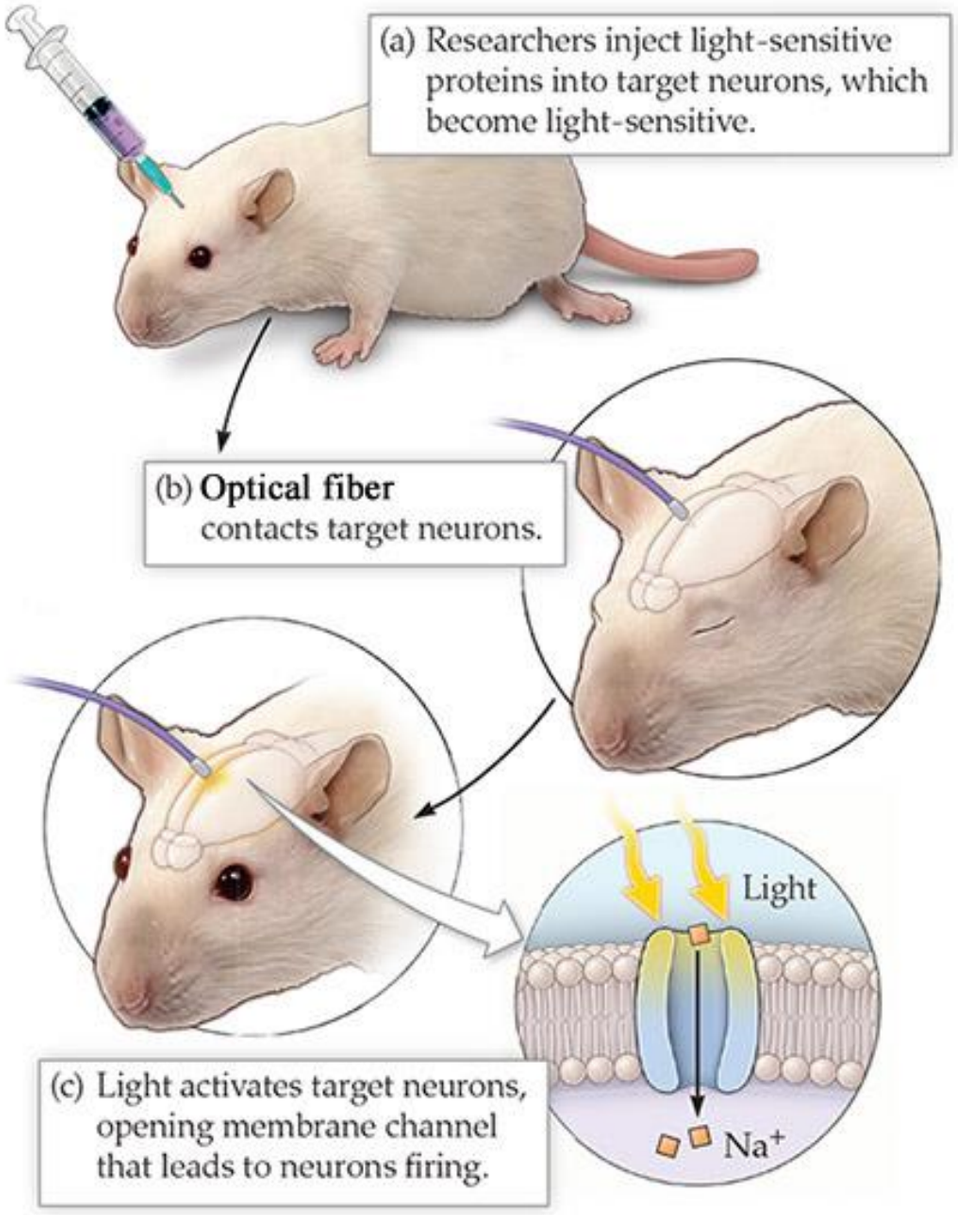


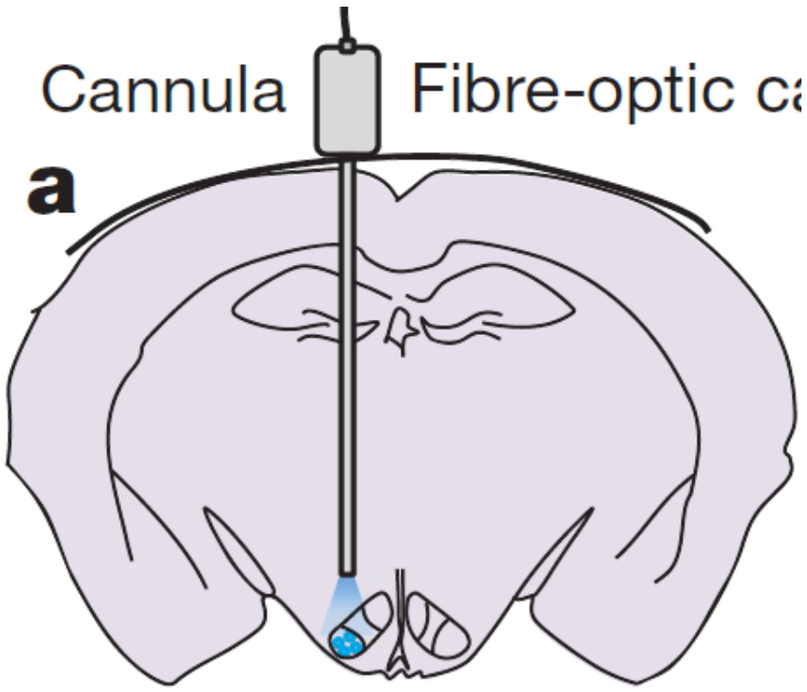


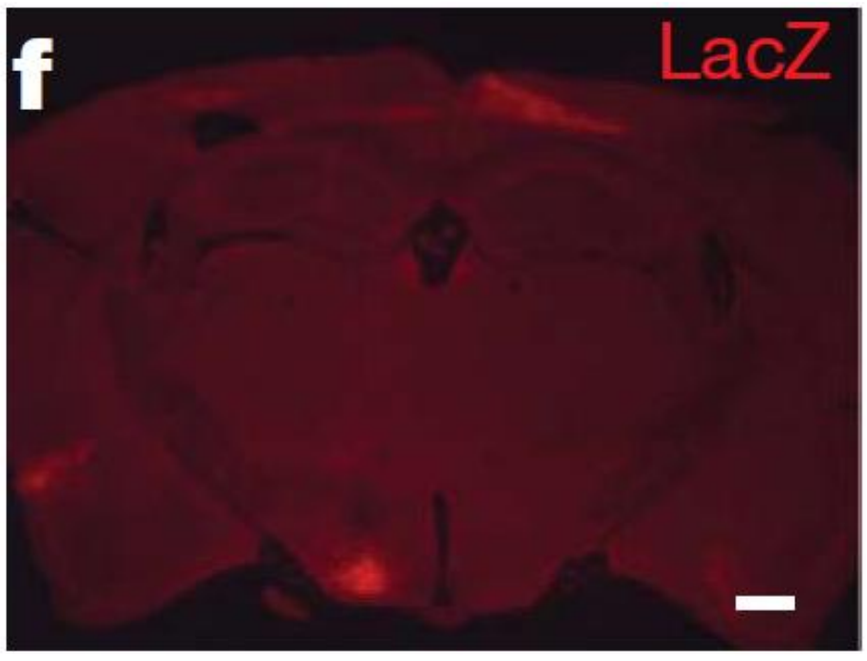
cFos+ cells in ventromedial hypothalamus

Anderson and colleagues

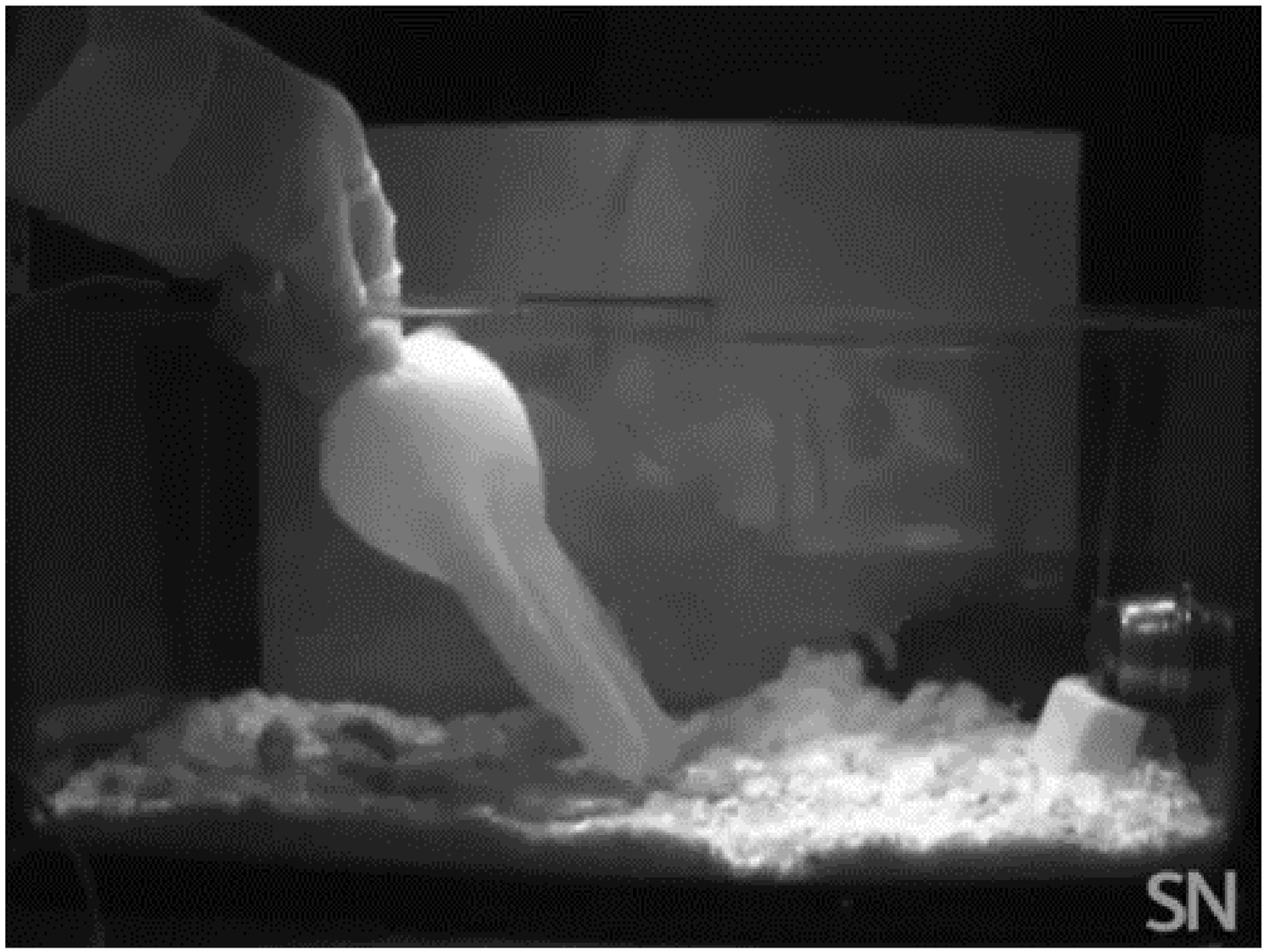
Aggression – Activating cells in ventromedial hypothalamus with optogenetics





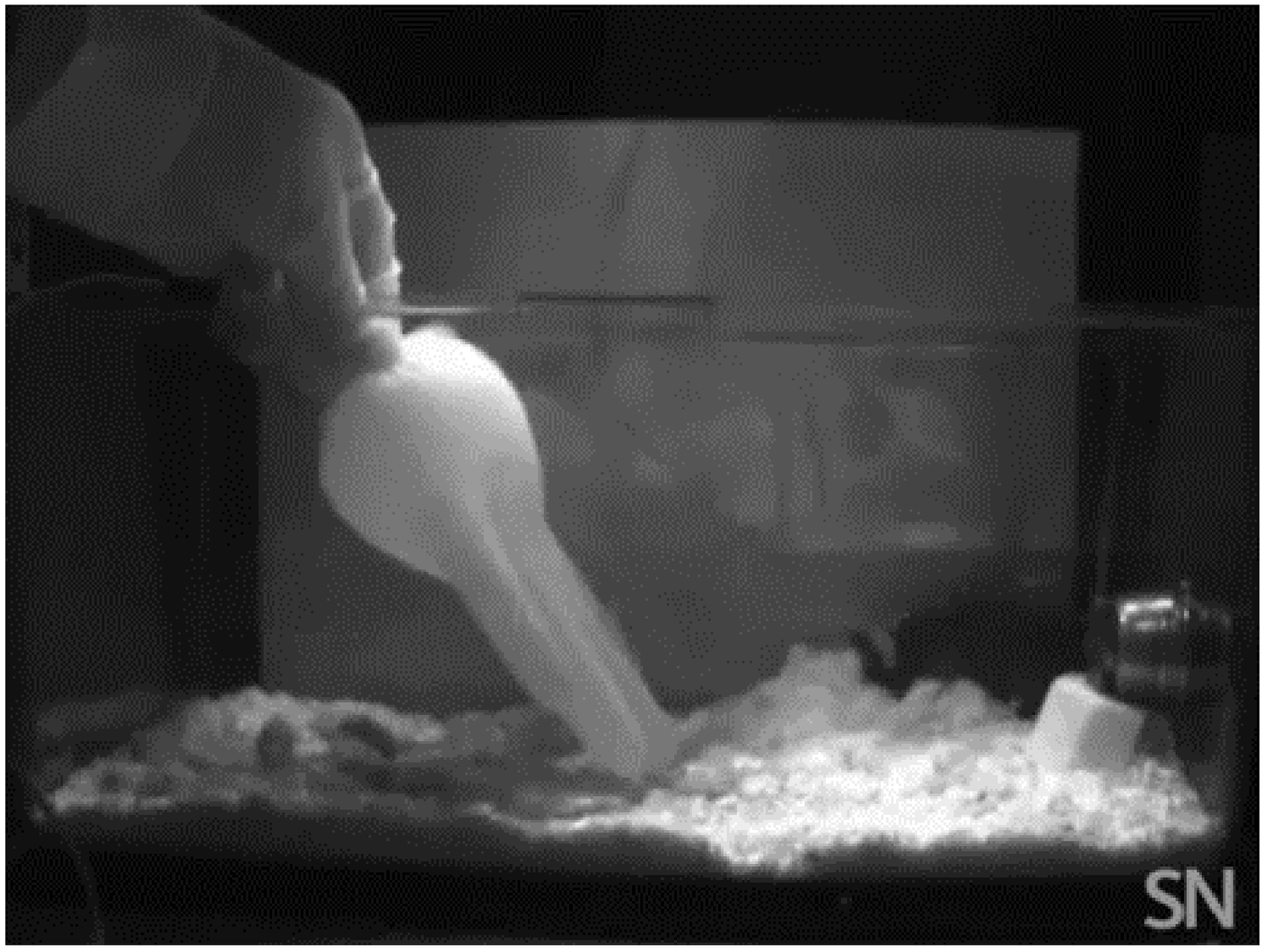


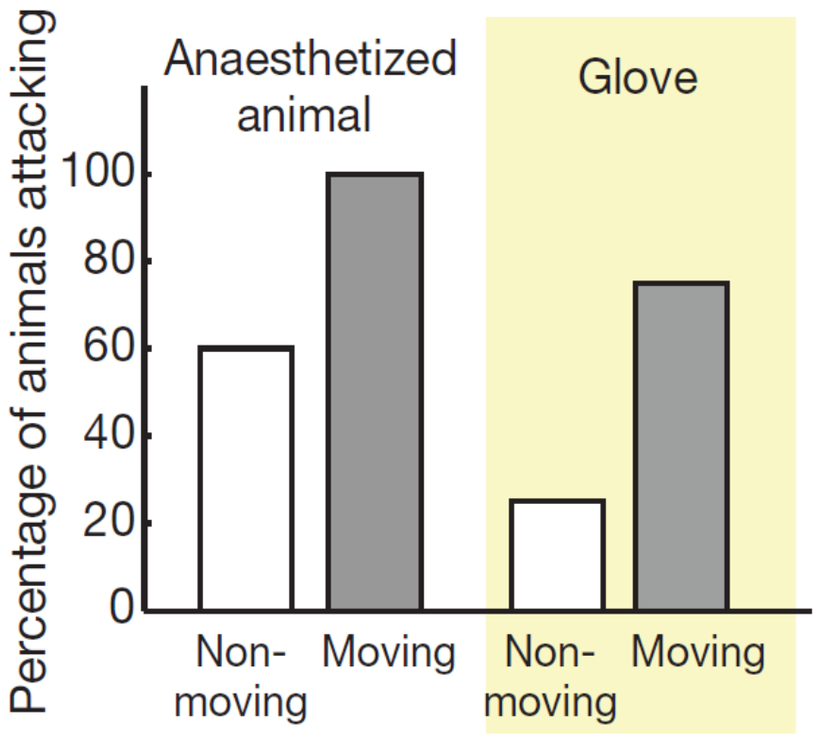
Aggression – Activating cells in ventromedial hypothalamus causes aggression



Anderson and colleagues

Aggression – Activating cells in ventromedial hypothalamus causes aggression





An interesting aspect was that they attacked moving objects more

Anderson and colleagues

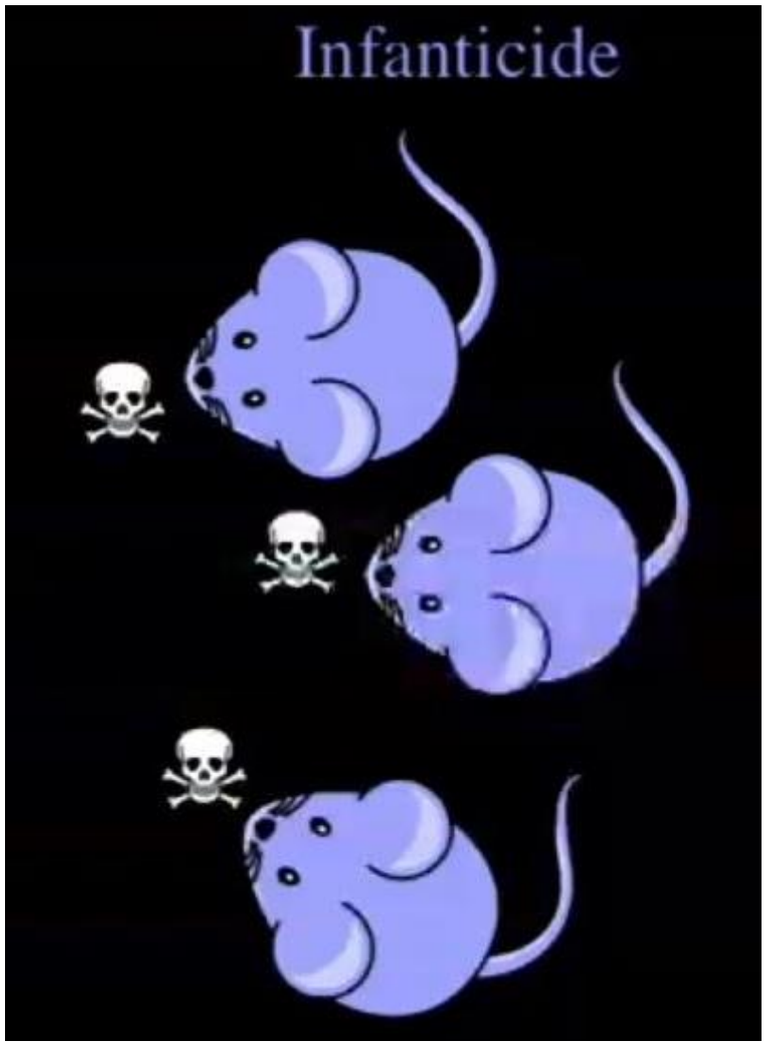
Aggression – Infanticide of others' pups



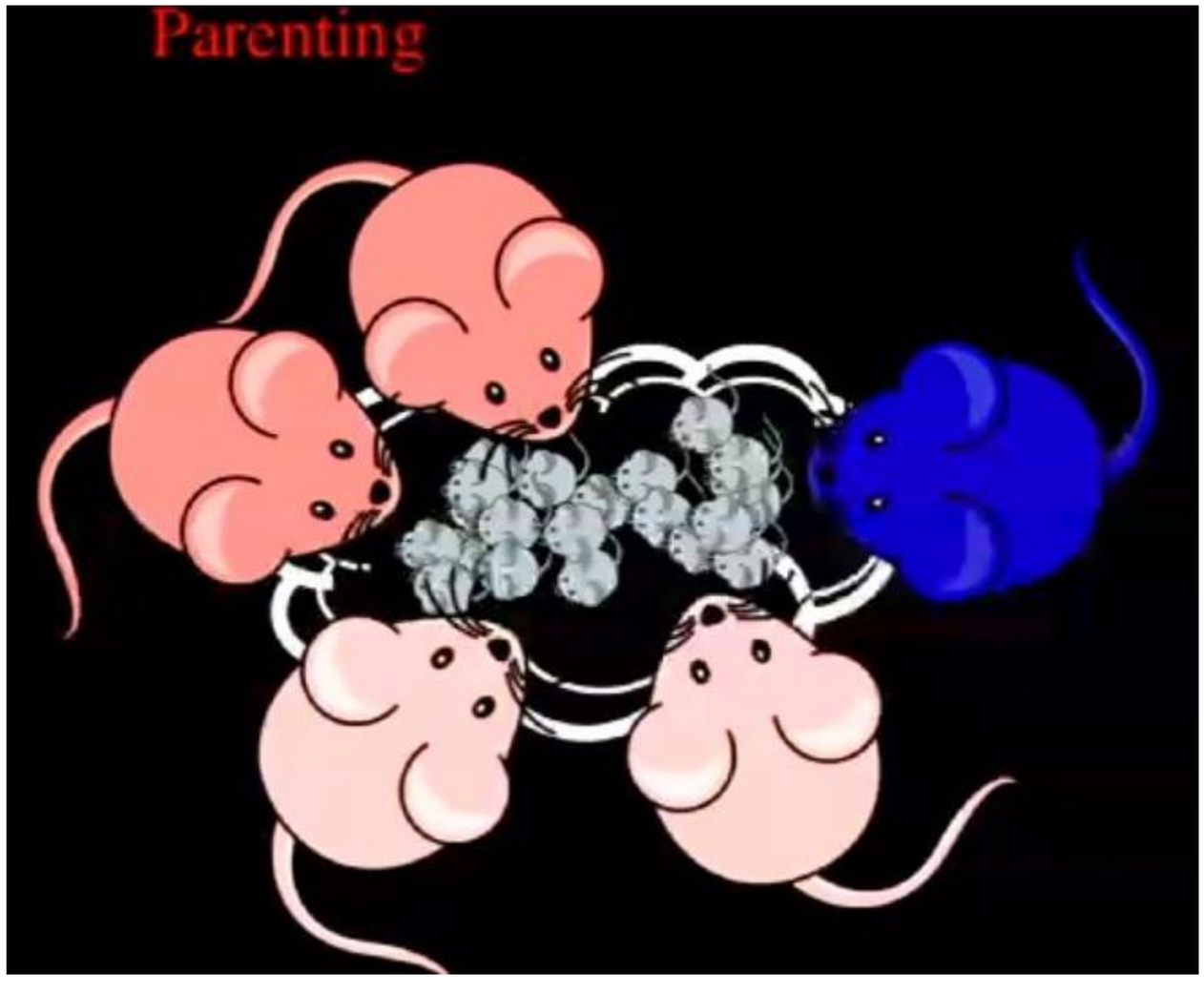




Parenting – Switch from infanticide to parenting



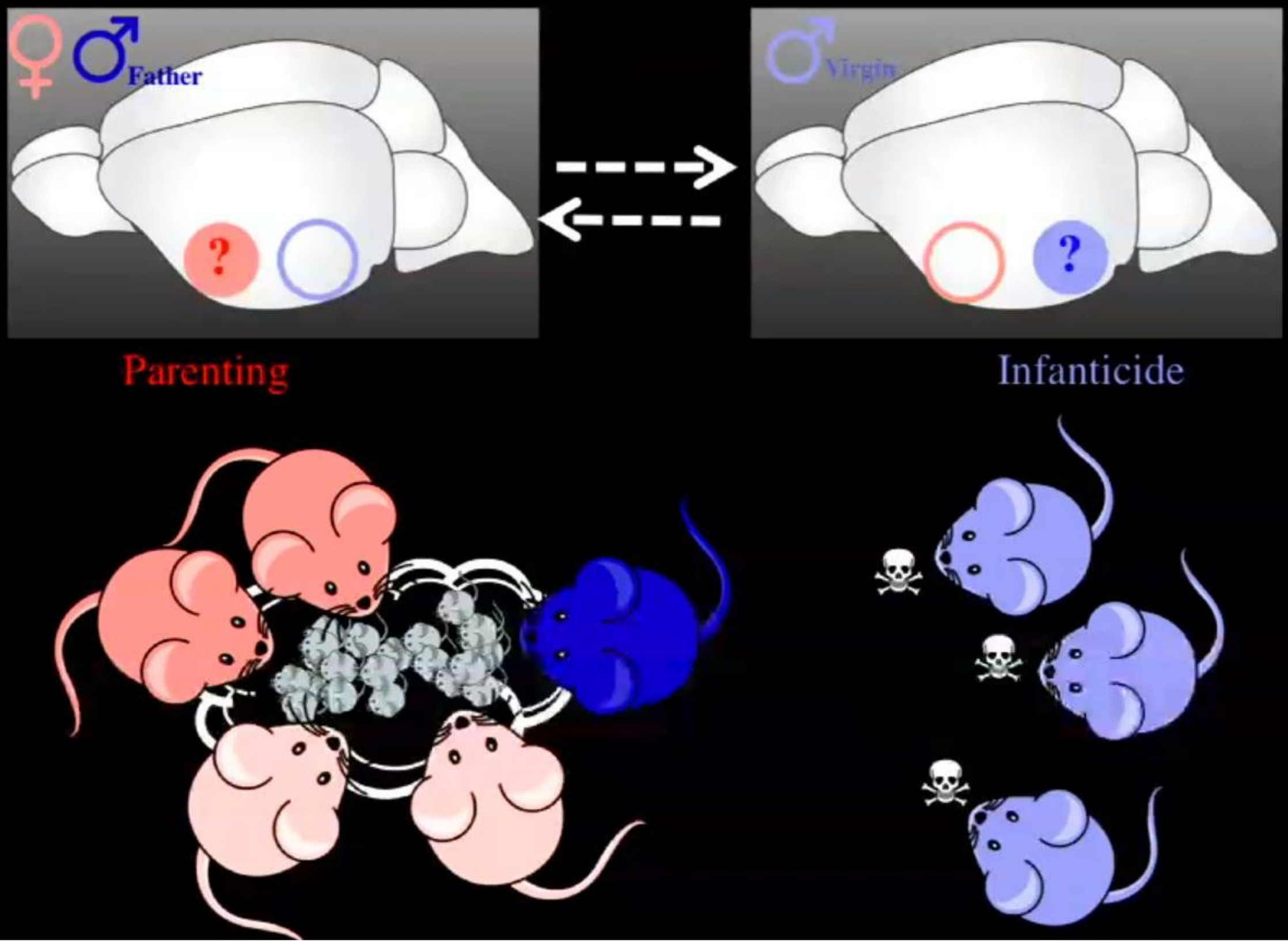
Male mice who have mated, switch to parenting and not be infanticidal ~3 weeks after mating (which is the gestation time in mice) – also to others' pups



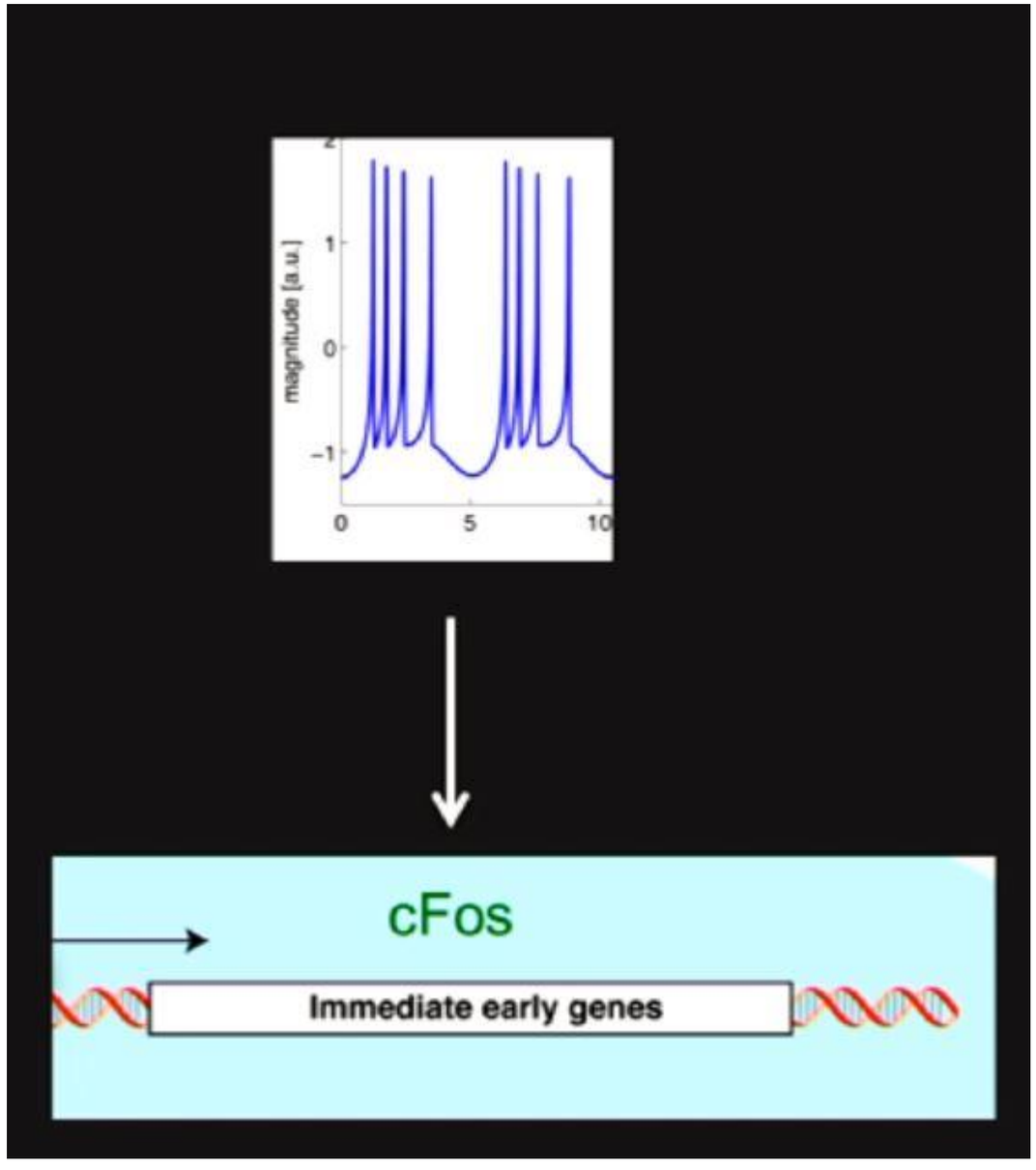
Male mice are normally infanticidal

Dulac and colleagues

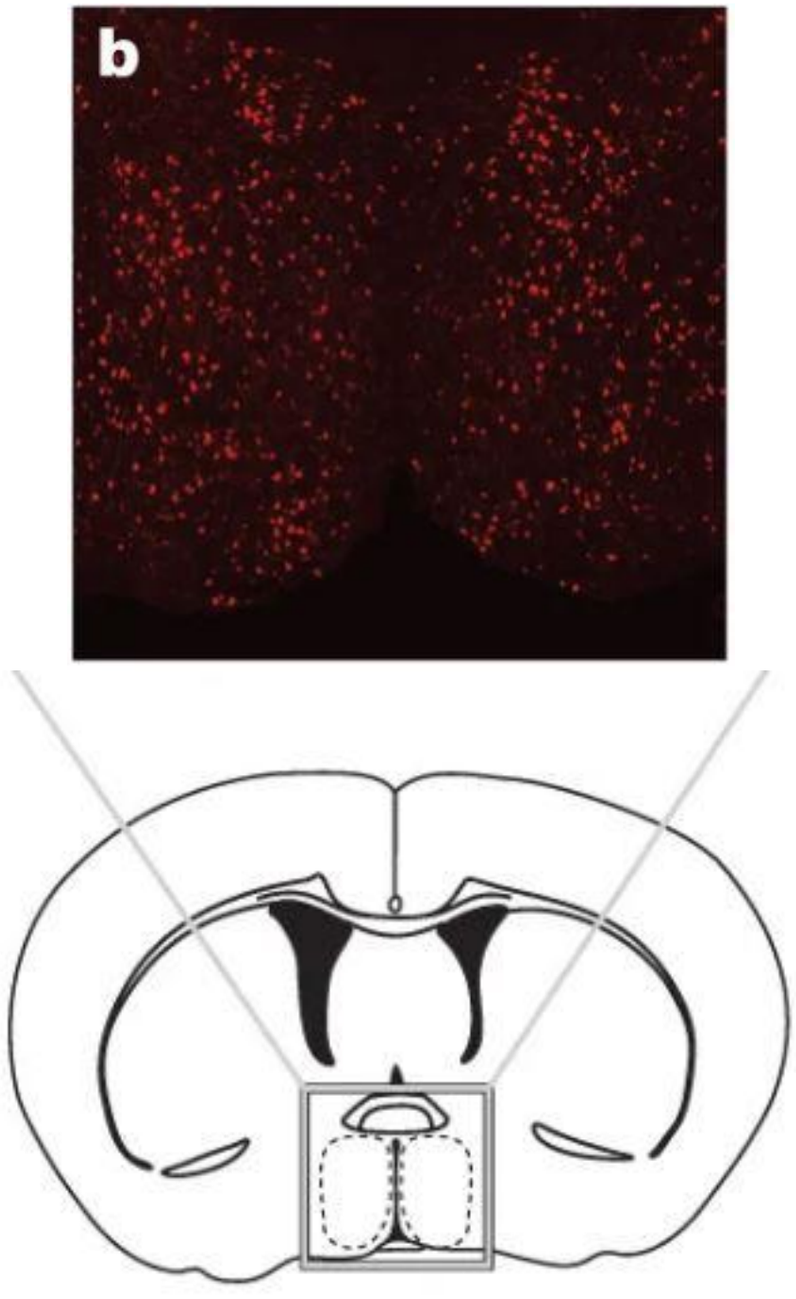
Parenting – What is changing in the brain?



Parenting – What is changing in the brain?

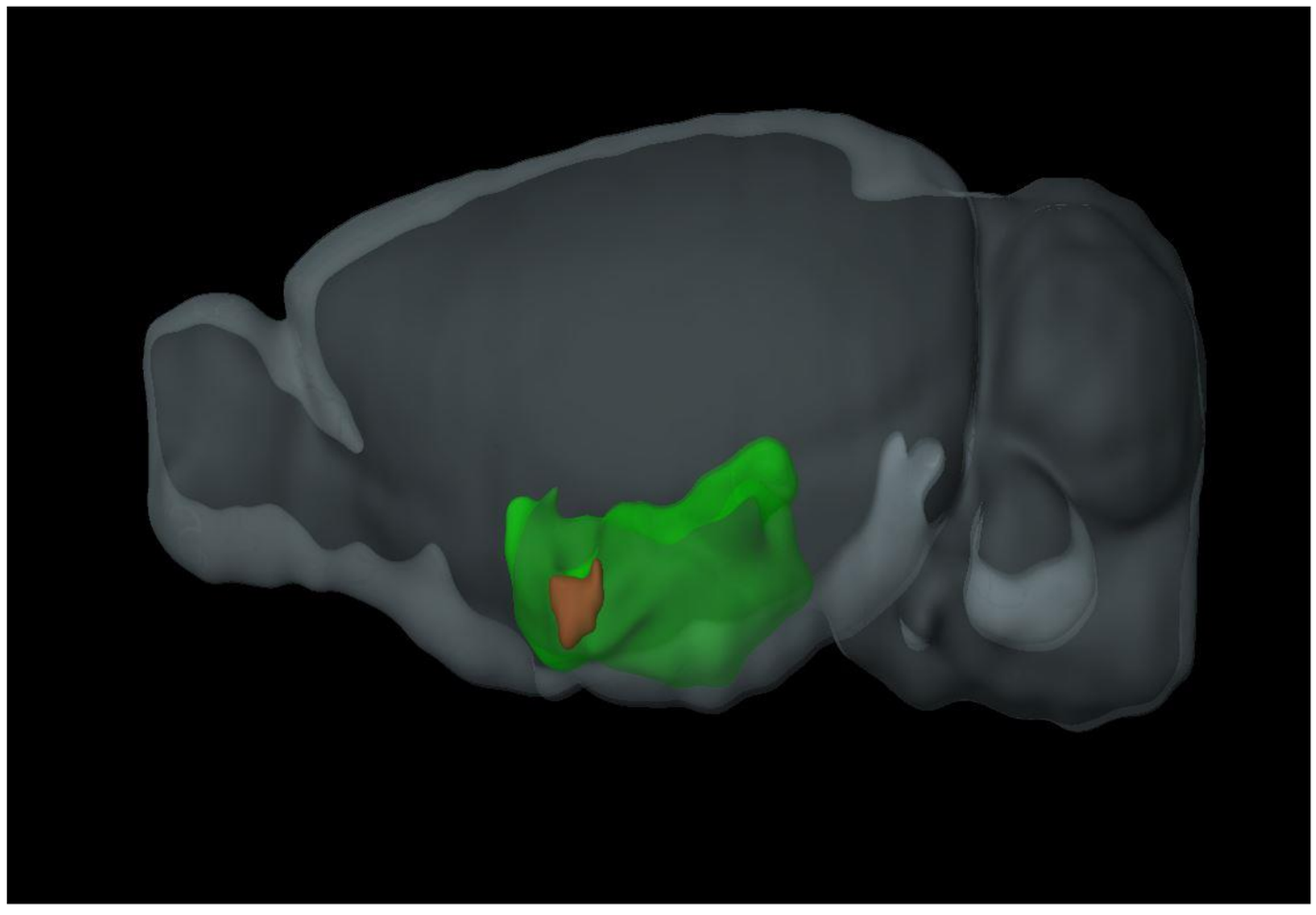


Dulac and colleagues



Medial preoptic area in hypothalamus

Parenting – What is changing in the brain?



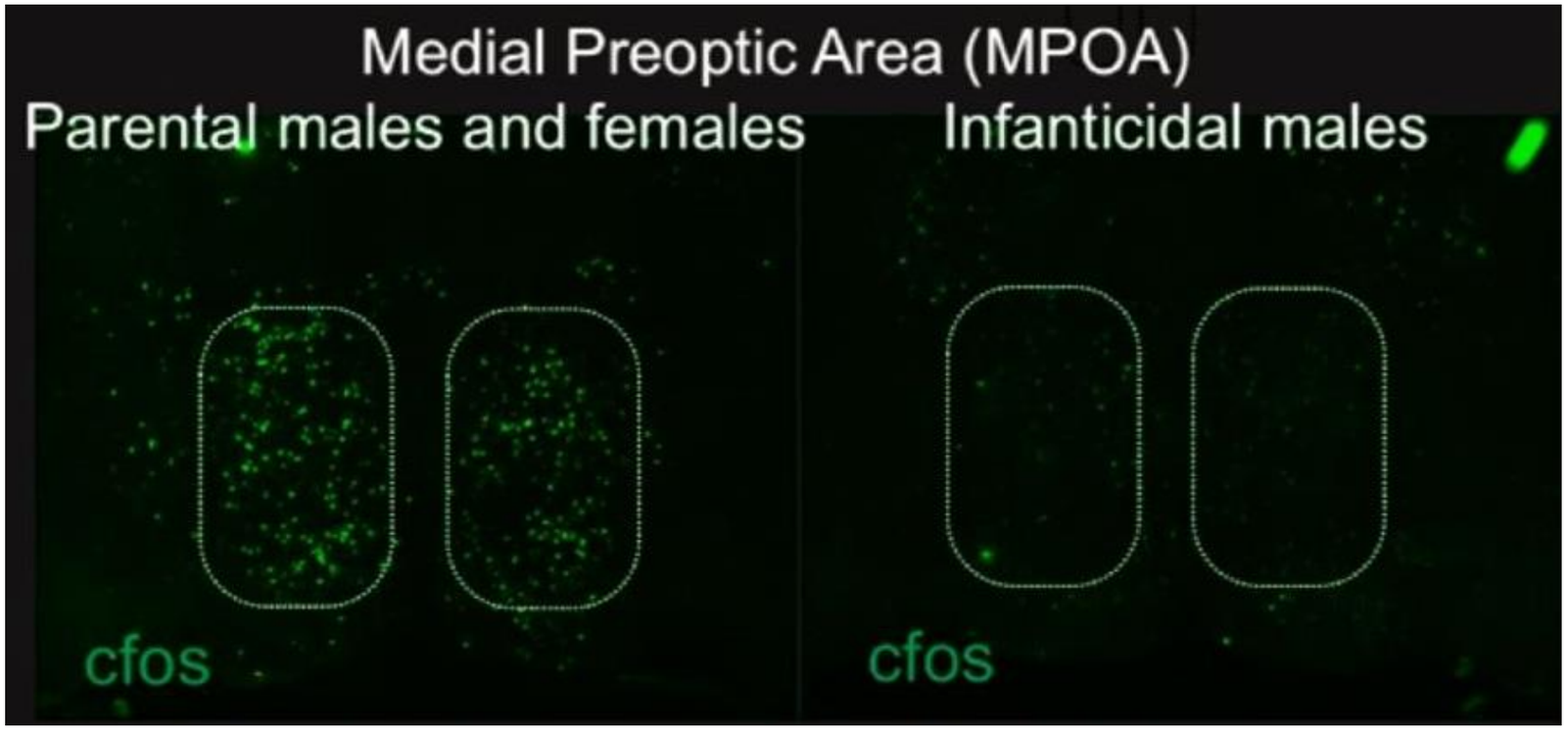
Hypothalamus

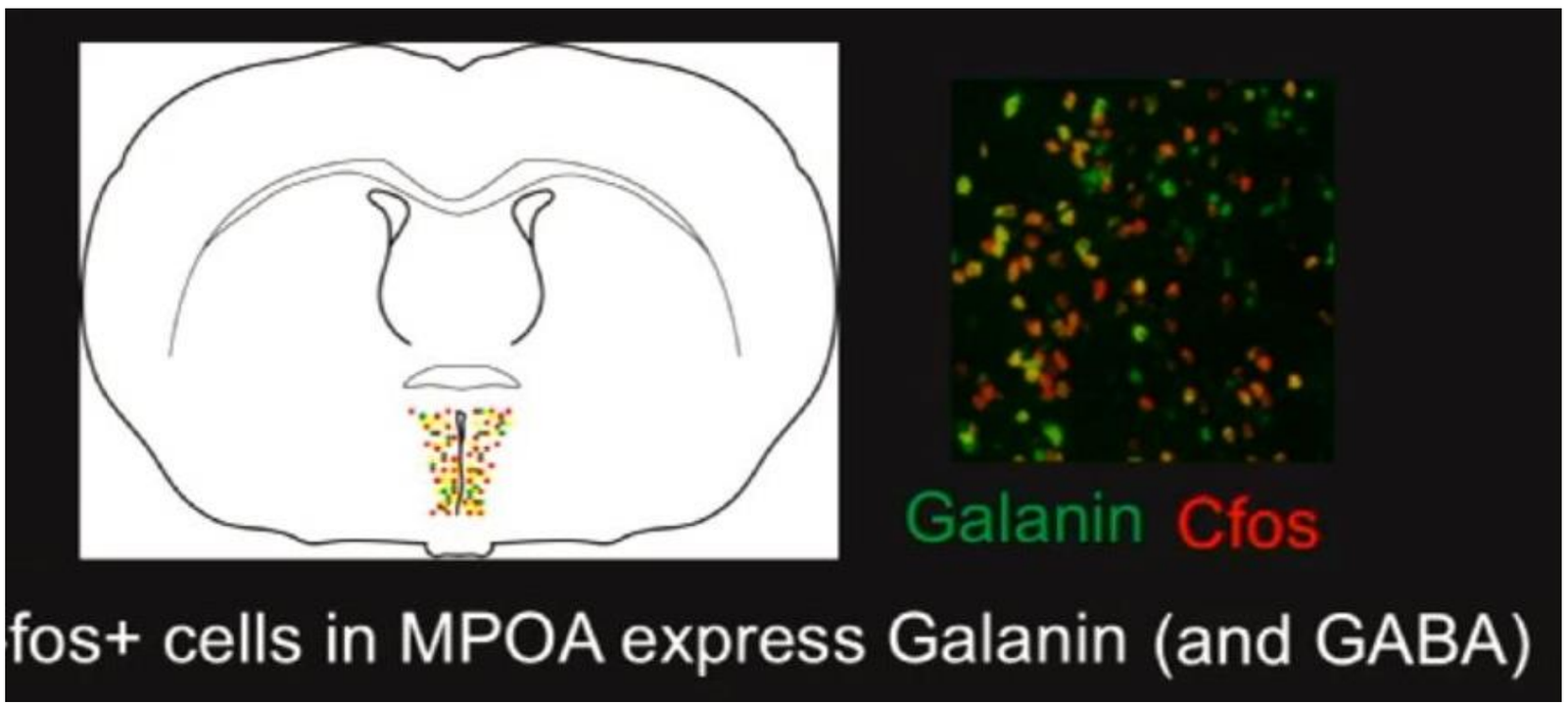
-Sleep
-Feeding
-Arousal
-Aggression
-Blood pressure
-Temperature Control
-Social hormones like oxytocin & vasopressin

Medial Preoptic Area (MPOA)

-Sleep
-Temperature control
-Sexual behaviors
-Parental behaviors

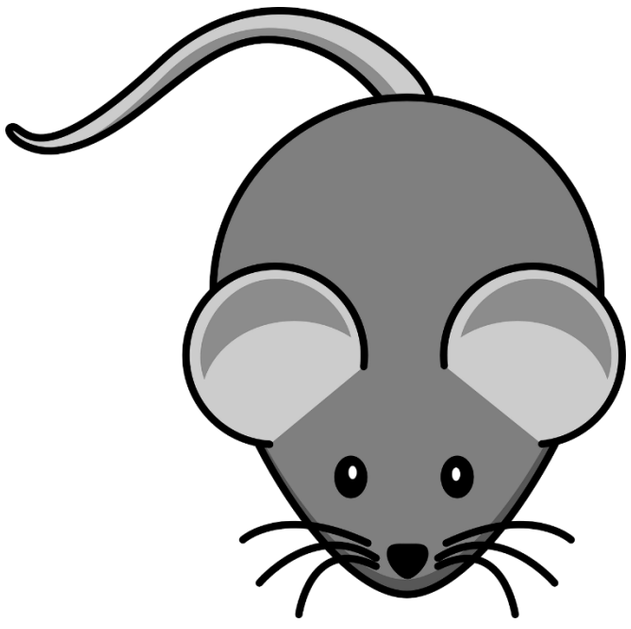
Parenting – Galanin+ neurons in MPOA activated in parents





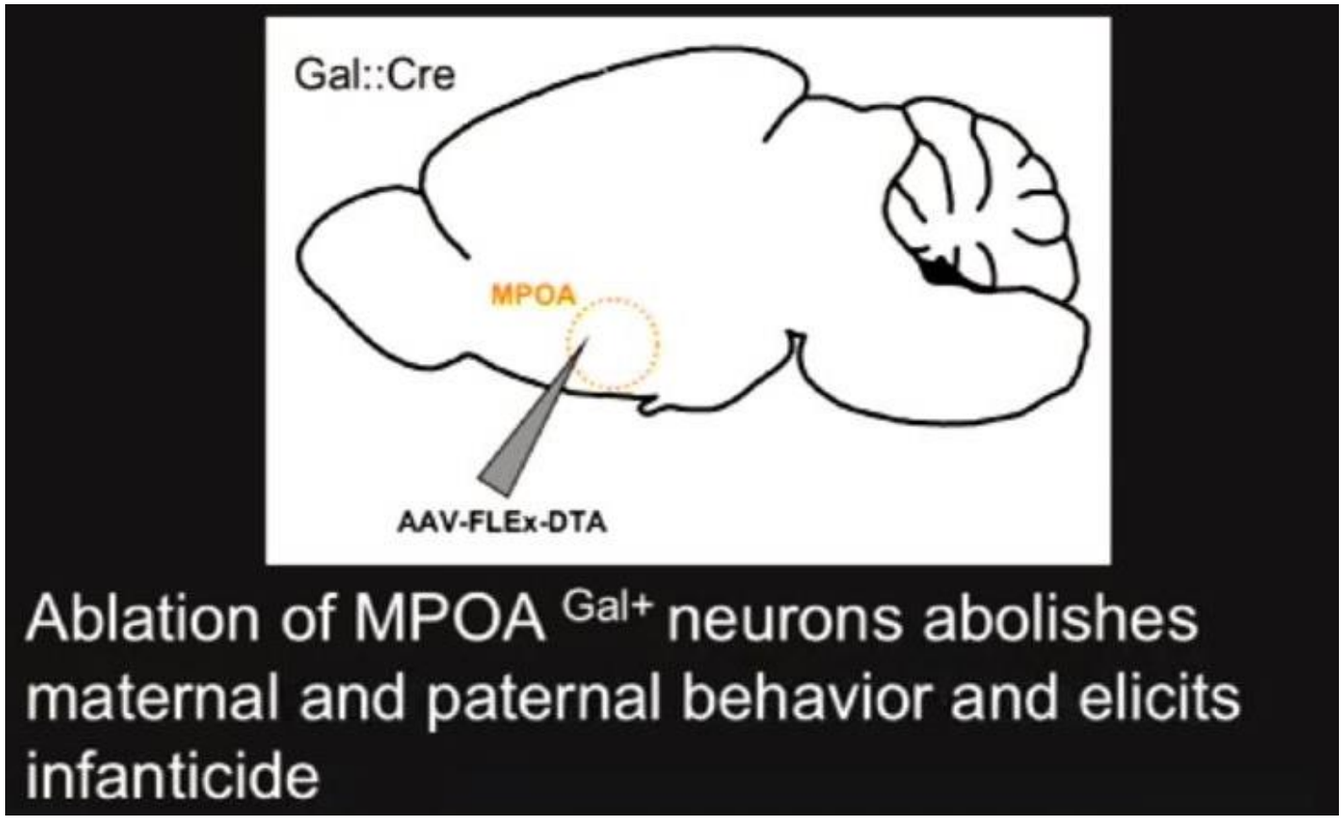
Dulac and colleagues

Parenting – Galanin+ neurons in MPOA



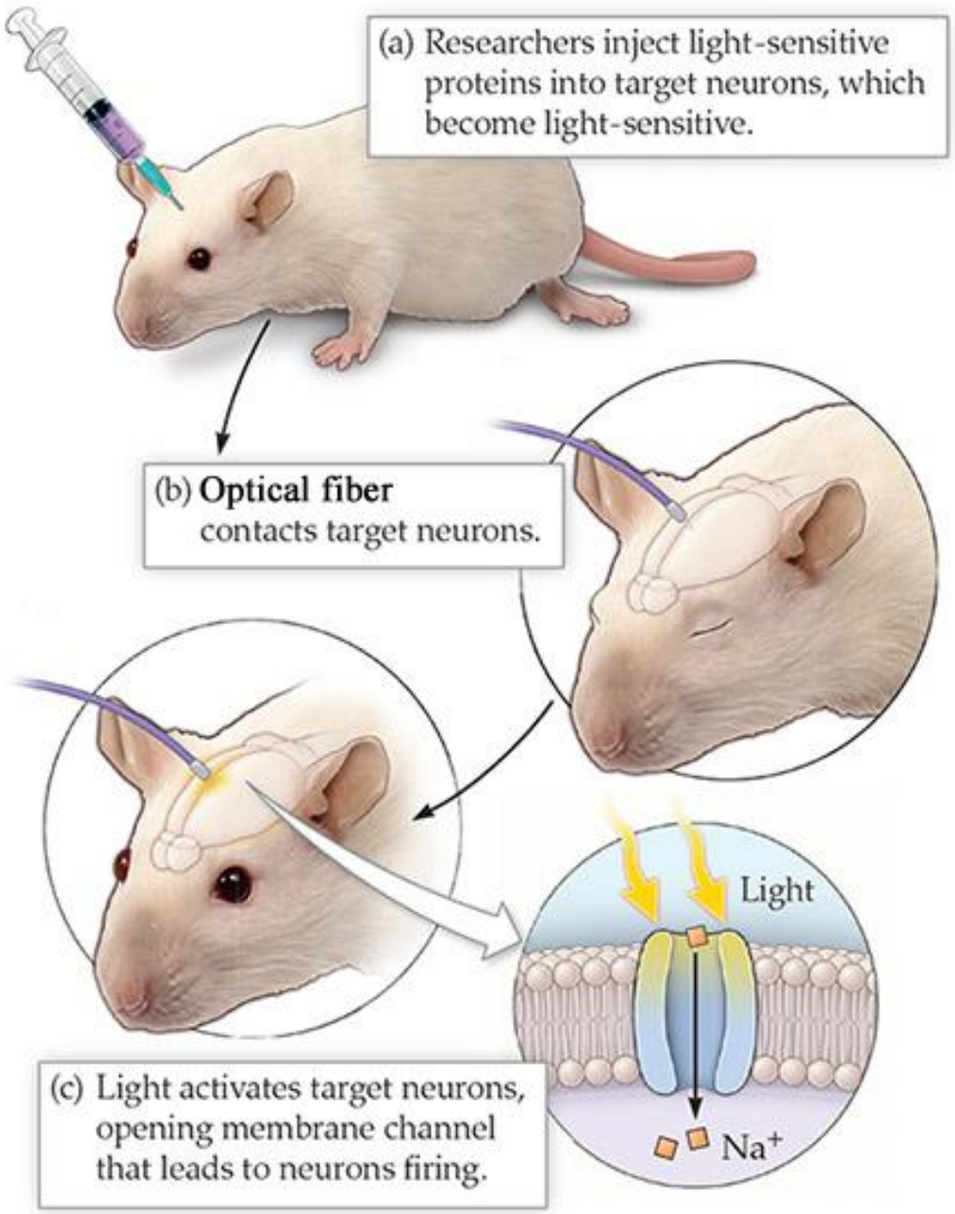
Cre mouse
To target only Gal+ neurons

Inject virus with toxin in MPOA
to target only Gal+ neurons

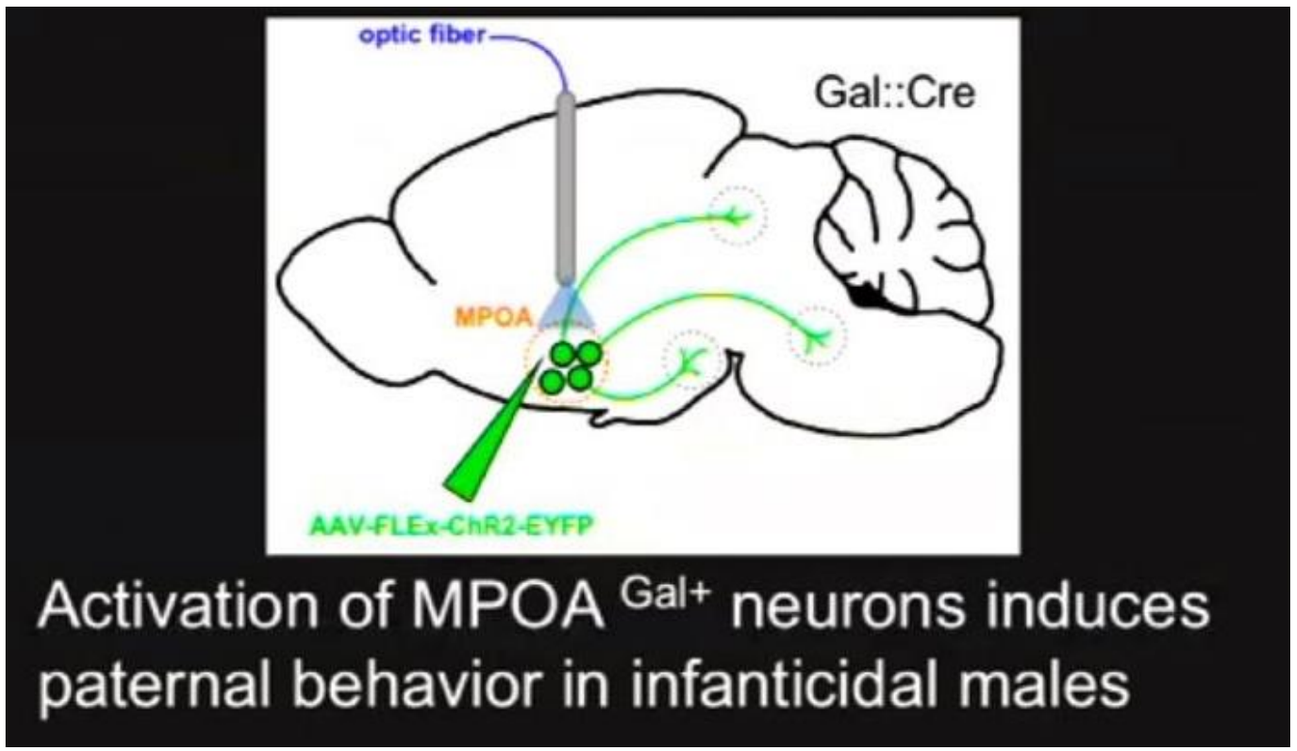


Dulac and colleagues

Parenting – Galanin+ neurons in MPOA

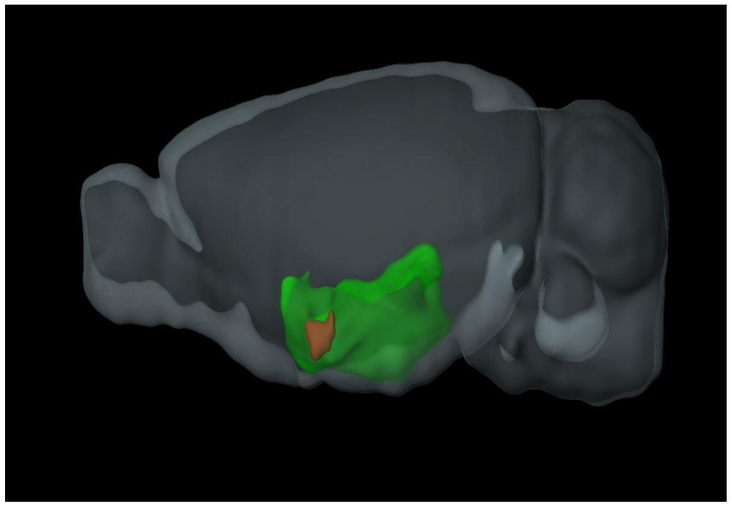


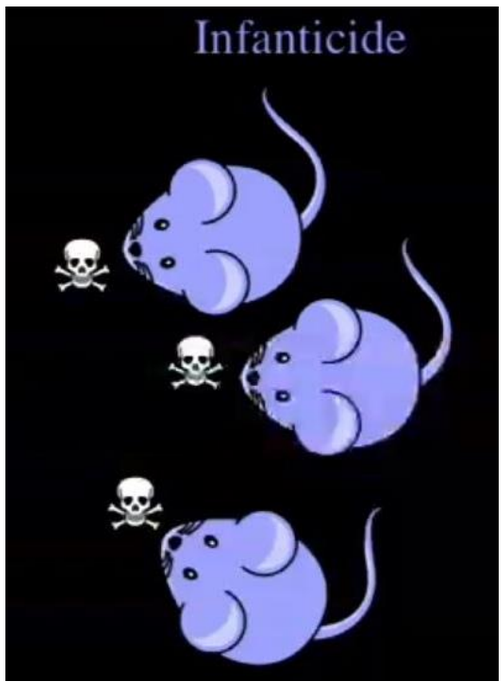
Inject virus with channelrhodopsin in MPOA to target only Gal+ neurons and then shine blue light to activate them

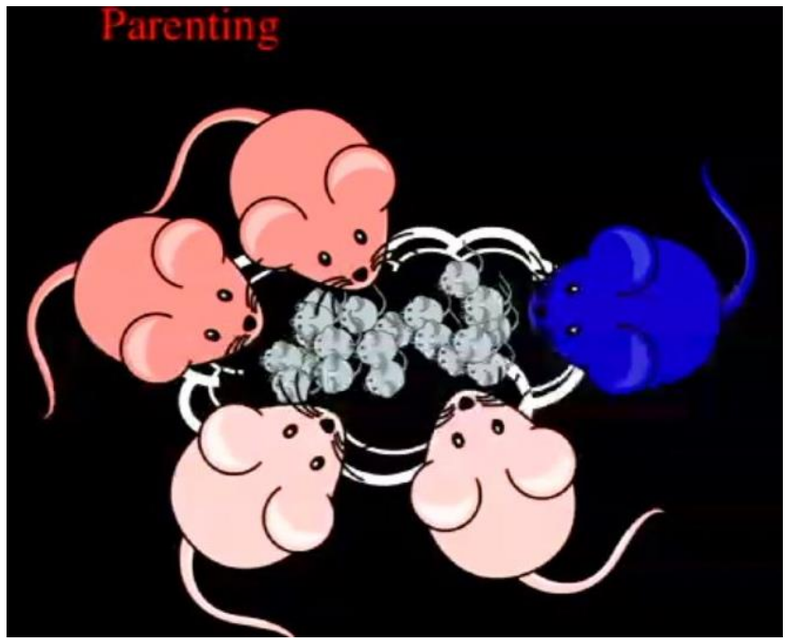


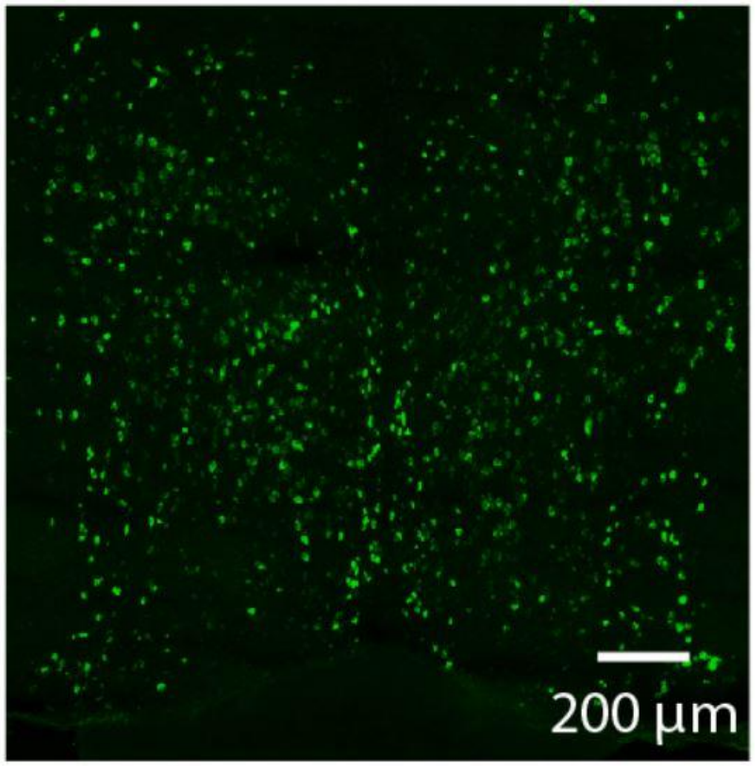
Dulac and colleagues

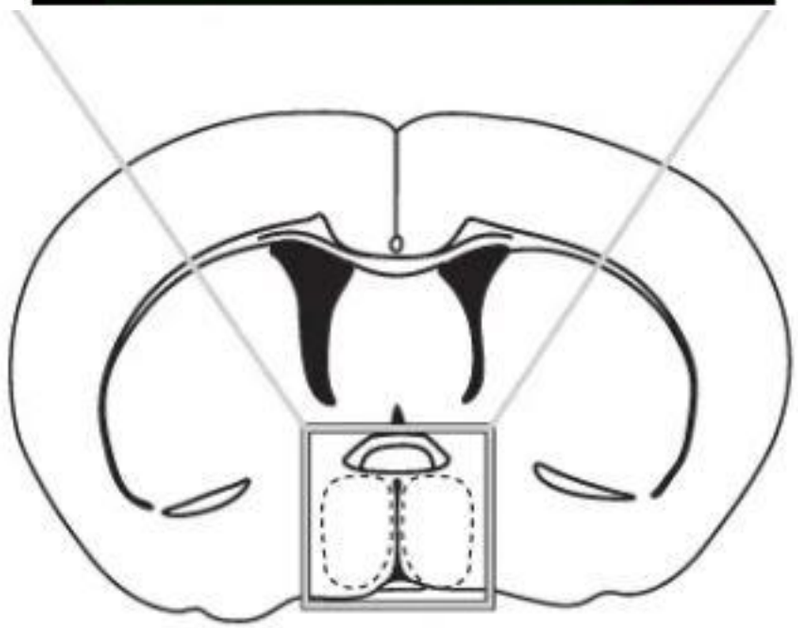
Parenting – Galanin+ neurons in MPOA





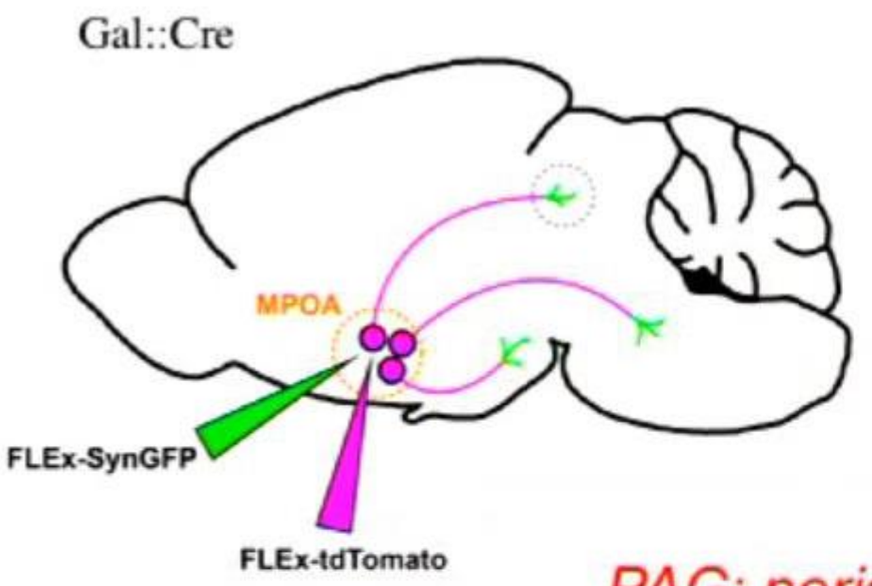


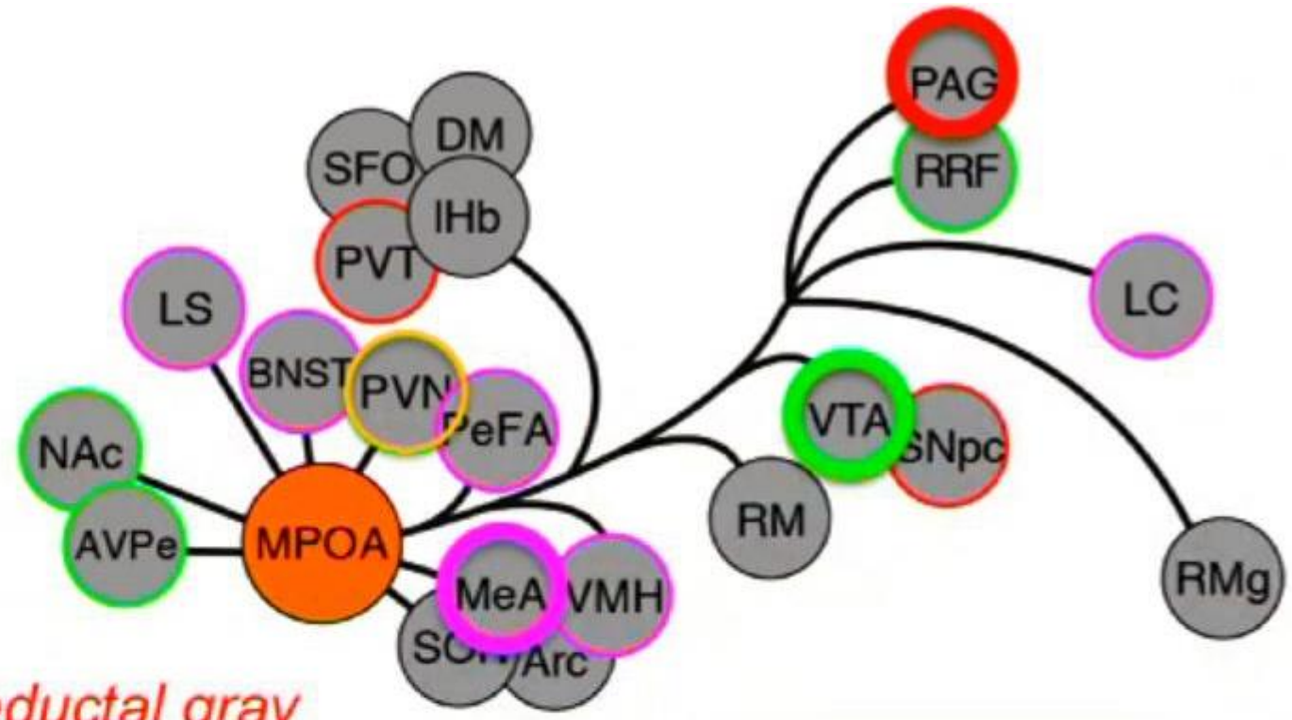




What other aspects of parenting do these neurons influence?

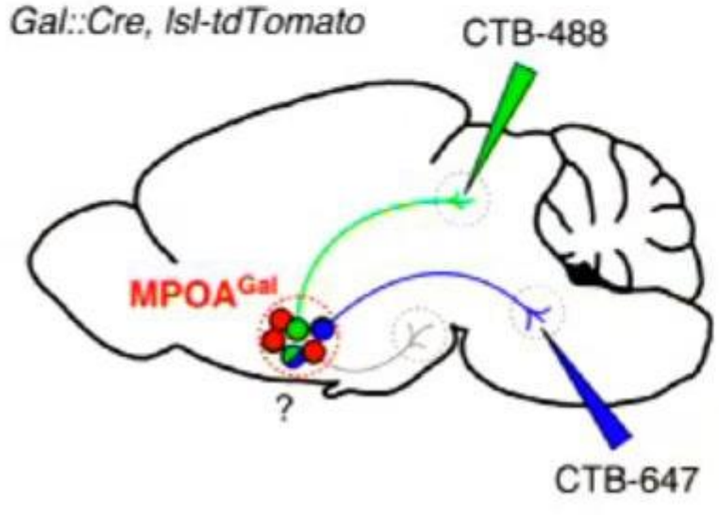
Parenting – Different Galanin+ neurons project to different areas

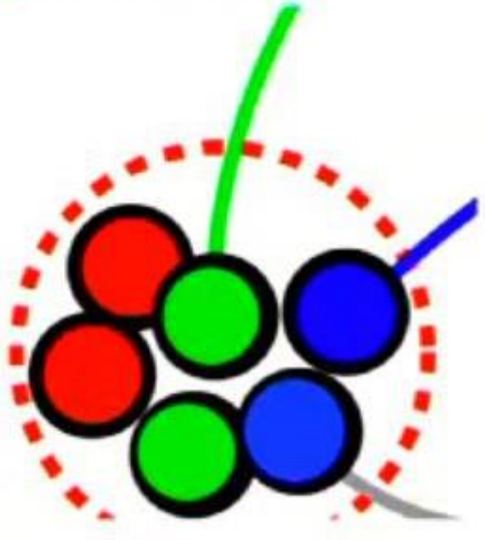




PAG: periacqueductal gray
VTA: ventral tegmental area
MeA: medial amygdala

motivational
motor
social behaviors
hormonal modulation

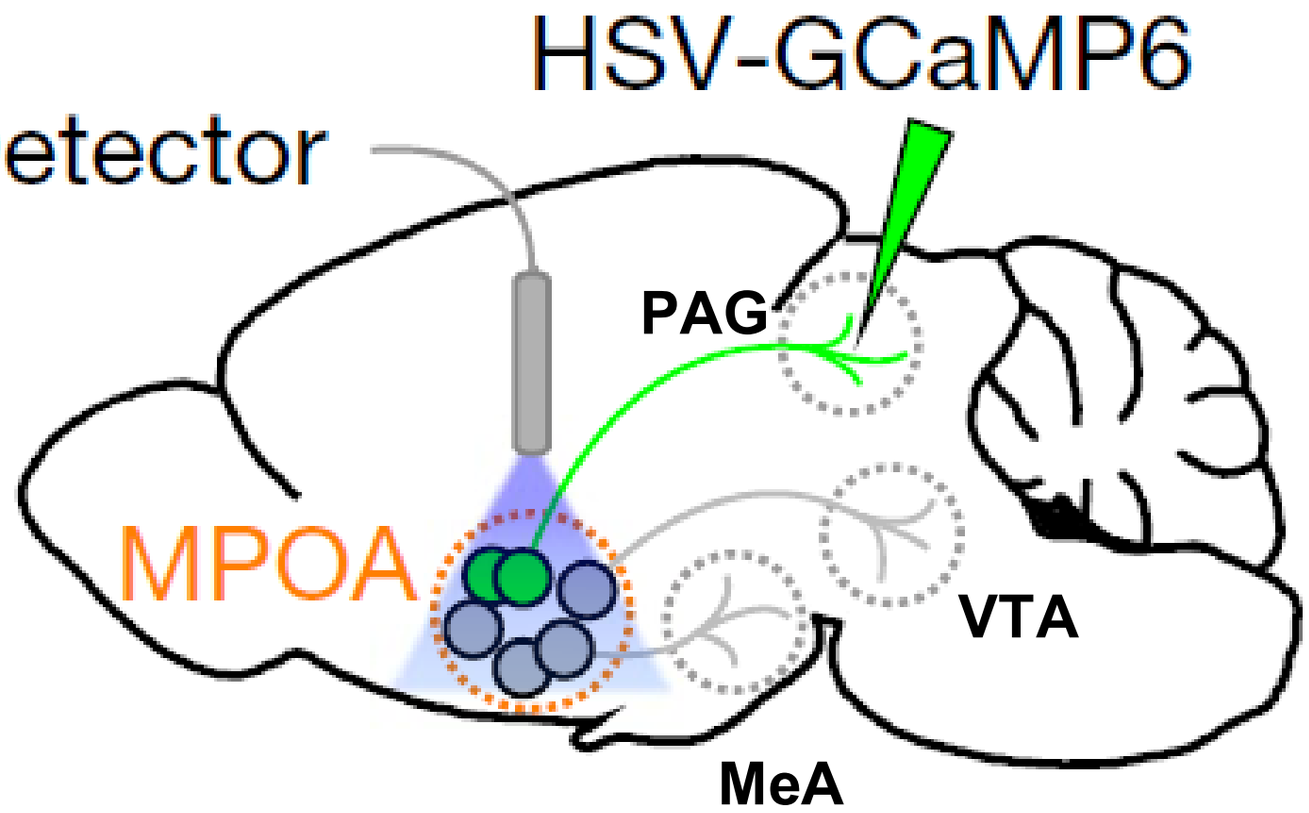




Gal neurons do not bifurcate

Dulac and colleagues

Parenting – Responses of Gal+ neurons projecting to different areas



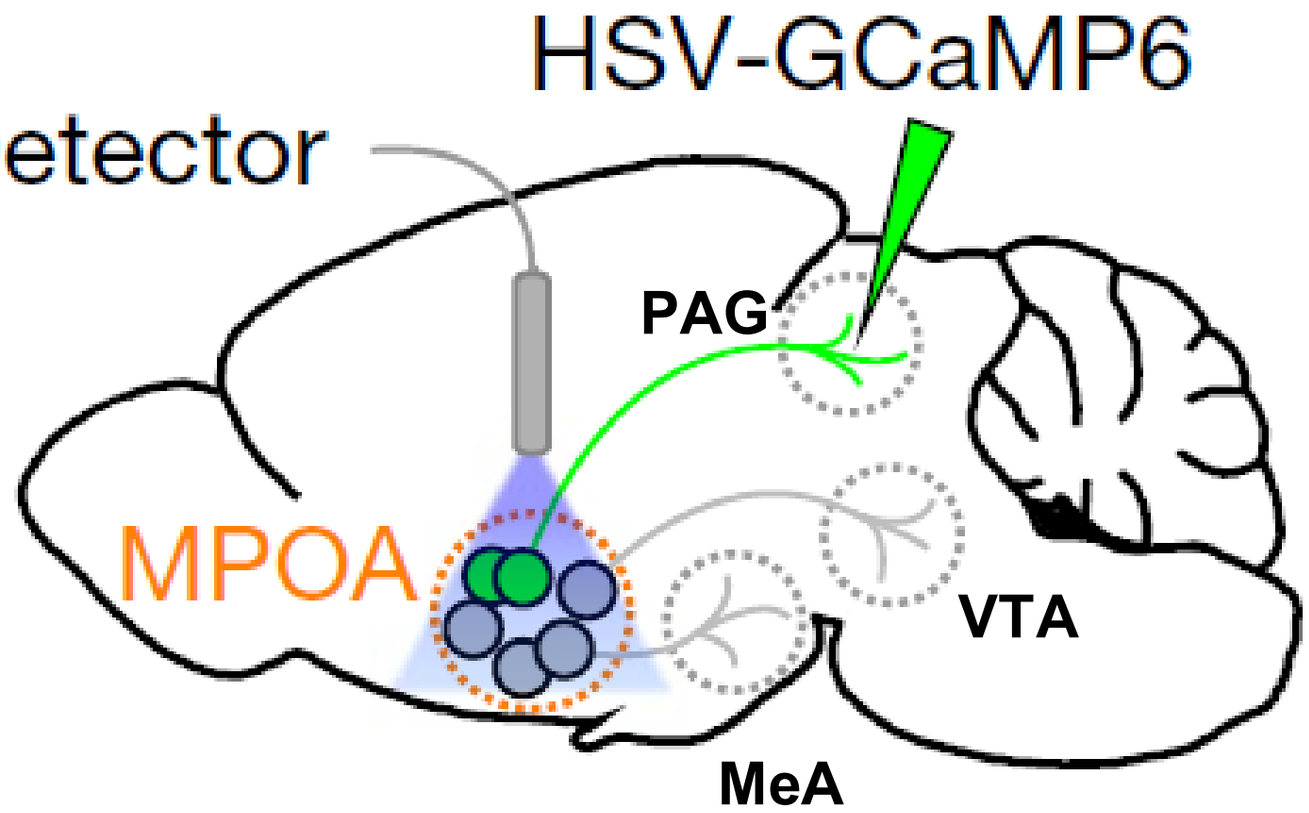
PAG – Periaqueductal Gray

VTA – Ventral Tegmental Area

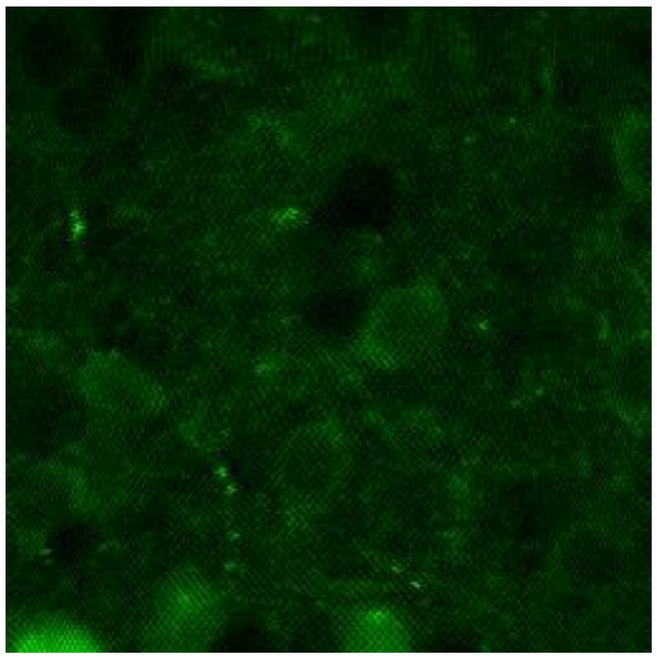
MeA – Medial Amygdala

Dulac and colleagues

Parenting – Responses of Gal+ neurons projecting to different areas



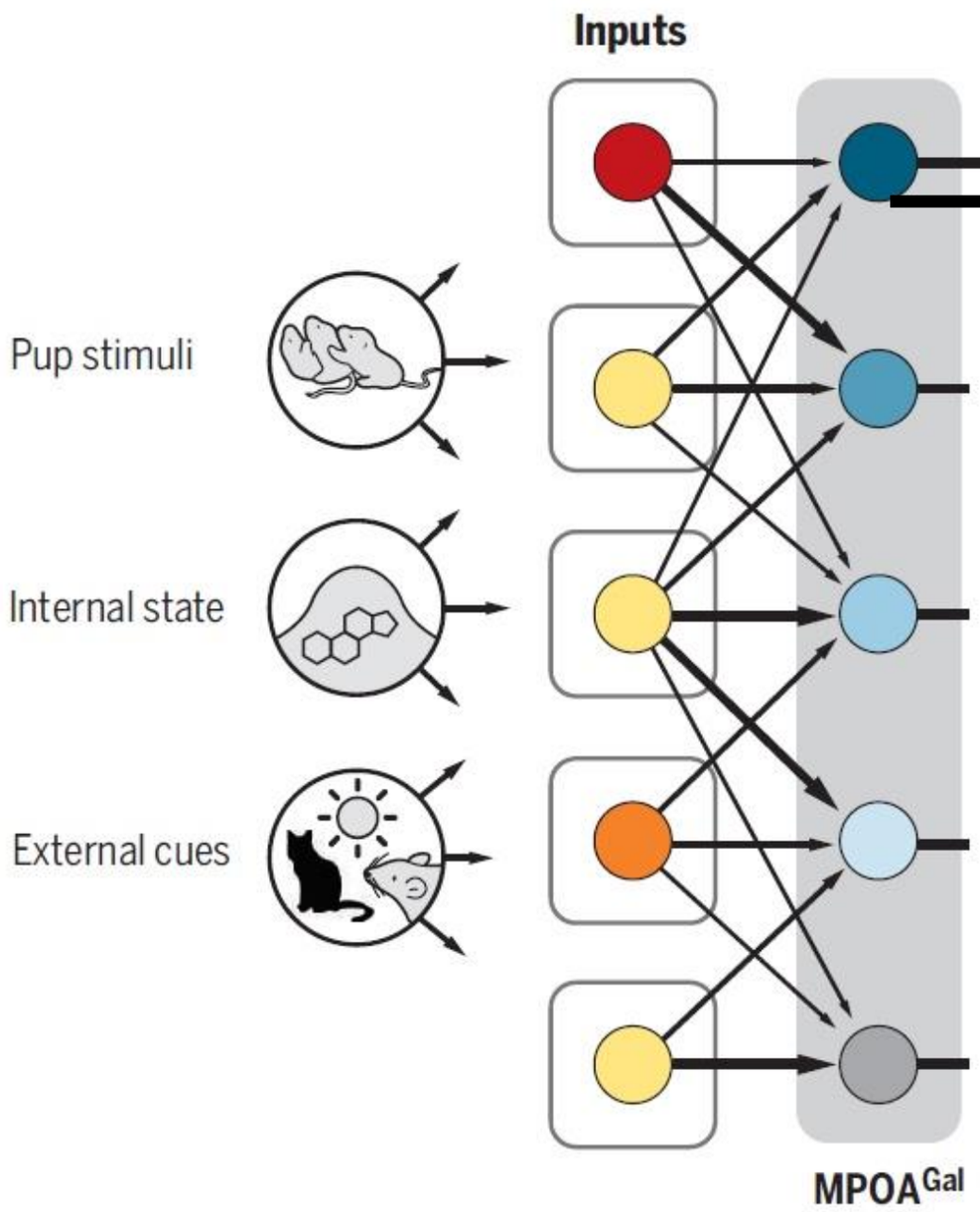
GCaMP6 – Ca^{2+} activity (spikes)



5x speed

Dulac and colleagues

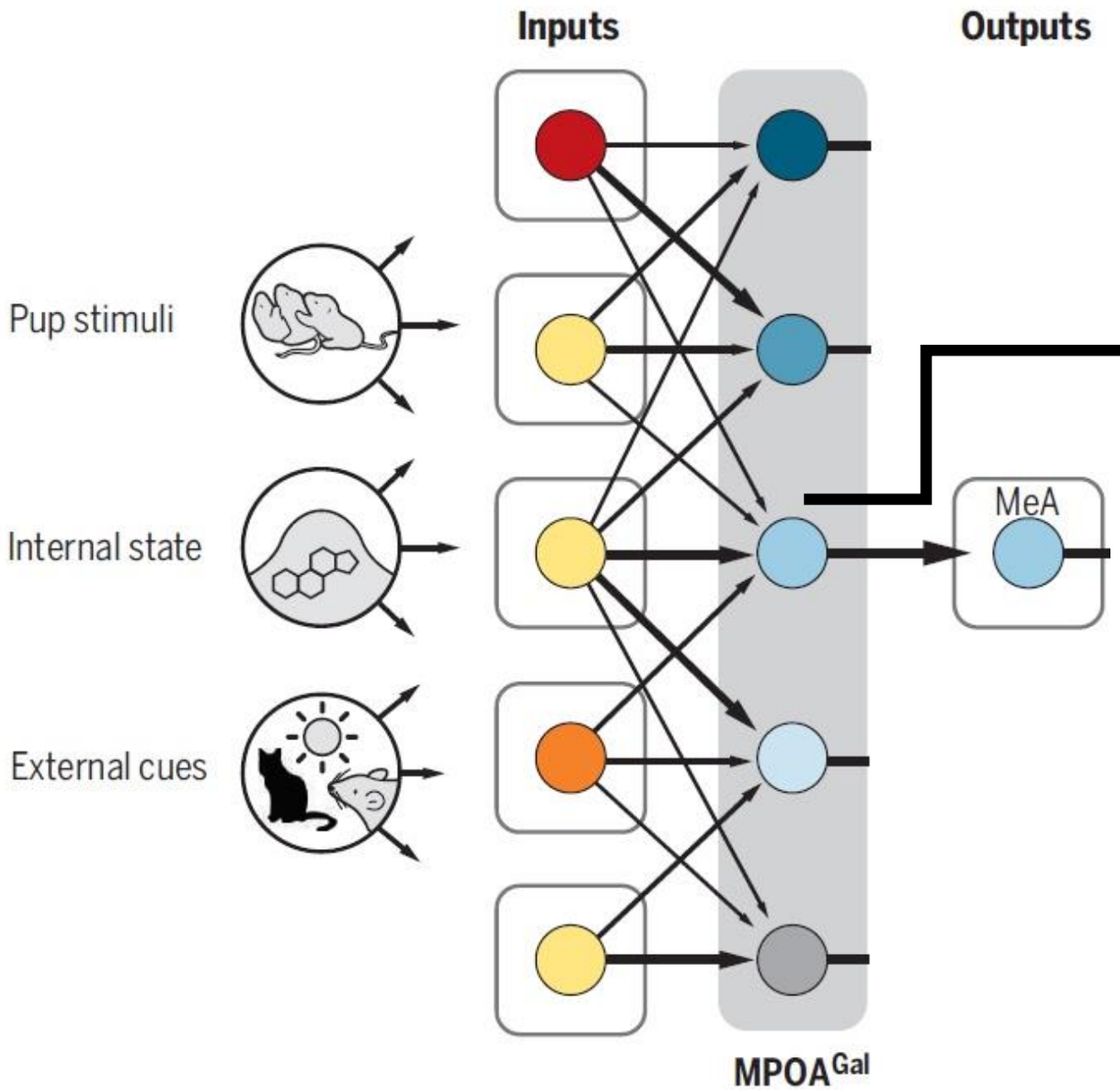
Parenting – Responses of Gal+ neurons projecting to PAG



Increased activity during pup grooming

Dulac and colleagues

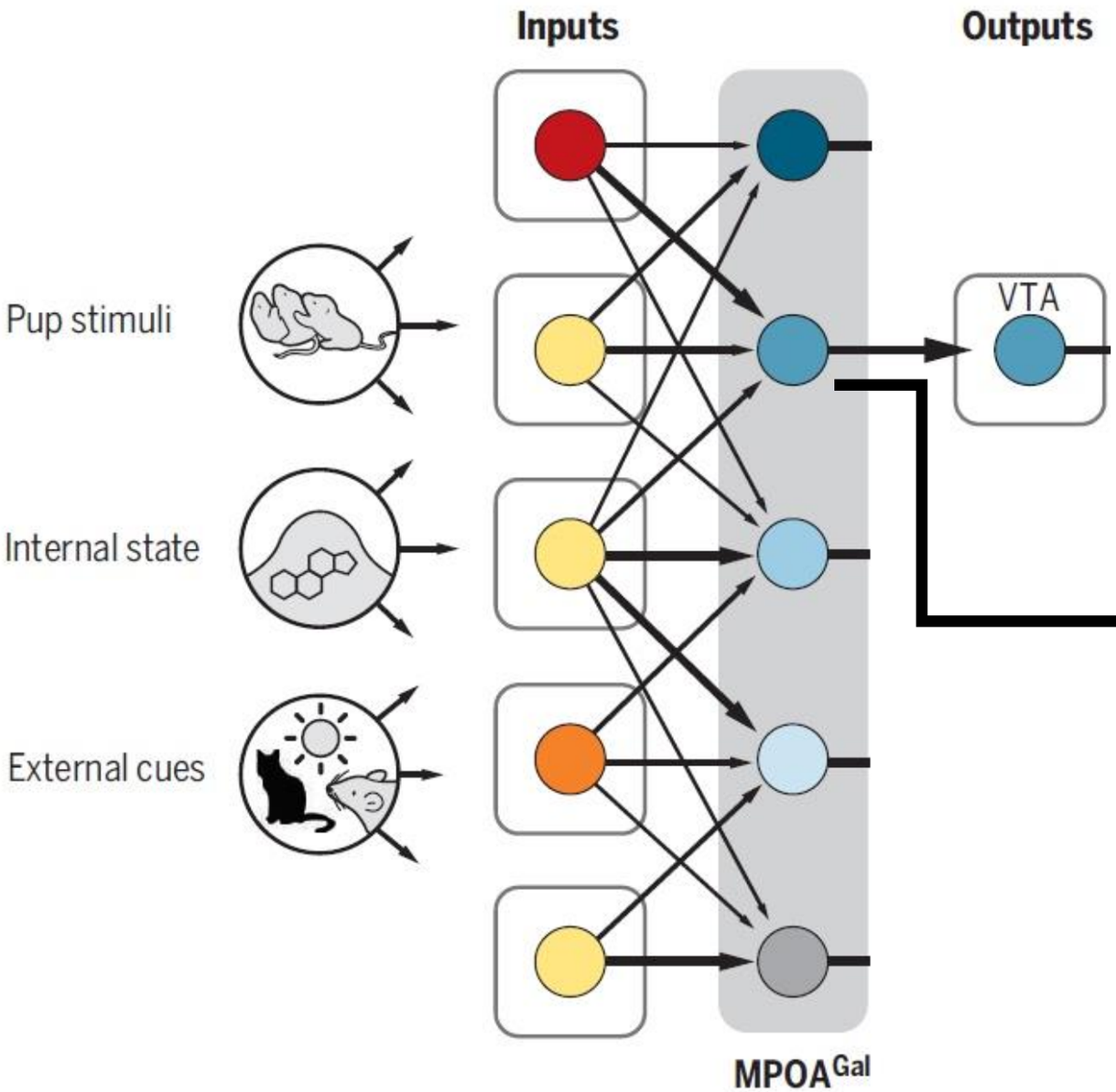
Parenting – Responses of Gal+ neurons projecting to MeA



Increased activity during sniffing or retrieving pups or entering nest with pups

Dulac and colleagues

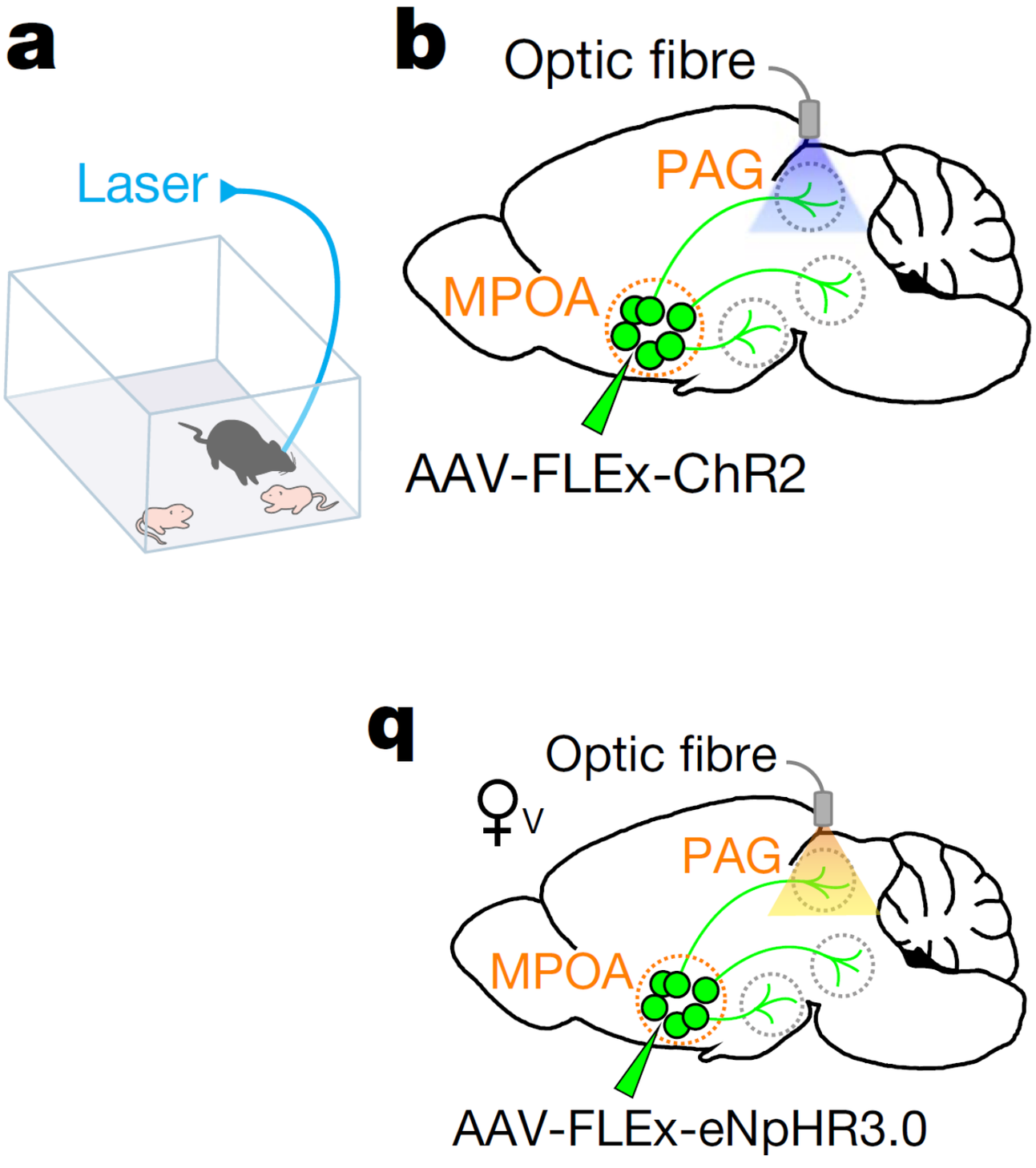
Parenting – Responses of Gal+ neurons projecting to VTA



Didn’t seem to do much

Dulac and colleagues

Parenting – Pup grooming



Activating PAG projecting neurons increases pup grooming

Inactivating PAG projecting neurons decreases pup grooming

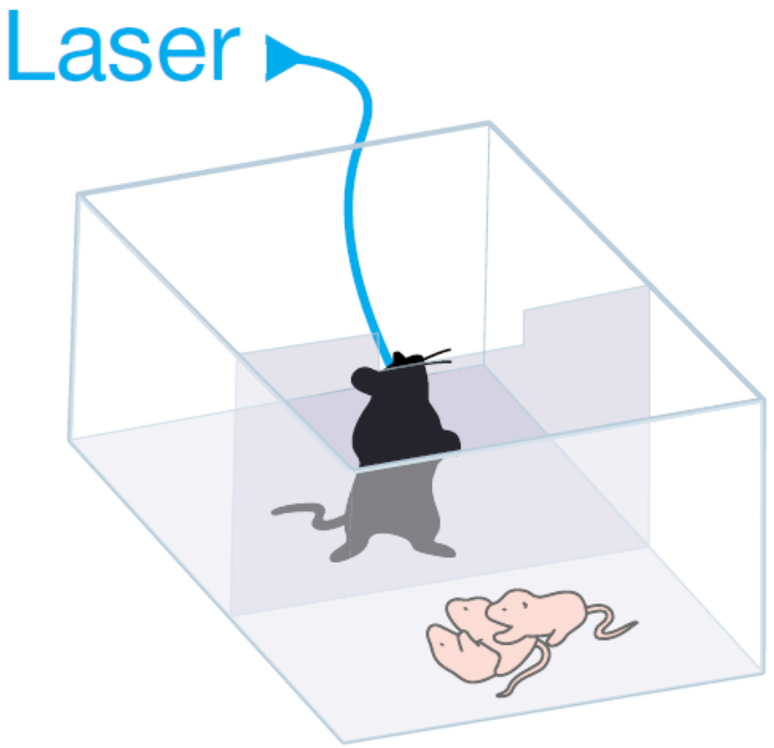
Dulac and colleagues

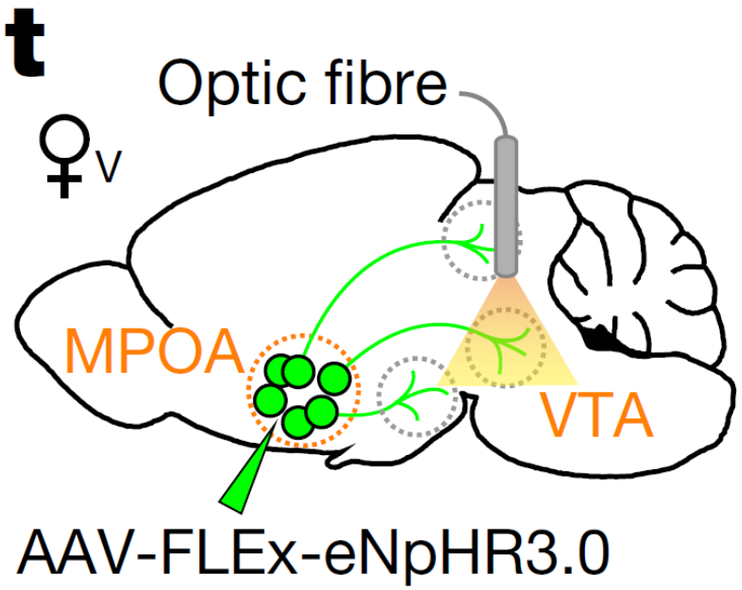
Parenting – Motivation to reach pups



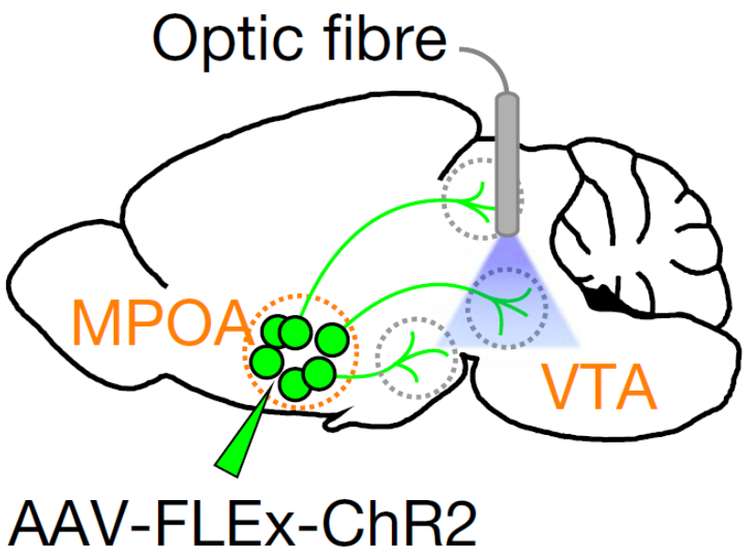
Dulac and colleagues

Parenting – Motivation to reach pups





Inactivating VTA projecting neurons decreased barrier crossing

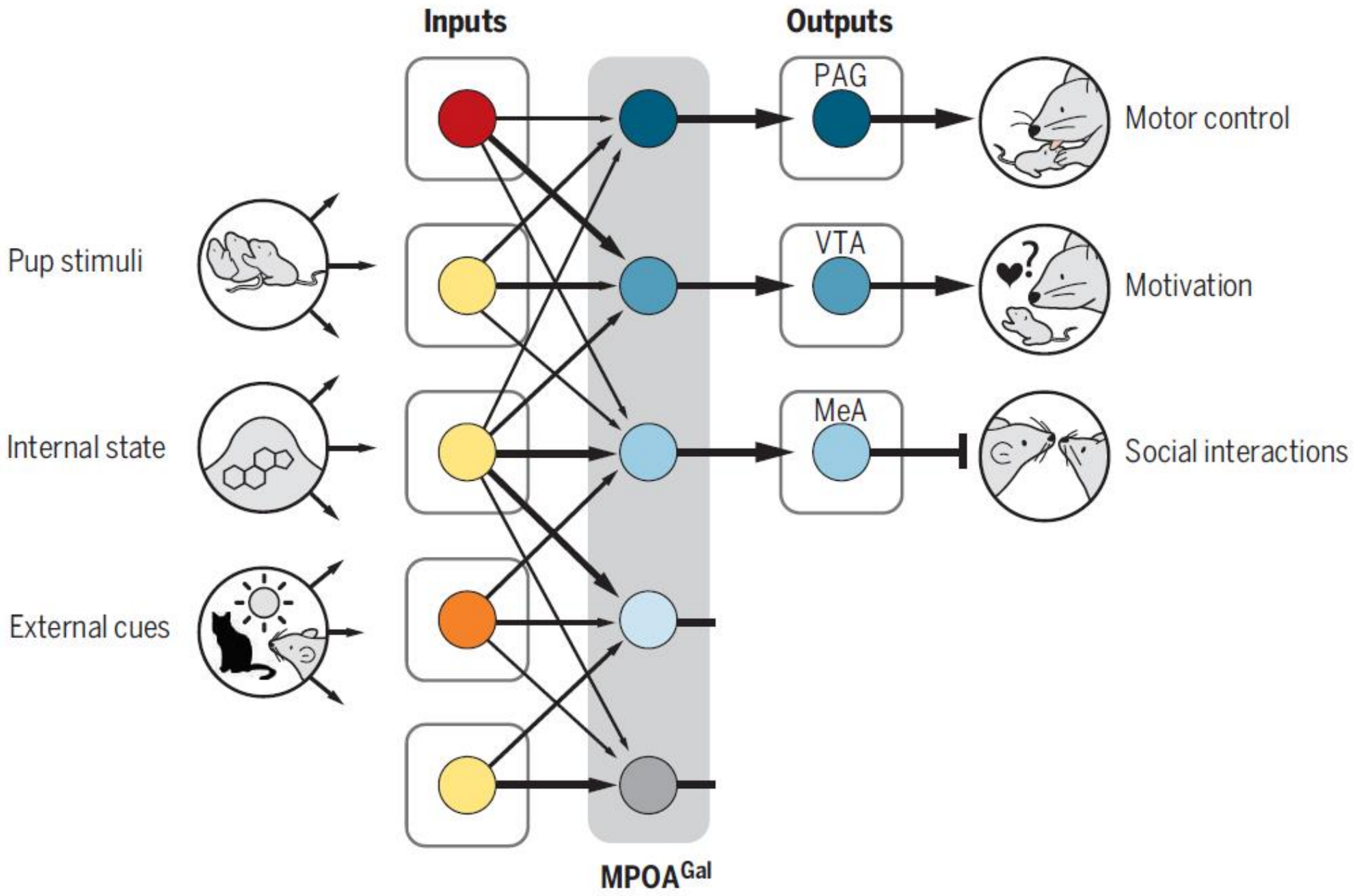


Activating VTA projecting neurons increased barrier crossing

But once they got there they might kill the pups, so it didn't make them parental overall

Dulac and colleagues

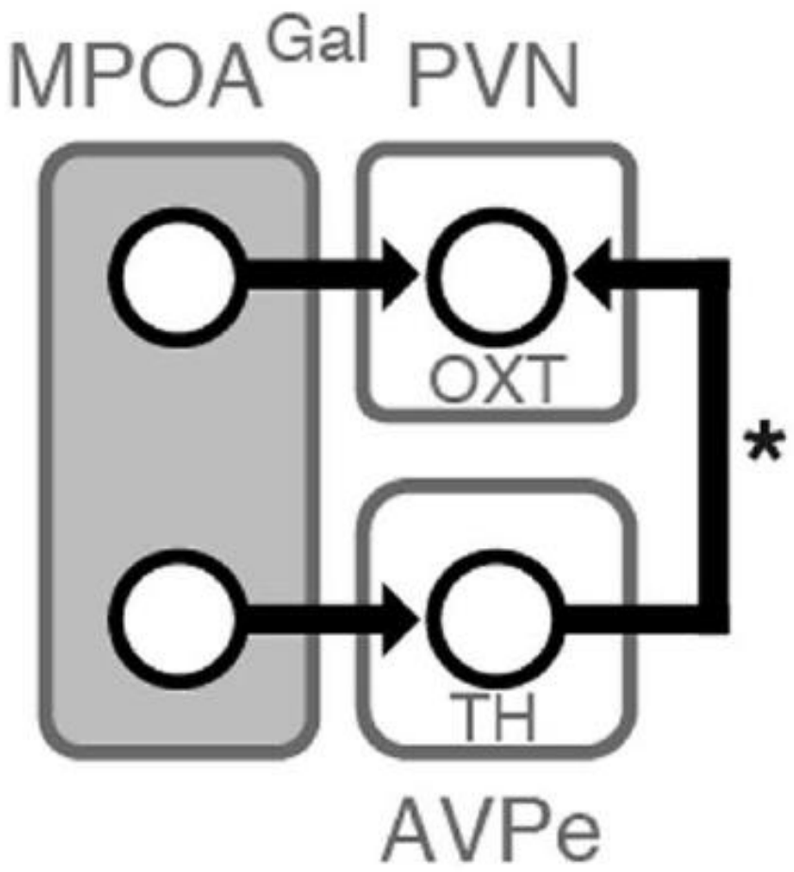
Parenting – Different aspects



Dulac and colleagues

Parenting – Gal+ neurons from MPOA also project to OXT+ neurons in PVN

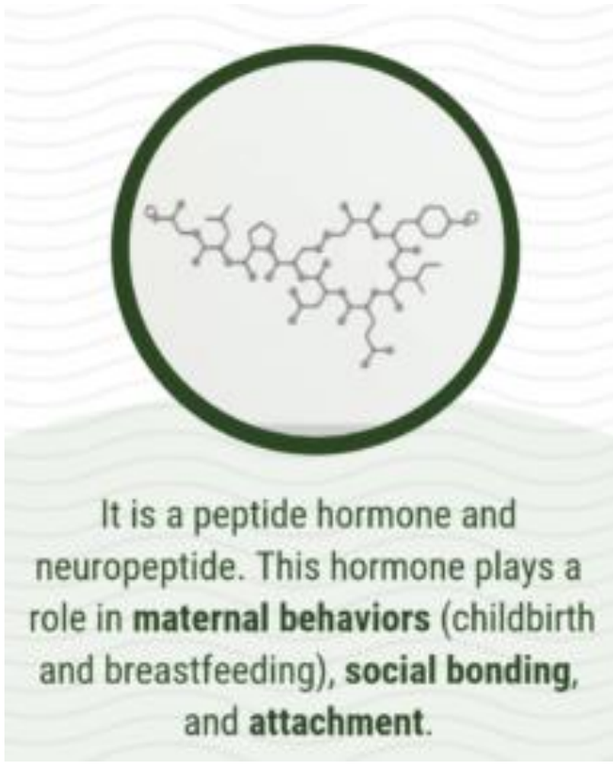
Periventricular Nucleus



Anterior Periventricular Nucleus

MPOA Gal+ neurons also project to PVN directly or indirectly

– they project to oxytocin⁺ neurons in the PVN



Oxytocin

Parenting – Carrying your young ones







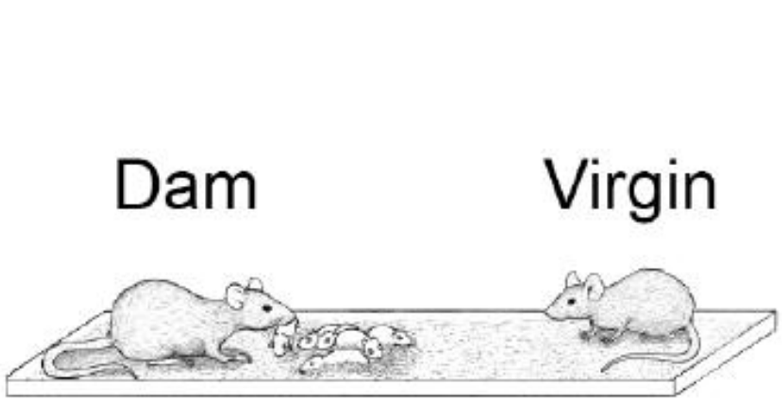
Parenting – Carrying your young ones

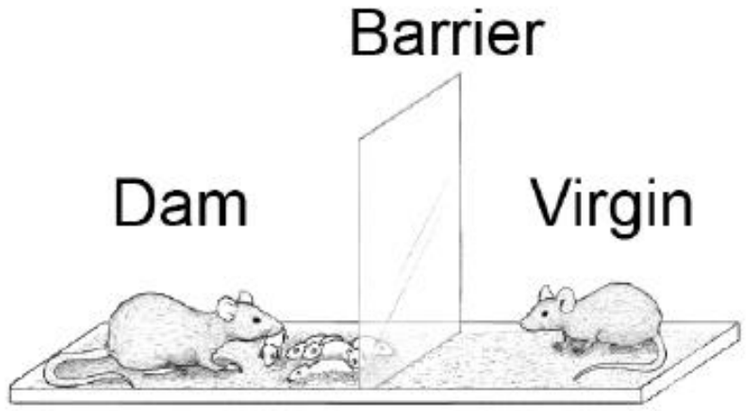


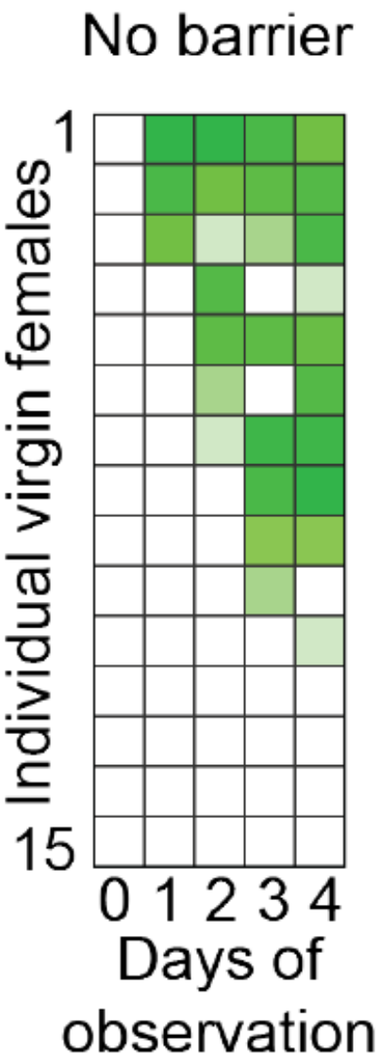


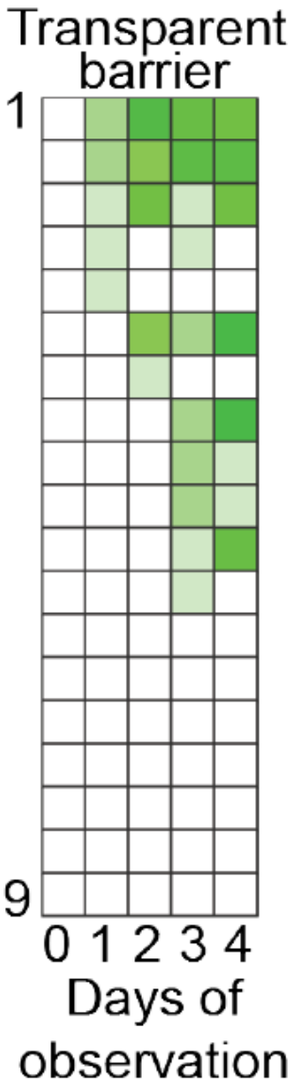


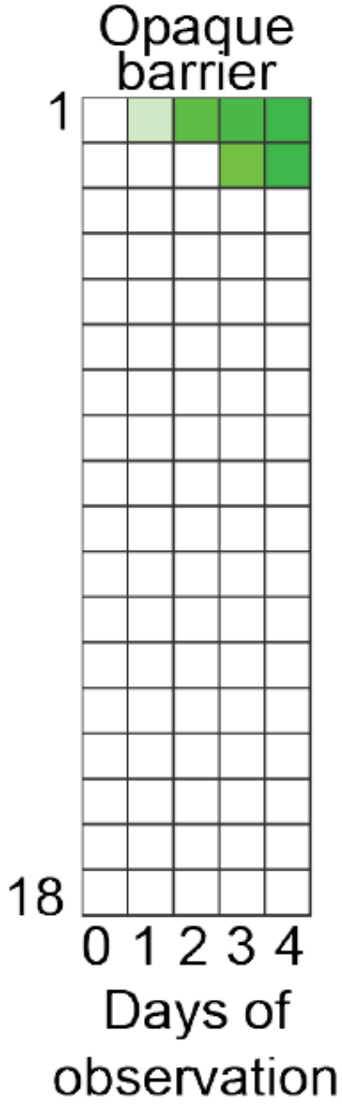
Parenting – Observational learning of pup retrieval in virgin female mice

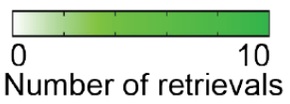






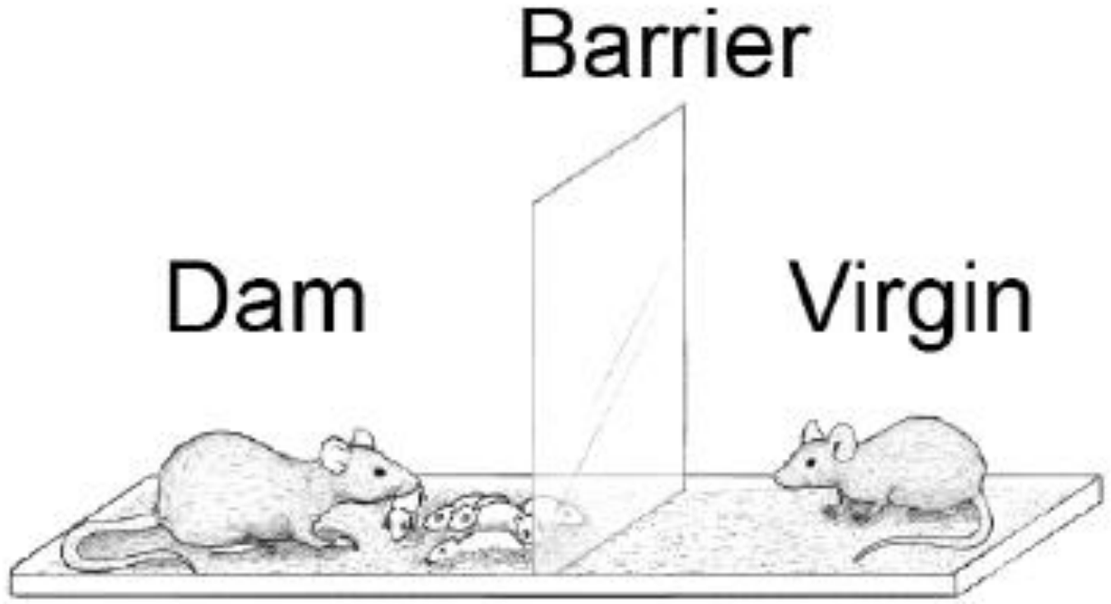


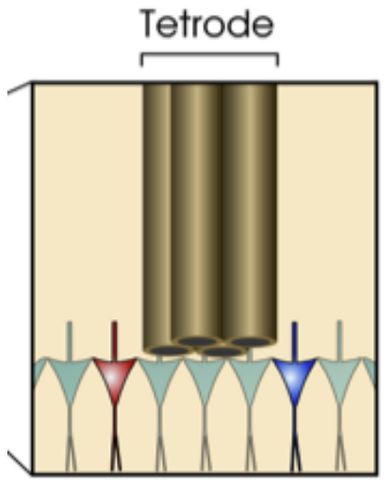


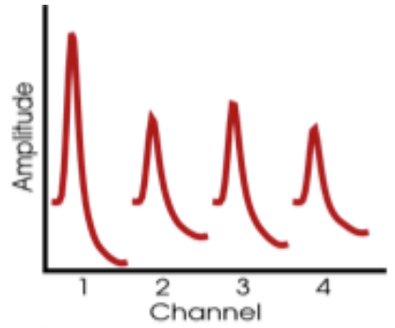


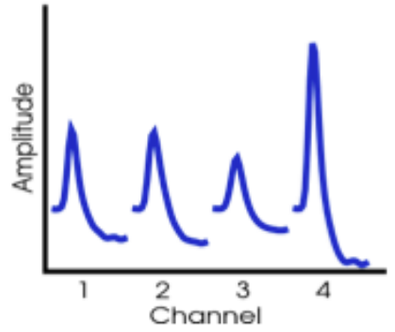
Froemke and colleagues

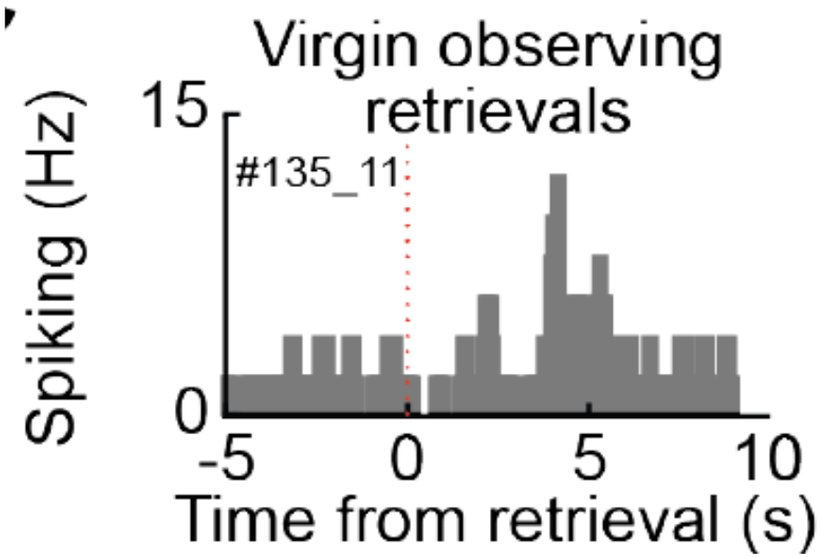
Parenting – OXT+ neurons in virgin females increased firing on observing retrieval of pups





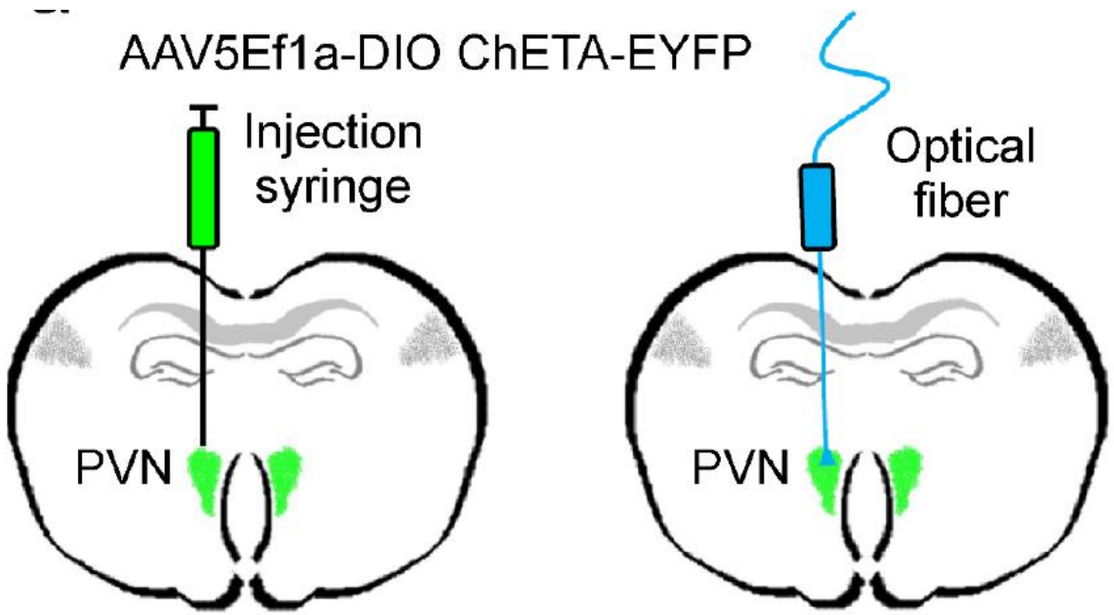


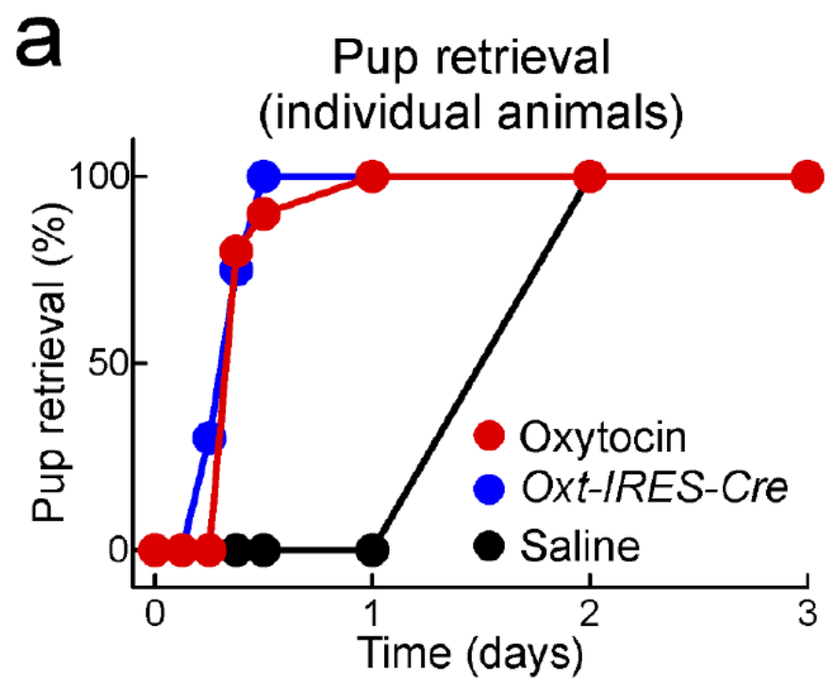




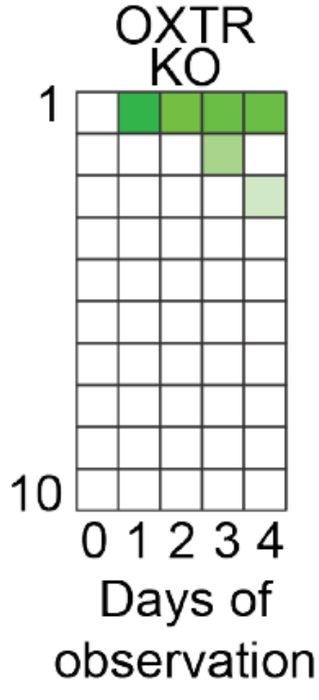
Froemke and colleagues

Parenting – Oxytocin neuron activation makes learning pup retrieval for virgin females faster





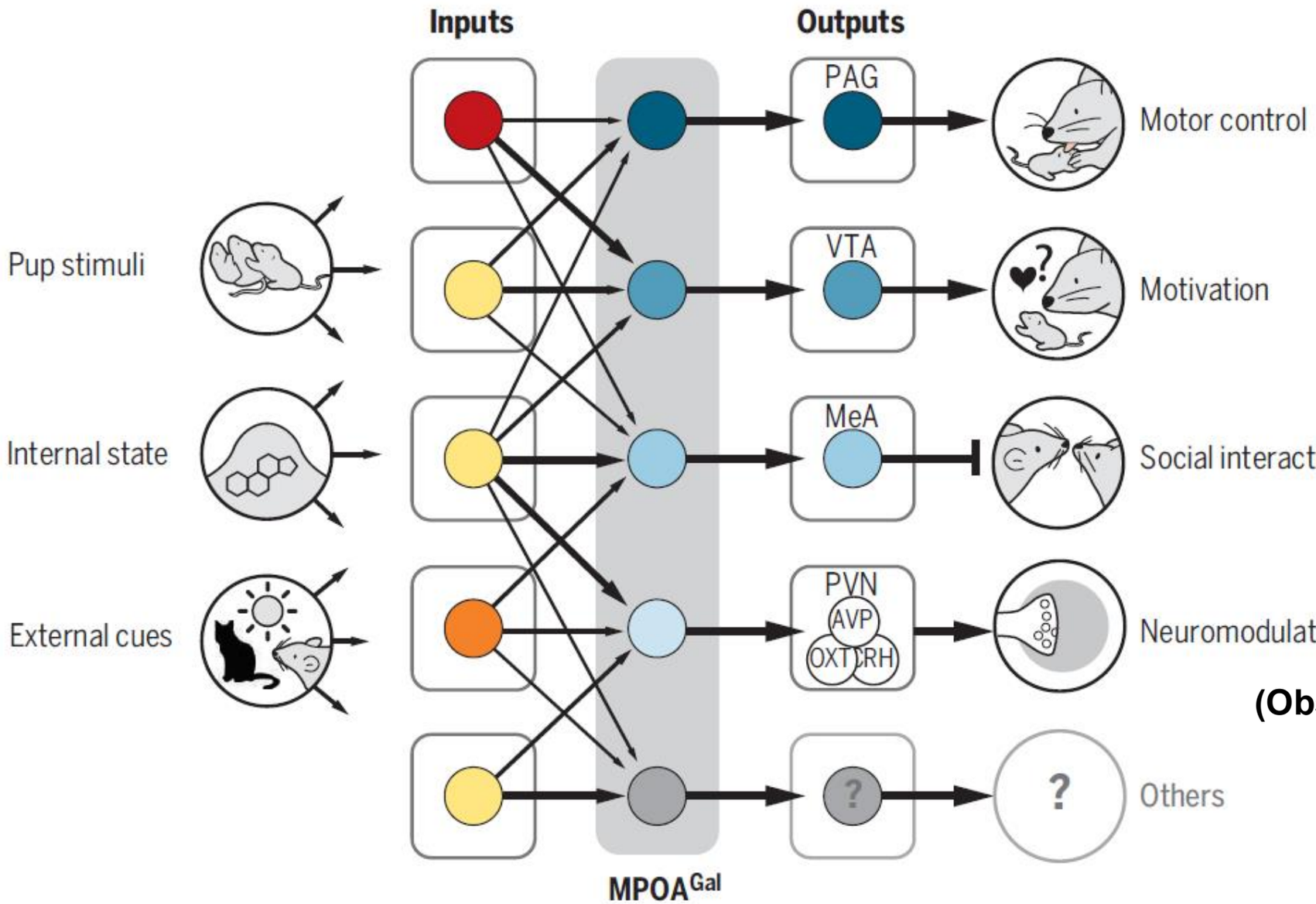




Oxytocin receptor knockout virgin mice do not learn pup retrieval

Froemke and colleagues

Deconstructing parenting



(Grooming)

(Reaching pups)

(Sniffing, retrieving)

(Observational learning of retrieval)

Questions

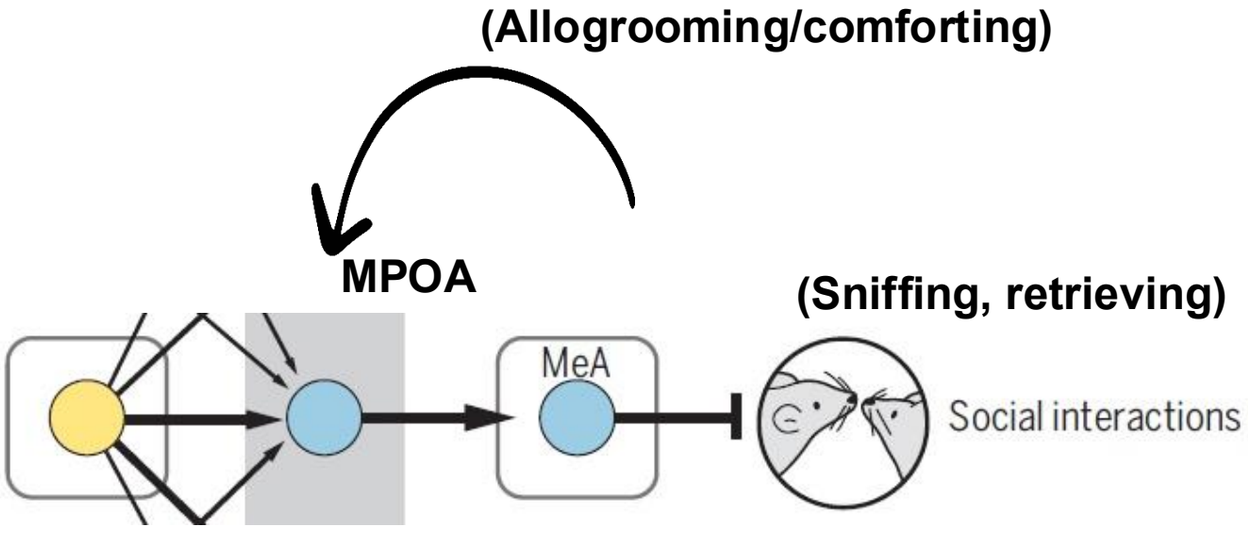
Prosocial behaviors

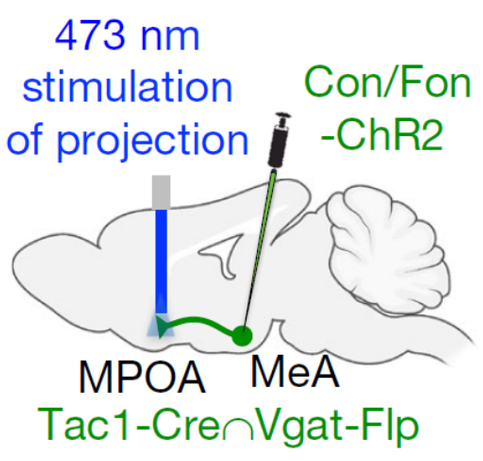
Positive Behavior – Allogrooming of stressed partner

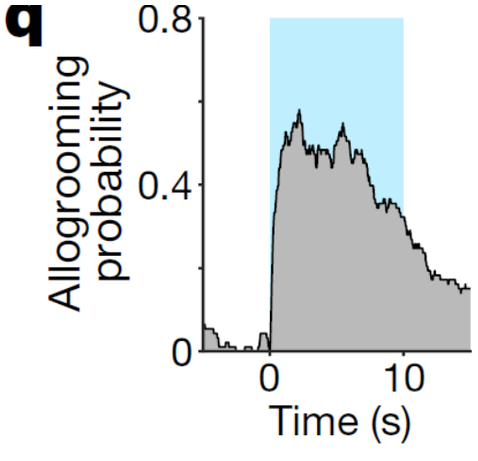


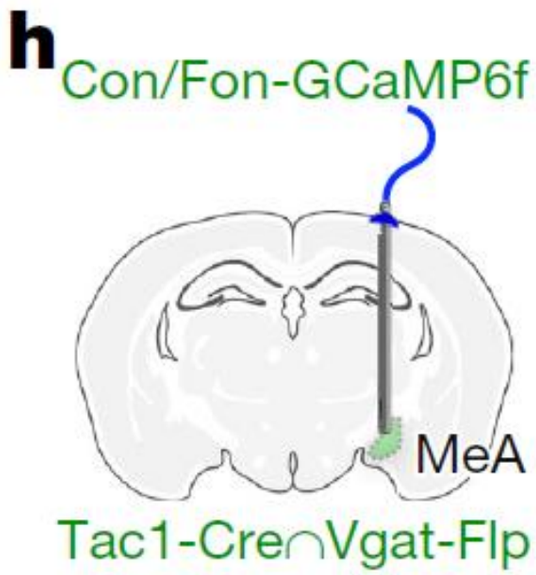
Hong and colleagues

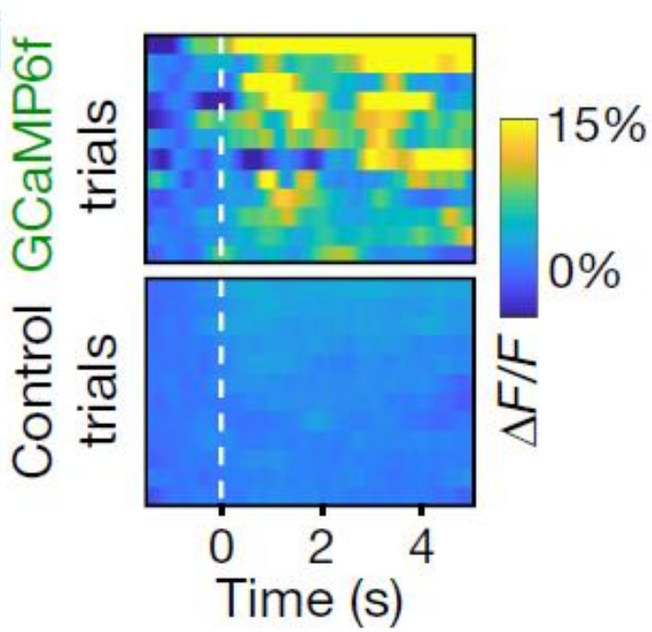
MEA to MPOA - Allogrooming

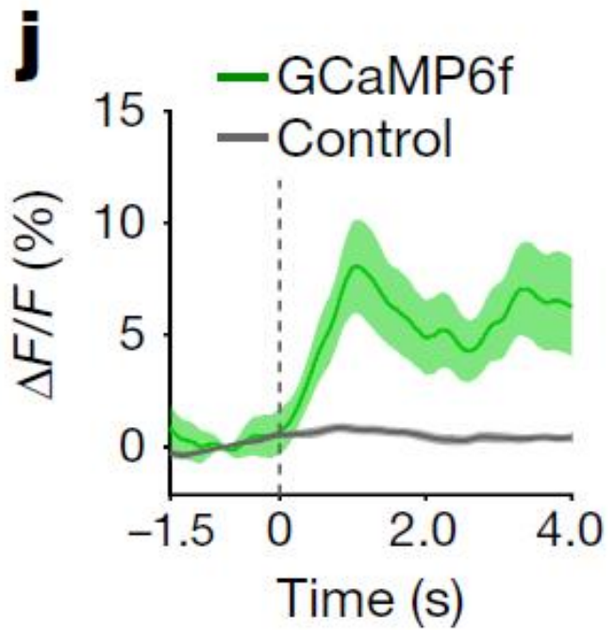




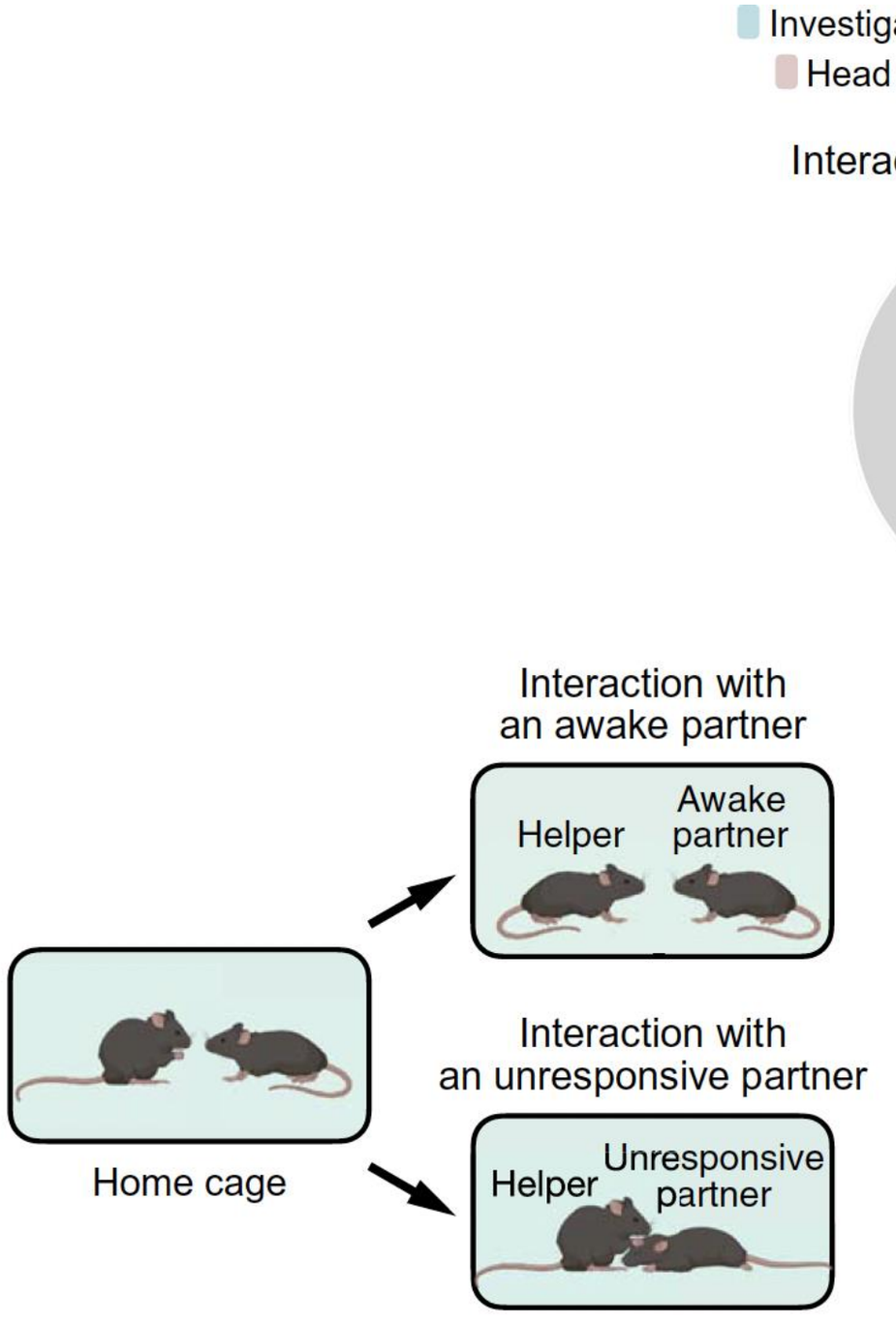


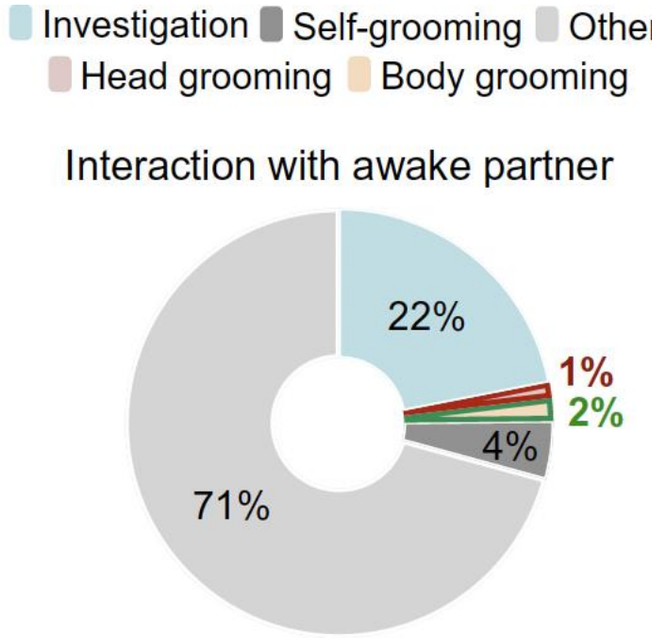


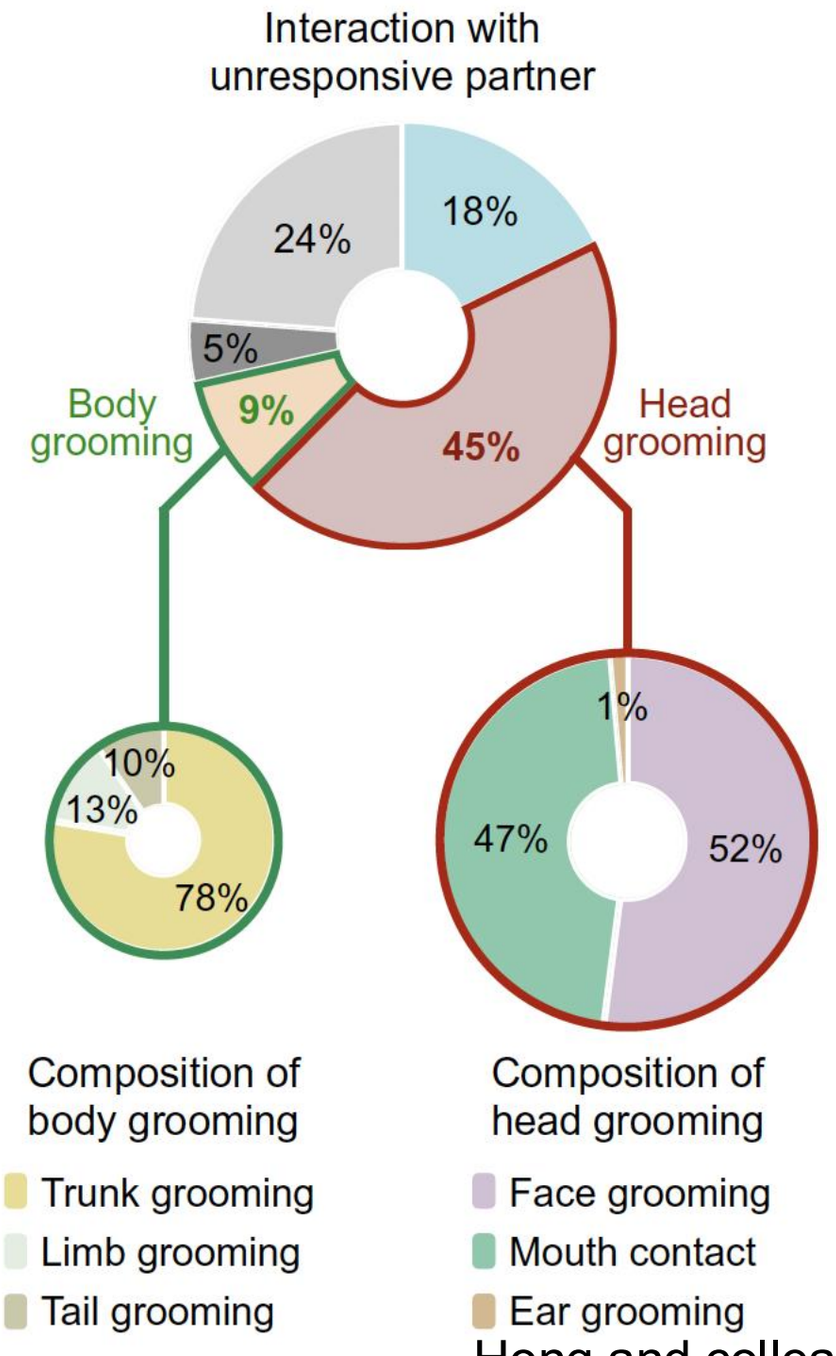




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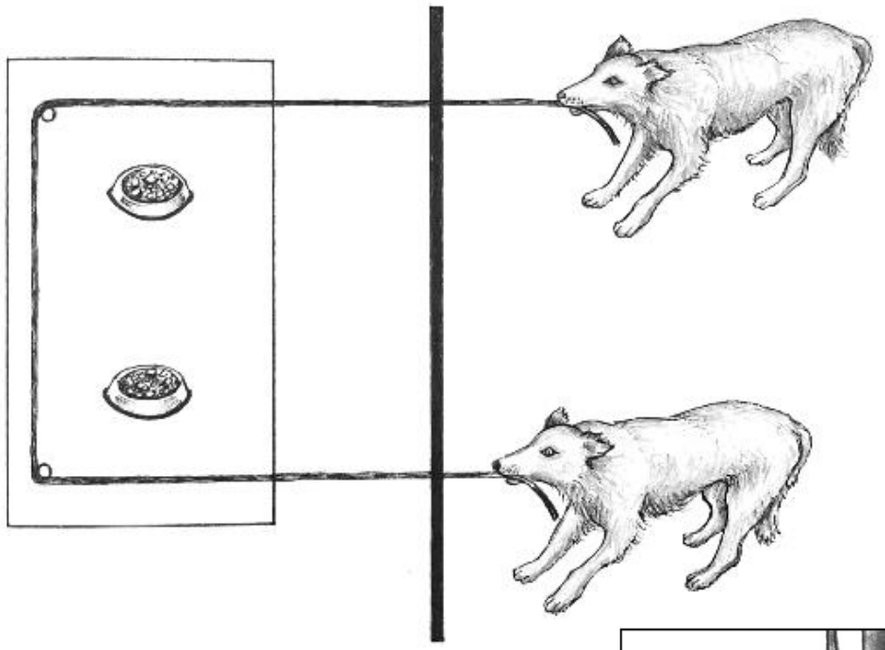
MeA neurons also involved in allogrooming unresponsive partner

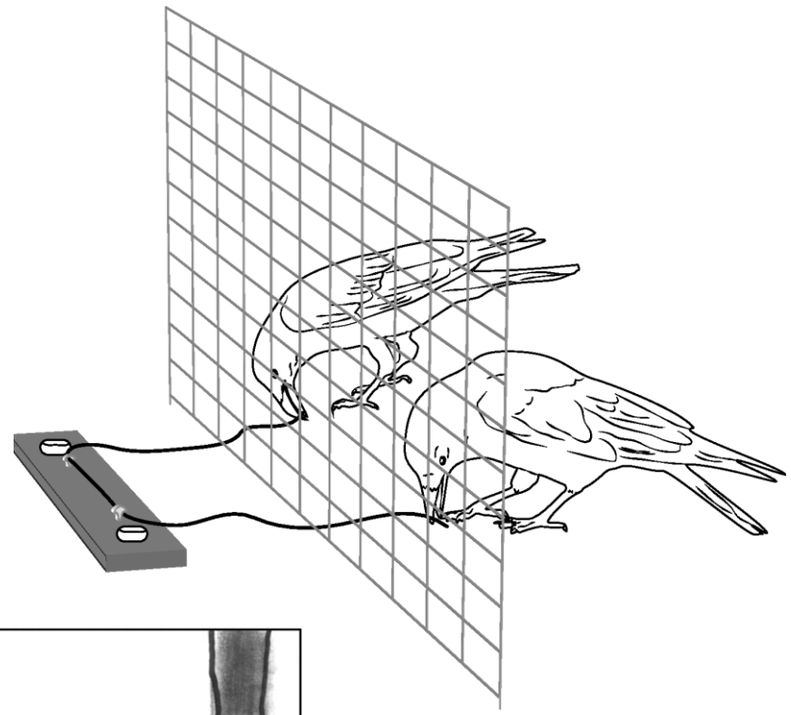
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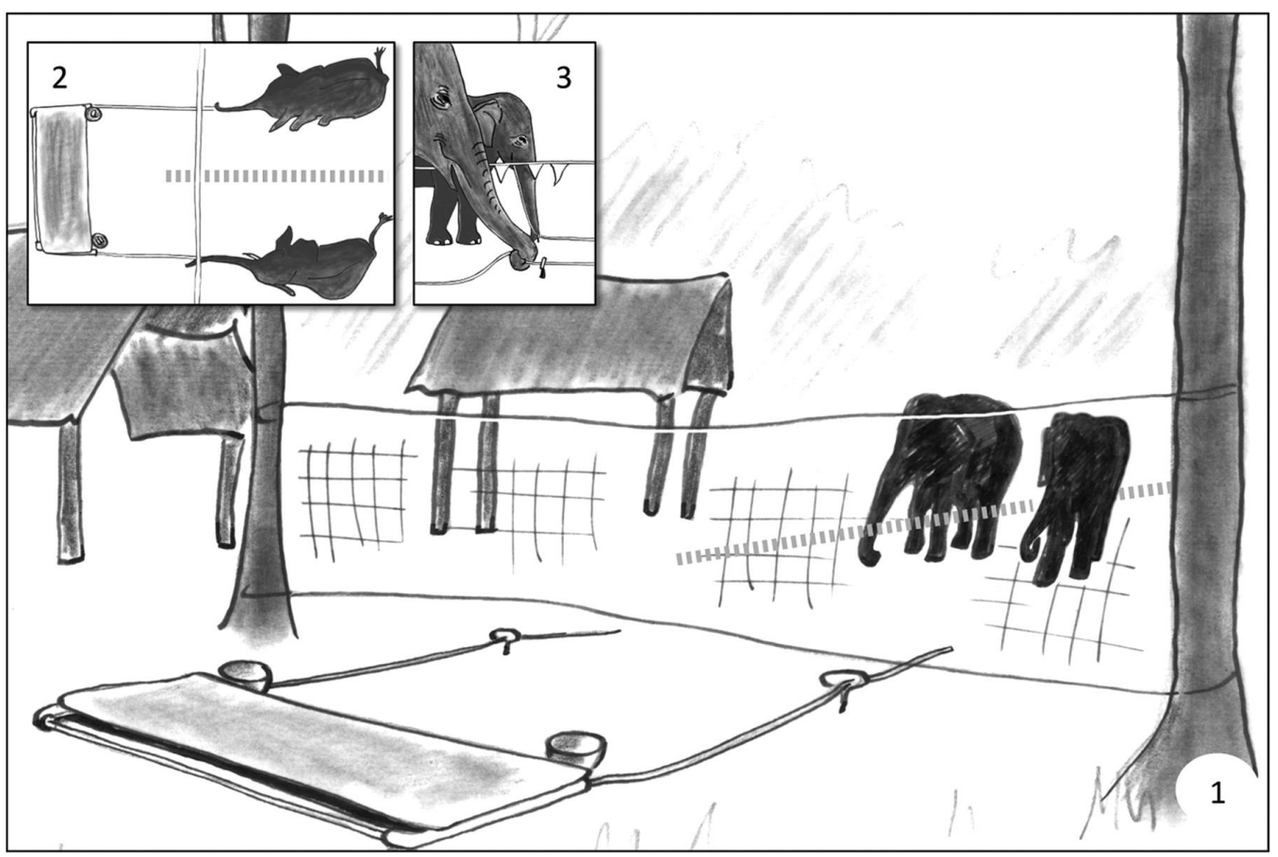
Cooperative Behavior – Chimps cooperating



Cooperative Behavior – Across species





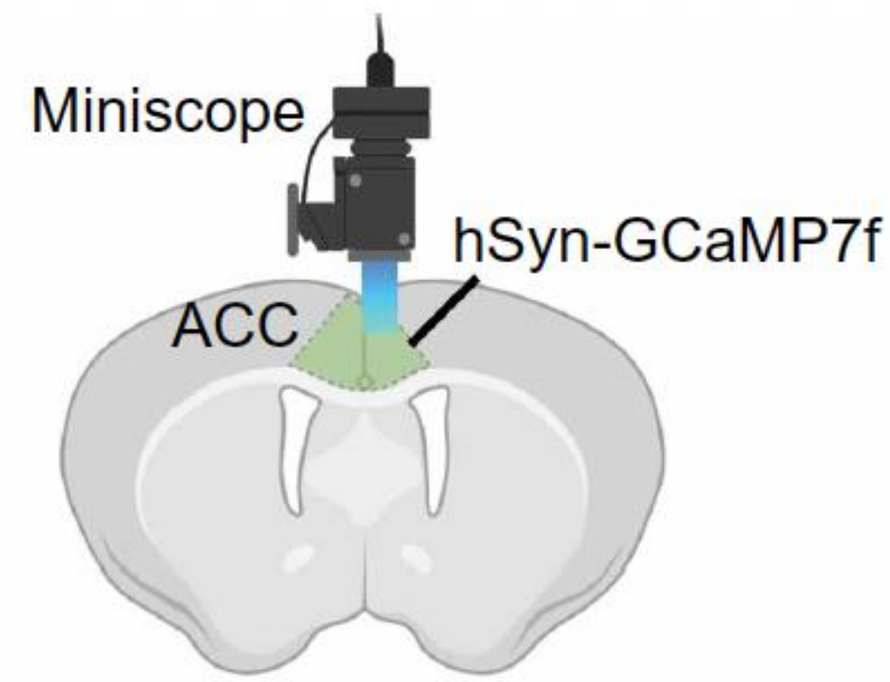


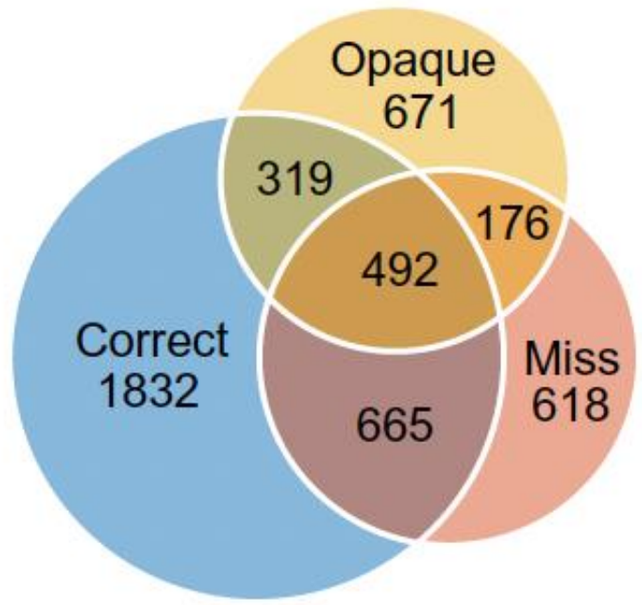
Cooperation in mice



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Neural responses in Anterior Cingulate Cortex (ACC)



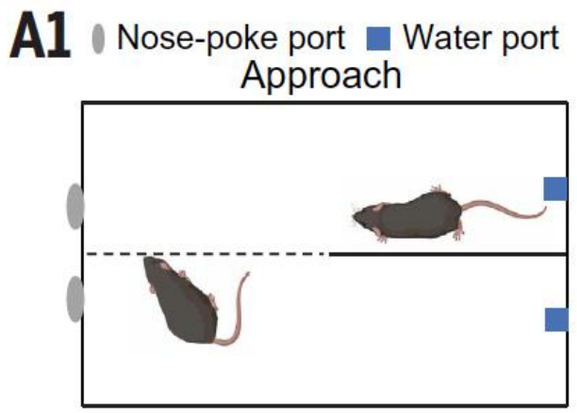


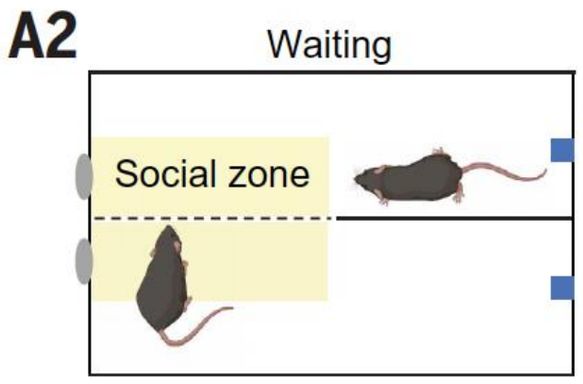
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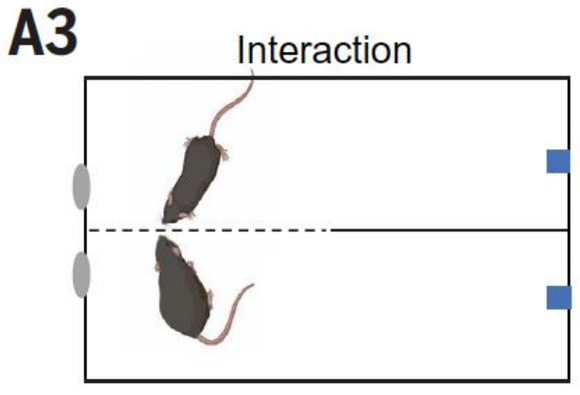
Different cooperation strategies

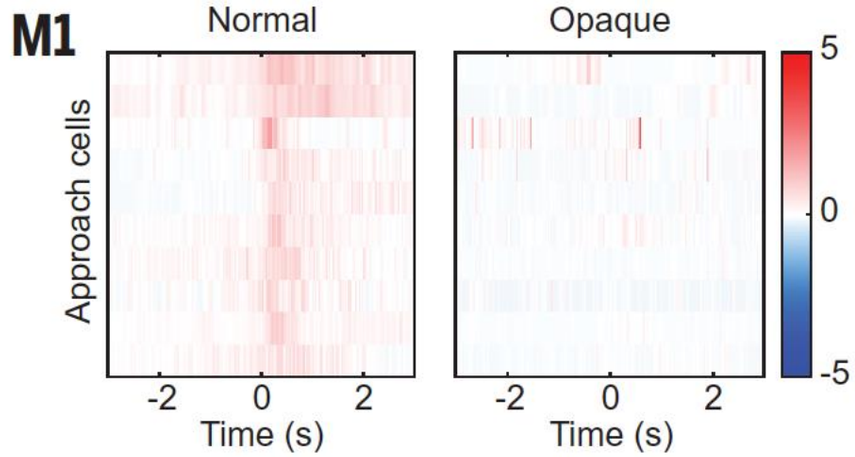


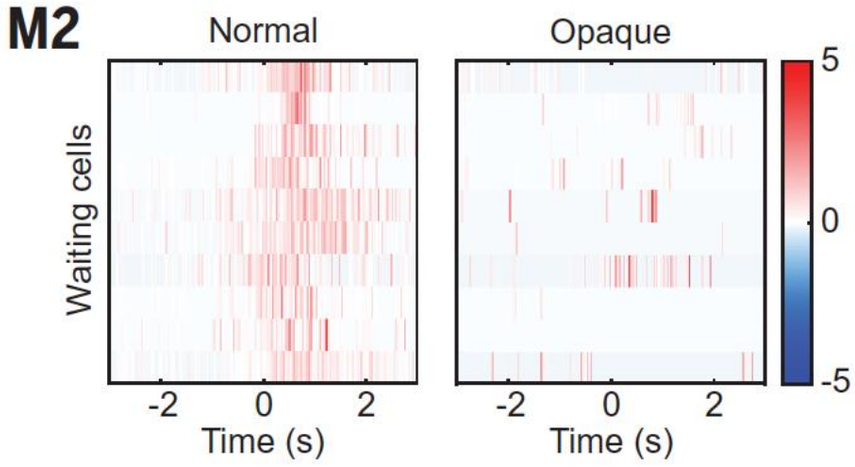
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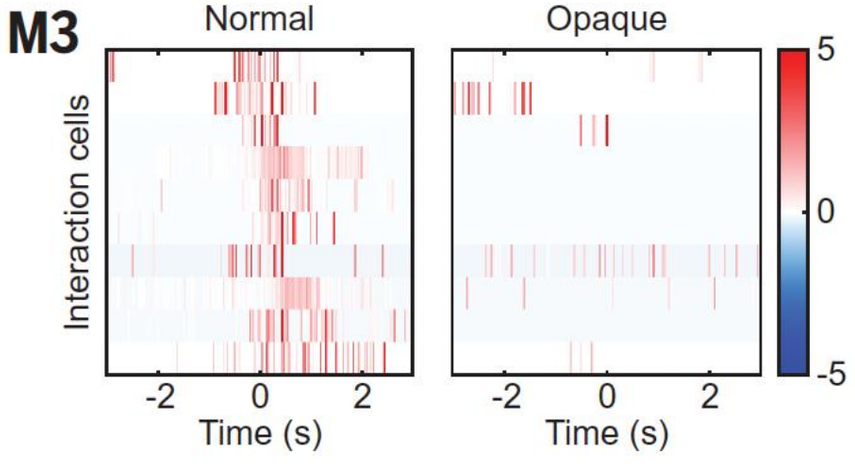






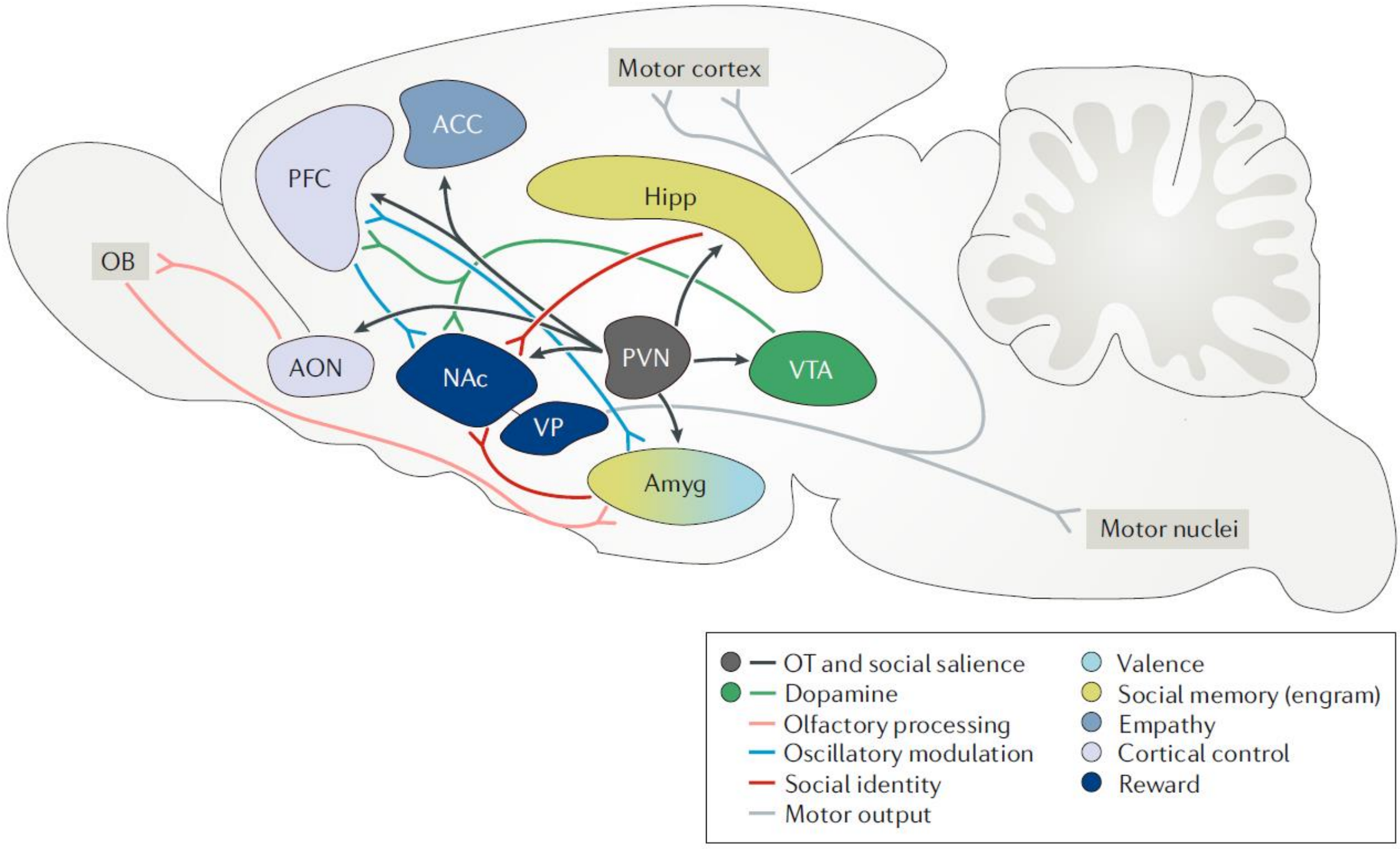






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Social representation and behavior in the brain



The future?

Neuroscience of social behaviors – What are we not looking at



Neuroscience of social behaviors – What are we <u>not</u> looking at



Group behaviors like hunting

…or communal defense

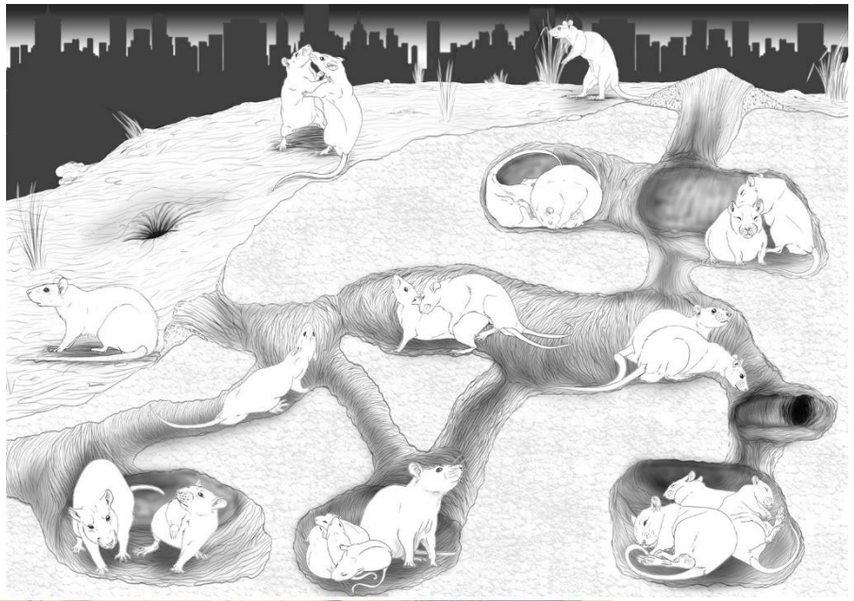


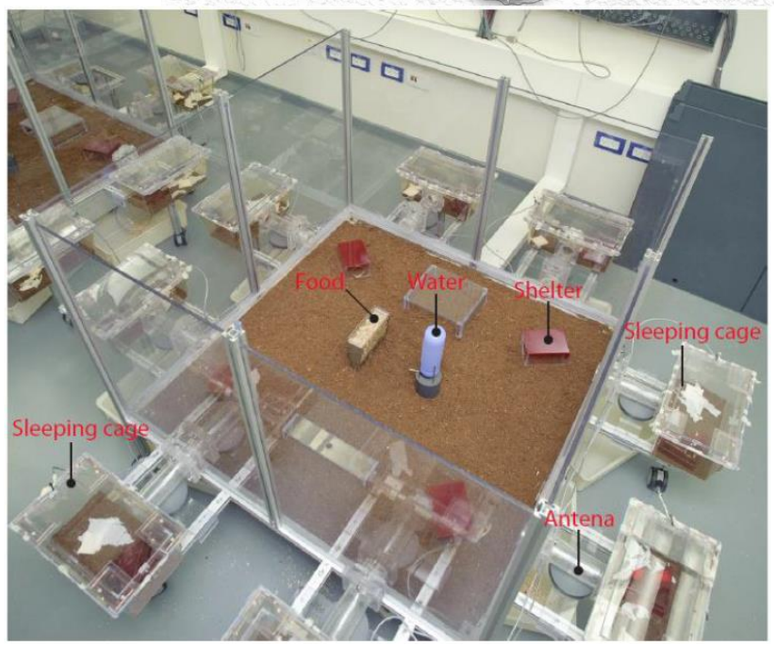
Neuroscience of social behaviors – What are we <u>not</u> looking at

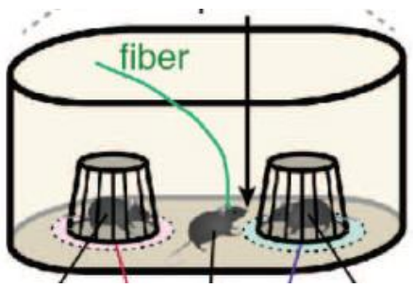


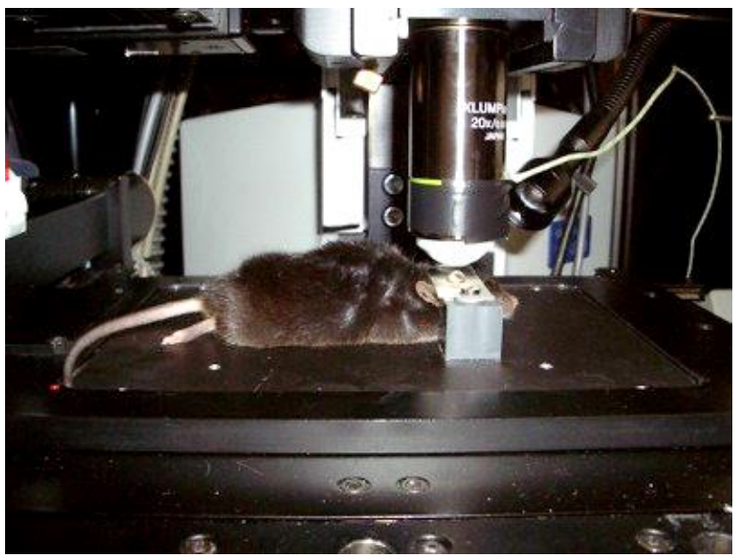
Altruistic behaviors

Neuroscience of social behaviors – Where they happen











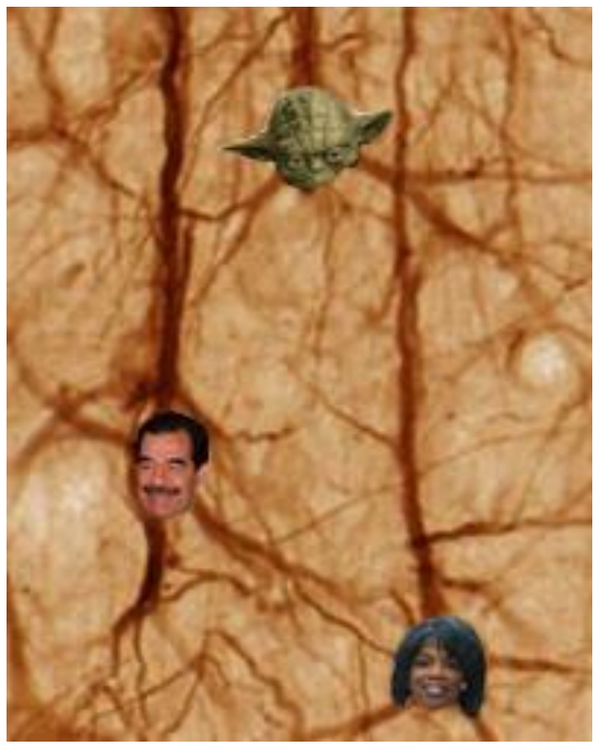
Summary

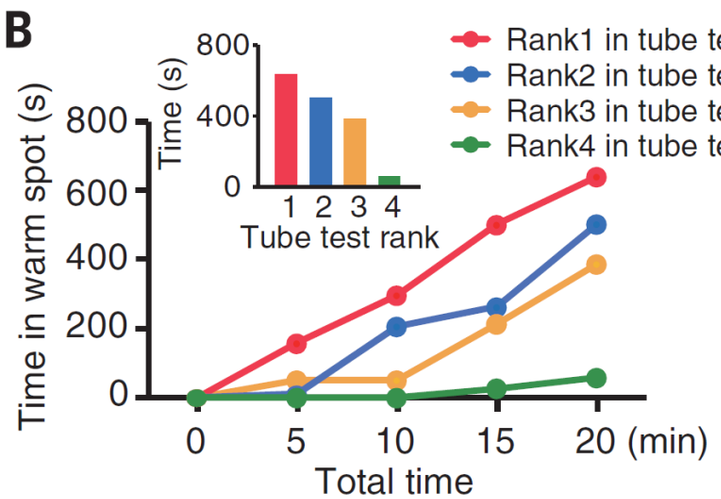











B
Time in warm spot (s)
800
600
400
200
0
0
5
10
15
20 (min)
Total time
Time (s)
800
400
0
1 2 3 4
Tube test rank
Rank1 in tube test
Rank2 in tube test
Rank3 in tube test
Rank4 in tube test

Questions